The benefit incidence of school vouchers: Evidence from private schools in India*

Mauricio Romero† Abhijeet Singh‡

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Abstract

The incidence of redistributive policies is central to whether they meet their stated goals. We study this in the context of one of the world's largest programs to improve social equity in schooling: a 25% quota in all Indian private schools for students from disadvantaged groups. We show that, although eligible groups are poorer than the state average, program benefits accrue disproportionately to better-off households: 7.4% of the program spending on tuition fees accrues to the bottom socioeconomic quintile, compared to 24.3% to the top quintile. This is consistent with selection within eligible groups at the application stage: program applicants are disproportionately drawn from more-educated and better-off households. Lottery-based estimates of the causal effect of receiving a quota seat show that three-quarters of lottery-winners would have attended a private school even without the voucher; the bulk of program spending is inframarginal for school choice. Finally, we use rich survey data to show that low application rates for poorer children reflect information constraints and application frictions. A randomized intervention providing information and application assistance substantially raises application rates but has similar estimated effects across socioeconomic groups. Our results illustrate how constraints facing potential applicants can make redistributive policies regressive in practice.

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†ITAM; J-PAL; E-mail: mtromero@itam.mx

‡Stockholm School of Economics; J-PAL; E-mail: abhijeet.singh@hhs.se

1 Introduction

Governments routinely support disadvantaged citizens in accessing essential services through targeted vouchers. However, many eligible individuals do not apply for social benefits (Currie, 2006), and the *de facto* incidence of program benefits may differ substantially from the program's intent. This may, in turn, dampen downstream effects on the intended policy outcome, namely expanding choice for poorer citizens in key sectors such as housing, healthcare, and education. Despite substantial research on interventions to improve application rates (Ko and Moffitt, 2024; Banerjee et al., 2024), the empirical importance of this link between benefit incidence, program effectiveness, and various application constraints remains unclear. However, it may be central to concerns of both redistribution and program effectiveness.

We study this link directly in the context of the world's largest school voucher policy, which provides targeted vouchers to attend private schools in India. Targeted school voucher programs are common worldwide (Urquiola, 2016) and resemble similar interventions in other domains. The Right to Education (RTE) Act, 2009, mandated a 25% quota in the student intake of all private schools for students from disadvantaged economic and/or caste backgrounds to reduce socioeconomic stratification. The state pays tuition fees (up to a cap) for students enrolled under this quota and schools are not allowed to charge top-up fees. In 2018/19, approximately 4 million students nationwide used a quota seat to attend private schools (Indus Action, 2019), which are common in India and, as in many countries, differentiated in price, quality, and amenities (Muralidharan and Sundararaman, 2015; Andrabi et al., 2025).

We study this policy in Chhattisgarh, a state with a population of ∼29 million that has implemented it since 2010. Eligible students in the state can apply for admission under the policy in either preschool grades (Nursery and Kindergarten) or in Grade 1, the formal start of primary schooling, depending on their age.[1] Program eligibility is determined by the age of the child and official documentation of household disadvantage, either on grounds of being poor ("economically weaker sections") or from a disadvantaged population group. Applications for the voucher are made online and seats are allocated using a centralized algorithm with lottery-based allocation of oversubscribed slots (Abdulkadiroğlu et al., 2017). We evaluate the benefit incidence of the policy and its effects on voucher recipients using administrative data on all eligible applications in 2019, primary survey data on voucher applicants, and a field experiment that relaxes application constraints for treated households.

We begin our analysis by computing the *de facto* incidence of policy benefits. We collected primary survey data on the socioeconomic characteristics of voucher applicants and mapped them to representative population-wide surveys. Using this, we show

[1] Applications for non-quota private school seats are decentralized. Students may seek admission for a non-quota seat in any private school and, if admitted, pay the regular tuition fees to attend.

that 7.4% of the program spending on school fees accrues to households in the bottom socioeconomic quintile, while 24.3% accrues to the top quintile. Thus, the *de facto* incidence of policy benefits is sharply regressive. This regressivity is explained by self-selection in who applies for the vouchers: although eligible groups *are*, on average, poorer than the state average household, membership in deprived groups is an imperfect proxy for socioeconomic disadvantage at the household level. Socioeconomic status varies widely within caste groups, and better-off households within eligible castes are more likely to apply — voucher applicants have significantly higher average parental education and household assets than both eligible groups and the general population.

Next, we use the lottery-based allocation of oversubscribed slots to estimate the effects of receiving a free seat on students' enrollment choices. Being allotted an RTE quota seat increases the probability of attending any private school by 25 percentage points; virtually all lottery-winning students attend private institutions, but so do ∼78% of applicants who were *not* assigned an RTE seat; the latter enroll as non-quota students who pay regular school fees. The extensive margin effect is concentrated in preschool grades, shifting some students from home care to preschool. By Grade 1, when enrollment is compulsory and near-universal, this effect is 13 percentage points.[2] Lottery winners attend schools that are, on average, more expensive and offer English-medium instruction, but much of the voucher spending is inframarginal for school choice — as an extreme, 54% of applicants who lost the lottery nevertheless attend their top-ranked school by paying full price (i.e., full inframarginality). These results are directly linked to regressive selection: private school attendance in representative state-level data rises sharply from ∼5% in the bottom decile to ∼64% in the top decile. Thus, the over-representation of wealthier households in the applicant pool raises distributional concerns *and* reduces the policy's effectiveness.

These muted effects are striking for a policy intended to substantially expand school choice for students from disadvantaged backgrounds. It is unsurprising that wealthier households, who would otherwise send their children to private schools, apply for free seats when eligible. The puzzle is why poorer households, who are more likely to be eligible for these seats, do not apply in greater numbers. In the third and final part of our analysis, we focus on barriers to application that may explain this regressive take-up.

Investigating barriers to applying requires a representative sample of households with children of eligible ages, not just those who applied. We collect this data through household listing in randomly sampled clusters in urban and rural areas of Raipur, the most populous district in Chhattisgarh. Survey evidence in our sample (N=1,059) suggests that frictions to applying are common and empirically important: for instance, we find that 76% of Scheduled Caste and Scheduled Tribe households with children

[2]These causal effects are only identified for the students who applied to oversubscribed schools, who only represent ∼1/3 of the applicants. We discuss the external validity of these results to all applicants in Section 5.3. Specifically, effects are similar across most lottery-risk groups, and rich vectors of household characteristics explain little heterogeneity.

in fee-charging private pre-primary or Grade 1 classes had *not* applied or heard of the policy. The frictions we investigate disproportionately affect poorer households.[3] The first is information: 65% of parents in the top decile had heard of the policy, whereas only 20% of parents in the bottom decile reported the same. The second is the ability to apply online: Nearly all high-socioeconomic-status households reported access to the internet at home, compared to only 9.1% of households in the bottom decile; low-SES households were also less likely to report having ever applied for a government benefit online and fewer of them can access support from others for doing so. Both these patterns hold across rural and urban areas, and with fixed effects for sampling clusters.

Following this survey, we implemented a randomized field experiment with households in our sample that had a mobile phone (N=914). We randomized ∼50% of households in each locality (N=459) to receive an intervention that provided information about the policy when applications were open and, if the household reported having the required documents to demonstrate eligibility, offered in-person assistance with completing the application. This intervention resembles common approaches to reducing application barriers delivered by NGOs and local governments and, reflecting both common practice and typical constraints in verifying eligibility, was randomized across the entire sample without screening for household characteristics from the baseline survey. We expected the intervention to boost application rates (our primary outcome). Further, we expected treatment effects to be larger for poorer households — since they are more likely to be eligible, to lack information about the policy, and to face barriers to applying online — and thus improve progressivity even though the intervention did not disproportionately target poorer households.

Assignment to the intervention boosted application rates by 9.4 percentage points, which is a 43% increase over a control mean of 22%. The increase in application rates is nearly identical to the proportion of randomly assigned households that we assisted in submitting applications. This suggests that information and application frictions deter applications in this setting, even a decade after the law was introduced, and our intervention did not crowd out alternative sources of application assistance. However, contrary to our priors, treatment gains were not larger for poorer households even though information constraints and barriers to online application were more severe. The proximate explanation for this pattern is a further regressive constraint: the administrative burden of certifying eligibility. Richer households were more likely to have all documents — 56% for the top decile compared to 24% for the bottom decile — which largely reflects difficulty in obtaining documents rather than "true" ineligibility. Thus, despite being willing to apply, poorer households were disproportionately less able to submit applications.

[3] An alternative explanation would be that poorer households do not want to enroll their children in private schooling. We discuss this extensively in Section 5.4 and in Appendix D, and show that low parental demand is not a constraint on filling quota seats with students from poorer households.

Our results provide new evidence on one of the world's most ambitious policies to promote social mobility through school education: if the program were fully implemented across all states, projections suggest it would cover 16 million direct beneficiaries in a year (Indus Action, 2019). Through potential effects on peers (Rao, 2019), this could affect the entirety of the private schooling sector, which accounted for ∼95 million children in K-12 education in 2020–21.[4] Despite the importance of this policy, however, empirical evidence on its effects has remained scarce.[5] We provide evidence on the incidence of policy benefits, effects on school choice for current beneficiaries, and on potential supplementary interventions that could boost its redistributive potential.[6]

This evidence speaks to global debates and relates closely to three strands of research on promoting social equity in education. First, our results relate directly to a large literature on policies that combine public funds with the private provision of schooling (e.g., Epple et al. (2017); Romero et al. (2020); Romero and Sandefur (2021); Cohodes and Parham (2021)). Although these programs are often motivated by the goal of providing access to underserved groups, the literature has largely focused on realized treatment effects among adopters. We focus, instead, on the *incidence* of policy benefits and to demonstrate that regressive take-up can substantially blunt the redistributive goals of even targeted voucher programs. Our results are closest to Walters (2018), who shows that disadvantaged students have low demand for charter schools, despite having larger gains from attending. Importantly, however, the mechanisms underlying the same policy outcome appear different: while Walters (2018) attributes undersubscription by disadvantaged households to low demand, the incidence of multiple constraints is more consequential in our setting. Understanding the causes of differential take-up is, thus, central to identifying appropriate interventions to boost progressivity.

Second, and relatedly, we contribute to the literature evaluating interventions to promote school integration (see, e.g., Bergman (2018); Campos and Kearns (2022)) and a broader literature studying social stratification in school systems.[7] While this literature often attributes stratification to residential sorting or heterogeneous preferences for school

[4] As a benchmark for scale, the ∼4 million students enrolled on RTE quotas in 2019 exceed total enrollment in US charter schools (∼3 million, (Cohodes and Parham, 2021)) or any voucher program internationally (Epple et al., 2017). Enrollment in Indian private schools exceeds total enrollment in most education systems.

[5] The voucher experiment in Muralidharan and Sundararaman (2015) was inspired by the (not-then-enacted) proposals for the RTE quotas. However, as a stand-alone intervention with substantial outreach through household visits, it did not study incidence or effects on school choice in steady-state (our primary focus). See also Damera (2017), who studies RTE quotas in Karnataka, and more recent work by Sahai (2025) and Agarwal (2024) on Madhya Pradesh and Mumbai who also examine effects on enrollment. These papers do not examine the incidence of public spending or the reasons for undersubscription.

[6] Consistent with our findings, Edara (2025) finds that while the policy improved private school participation for Scheduled Castes and Other Backward Classes, it led to only small reductions in the uneven distribution of students by caste across schools, with no substantial changes in overall private-public segregation.

[7] See, for instance, a vast literature on racial and socioeconomic segregation in US schools (e.g., Clotfelter (2011); Reardon and Owens (2014)) and in other settings (e.g., the UK (Jenkins et al., 2008), Chile (Hsieh and Urquiola, 2006), and Scandinavia (Söderström and Uusitalo, 2010)).

quality, our findings highlight the critical role of administrative and informational frictions in generating exclusion. They also flag possible constraints on current school integration initiatives and measures to address them, which may be feasible even within current policy rules. In our setting, these improvements may benefit poor students directly through skills with labor market returns (Muralidharan and Sundararaman, 2015; Singh, 2015), and richer students through peer effects (Rao, 2019).

Third, we relate to an extensive literature on affirmative action in education (e.g., Arcidiacono and Lovenheim (2016); Bagde et al. (2016); Bleemer (2022); Otero et al. (2021); Mello (2021)). In contrast to the exclusive focus on college admissions in this work, we provide unique evidence at *school entry age*. This distinction is important because concerns of academic mismatch and controversy about the fairness of 'non-merit' admissions criteria are both more muted at the start of formal education than in college. Our results also relate to the 'creamy layer' problem — a central concern in the economics of affirmative action whereby benefits are captured by the most advantaged members of the target group (Sowell, 2004; Schotte et al., 2023; Arcidiacono et al., 2015). However, unlike affirmative action policies in higher education in India, where there is limited evidence of this creamy layer problem (Bertrand et al., 2010; Frisancho and Krishna, 2016), we find beneficiaries come from the top quintiles of the socioeconomic distribution.[8]

Finally, beyond education, we also directly contribute to the literature on the take-up of social programs across sectors, and specifically to the role of application frictions. Our results contrast with influential studies in development economics that find that self-targeting and/or ordeals improve progressivity (e.g., Alatas et al. (2016)), or that low adoption rates are caused by low demand (Mobarak et al., 2012; Cole et al., 2013); instead, they resemble results from studies of application frictions in social programs in the US.[9] Results from our randomized intervention, highlighting multiple constraints, are perhaps closest to Banerjee et al. (2021), who find similar results with interventions to broaden enrollment in universal health insurance programs in Indonesia. However, we show that the incidence of multiple constraints affects not just overall application rates but also the *composition* within the eligible groups (which is central to redistributive programs and, here, also leads directly to inframarginality of program spending). Thus, optimal interventions to address regressive selection will likely differ from those focused solely on increasing program take-up, a distinction that is critical yet underemphasized.

[8]While the 'creamy layer' problem is a common criticism of affirmative action policies around the world, well-identified evidence of its existence is scant, probably because of the data requirements to study this question. However, evidence from higher education in the U.S. (Bleemer, 2023; Arcidiacono et al., 2022) suggests this is a relevant issue, whereas evidence from higher education in Brazil (Francis and Tannuri-Pianto, 2012; Melo, 2025) suggests it is not.

[9]For US-based evidence on the importance of application frictions, see, e.g., Currie (2006); Finkelstein and Notowidigdo (2019); Deshpande and Li (2019); Bhargava and Manoli (2015). In education, focusing on college financial aid, see especially Bettinger et al. (2012) and Dynarski et al. (2021). For evidence from developing countries on self-targeting and ordeals, see, e.g., Ravallion (1991); Besley and Coate (1992); Alatas et al. (2016). The framework in Finkelstein and Notowidigdo (2019) can reconcile these contrasting results.

We discuss the design of such interventions in Section 7.

2 Policy Background and Context

2.1 Private schooling in India

Fee-charging private schools in India account for ∼31% of primary school enrollment in rural areas and ∼50% in urban areas (Pratham, 2019; Kingdon, 2020). At the primary level, private schools typically have higher enrollment and more teachers than government schools. They are also more likely to include an integrated preschool section, use English as a medium of instruction, and offer single-grade classrooms and subject-specific teachers. Teachers in private schools are typically younger, less likely to have completed formal teacher training, and are paid substantially less than teachers in government schools. However, they are less likely to be absent from school or engaged in non-teaching activities. Kingdon (2020) provides a review of the private schooling sector in India.

Evidence on relative differences in quality is scarcer. Cross-sectionally, students in private schools score higher than those in public schools, but this difference is confounded by selection: private school students are more likely to be from higher SES households, urban, and male, and likely also selected on unobserved characteristics. Muralidharan and Sundararaman (2015) provide the best estimates of the causal effects of attending a private school in rural Andhra Pradesh and find that there are large effects on English and the national language (Hindi) with no evidence of effects in math and the local language; similar results are shown by Singh (2015) using value-added models in the same state. More recently, Berg et al. (2025) show in rural Tamil Nadu, a neighboring state, that while private and government schools have similar average test score value-added in math and lower in Tamil in primary school grades, there is a stark productivity difference at pre-primary ages: private preschools outperform government-run institutions by about 0.6–0.7σ in both math and local language.[10] These results refer only to average differences across sectors. There is no causal evidence on the distribution of school-level productivity *within* the private and public sectors in India. The closest evidence from a similar setting is provided by Andrabi et al. (2025) from three districts in rural Punjab in Pakistan. They find substantial heterogeneity in schools' within-sector productivity. Private schools have a higher average value-added, but there is substantial overlap across sectors.

2.2 Quotas in the Right to Education Act 2009

The Right to Education (RTE) Act, enacted in 2009 by the Parliament, sets the regulatory framework for compulsory schooling in India. Clause 12(1)(c) of the Act requires all fee-charging private schools, except minority-run institutions, to *"admit at least*

[10]While Berg et al. (2025) do not provide a decomposition of the sources of the higher private sector value-added in preschools, a likely reason is that public preschools spend only about 38 minutes per day on early childhood education (Ganimian et al., 2024).

25% of the strength of class I, children belonging to weaker section and children belonging to disadvantaged group from the neighborhood and provide them free and compulsory education till completion of elementary education. Further, where the school admits children at pre-primary level, such admissions will be made at that level". Here, "weaker section" typically refers to income-poor households and "disadvantaged groups" to castes and tribal groups that were historically discriminated against.[11] The government reimburses private schools for tuition fees of quota students admitted at notified levels.

This clause combines features of distinct classes of educational policy. Echoing school integration reforms elsewhere, program guidelines stress "the need for moving towards composite classrooms with children from diverse backgrounds, rather than homogeneous and exclusivist schools". Operationally, it combines quota-based affirmative action (which has a long history in India) with provisions that resemble voucher programs elsewhere and charter schools. While similar in many respects, it differs from typical voucher programs in that the offer of a quota seat is school-specific and not transportable to other schools.[12] It differs from charter schools in that lottery-based enrollment and government reimbursement of fees only apply to quota seats — private schools independently choose who to admit on non-quota seats and the fee levels they charge.[13]

Individual states in India retain substantial *de facto* power to decide how to implement the policy, including defining admissions procedures, reimbursement rules, and the precise definition of eligible groups. This provision has been contentious with multiple court challenges and, in practice, adoption has been partial and staggered across states and remains unimplemented in some.[14] In 2018–19, ∼4 million students enrolled in an RTE quota seat; full national implementation would cover an estimated ∼16 million children annually (Indus Action, 2019).

[11]The quota additionally covers children with medically-attested disabilities, orphaned children, and HIV-positive children. These latter groups account for a small share (1%) of applicants nationwide (Indus Action and Georgetown University, 2025).

[12]Following policy advocacy, this is now changing in some states: e.g., from 2025, the state of Odisha has allowed transfers across private schools without losing the RTE quota seat. In most states, including Chhattisgarh, the voucher is school-specific.

[13]While unique in its scale and application to all private schools regardless of funding source, the RTE quota has parallels in other developing countries. Pakistan's Punjab Free and Compulsory Education Act (2014) requires private schools to admit 10% of students from disadvantaged neighborhoods, though implementation remains low (Provincial Assembly of the Punjab, 2014). Nepal's Compulsory and Free Education Act (2018) requires private schools to allocate 10-15% of seats as scholarships for economically disadvantaged students (Government of Nepal, 2018). Chile's Ley de Inclusión Escolar (2015) requires publicly subsidized private schools (which enroll over half of Chilean students) to reserve at least 15% of vacancies for students from the most vulnerable households (Congreso Nacional de Chile, 2015). Mexico requires all accredited private schools to offer scholarships to at least 5% of enrolled students (Cámara de Diputados del H. Congreso de la Unión, 2019).

[14]The Supreme Court affirmed the constitutionality of the provision in 2012. However, as with many desegregation policies elsewhere, universal adoption did not immediately follow the ruling.

2.3 Quotas in Chhattisgarh: context and lottery design

Our study is based in Chhattisgarh state, which had a population of ∼29.4 million in 2020. ∼40% of the population was estimated to be below the poverty line in 2011 (compared to ∼22% nationally) and the state ranked 21st (out of 28) in its achievements of the UN's Sustainable Development Goals in 2019 (NITI Aayog, 2020).

Chhattisgarh has implemented the RTE quota since 2010. The government reimburses school fees for students admitted under the quota up to an INR 7,000 cap and provides student grants for books and uniforms. Schools cannot charge top-up fees (even if the school fees exceed the reimbursement cap).

Eligibility for the program combines two conditions. The first relates to students' age: students must be aged between 3–4 years to apply for admission in Nursery, 4–5 years for admission in kindergarten (KG), and 5–6.5 years for admission in Grade 1. The second requirement is to provide official documentation of household disadvantage, either on the basis of being income-poor ("economically weaker sections") or from a historically disadvantaged population group ("disadvantaged group"). In Chhattisgarh, demonstrating economic disadvantage requires having access to a card certifying "Below Poverty Line" status as determined in administrative surveys.[15] Documenting membership of a "disadvantaged group" requires showing "caste certificates" issued by the state government that certify the household as belonging to Scheduled Castes (SC) or Scheduled Tribes (ST).[16]

Applications for the program are open for roughly four weeks in February–April of each year — the exact dates vary across years — for enrollment in the subsequent school year, starting in July. These applications are made through a centralized application portal that shows parents which schools have seats available (based on where they live) and requires them to submit a rank-ordered list. Parents can do this on home devices or at internet cafes and government-authorized "choice centers", which are effectively private internet cafes registered and trained to help citizens apply for benefits through government websites for a fee.

The allocation of applicants to schools is also centralized. In 2019, the state used an Immediate Acceptance algorithm to allocate quota seats.[17] Specifically:

[15]This status was based on proxy-means tests using data collected by administrative surveys in 2002 (rural) and 2007 (urban) and/or the socioeconomic Caste Census in 2011. The precise documents used to demonstrate eligibility have undergone modest changes over the years.

[16]These groups are recognized by the Constitution of India as historically disadvantaged. They are entitled to quotas in political representation, education, and employment. 43.4% of the population in Chhattisgarh in 2011 belonged to these groups compared to about one-quarter nationally.

[17]Compliance with the 25% rule is enforced through the centralized portal: each school declares its entry-level capacity, the state designates 25% of these seats as RTE quota seats, and these are filled exclusively through the allocation algorithm described below. Schools cannot count disadvantaged students admitted outside the portal toward the quota. When applications to a school fall short of its declared quota capacity, the seats simply remain vacant.

1. Parents rank as many private schools in their catchment area as they want, in their order of preference. They may apply for their child's admission to one of three grades — Nursery, Kindergarten (KG), or Grade 1 — based on their child's date of birth.
2. All students are assigned to their first-preference school if it is not oversubscribed. No priority is given to students with enrolled siblings, living nearby, or otherwise.
3. Students whose first-preference school is oversubscribed enter a lottery (separate for each grade). Each child is in only one school-grade lottery per round.
4. Schools with filled quotas and allocated students are removed.
5. Steps 2–3 are repeated for unassigned students, treating the next school in their preference list that is not full as their "first preference", until either all students are assigned, all schools are filled, or there is no possible match.

The algorithm used for allocations was never mentioned in publicly available documents, beyond the stipulation that allocations would be lottery-based. Further, most parents only apply for a single private school: thus, despite the algorithm not being strategy-proof, we do not expect algorithm-related strategic considerations to be important in this setting.[18]

3 Data

Our analysis combines several datasets from administrative sources, official population-level censuses and surveys, and primary data collected by the research team. We discuss these in three distinct groups: (i) statewide data pertaining to RTE applications in 2019, (ii) state-level representative survey data, which we use to map applicants to the broader population in the state, and (iii) data collected for a field experiment we conducted to understand barriers to applications. Figure A.1 has a summary of each dataset and a timeline. Table 1 below provides an overview of how each dataset is then utilized in the empirical investigations.

[18]We do not use the application data to infer preferences, only as a source of conditional random allocation of quota seats. In 2020, partly in response to feedback from our research team, the state adopted the (strategy-proof) Random Serial Dictatorship mechanism to allocate seats.

Table 1: Analysis framework

Question	Sample	Method
Who receives the benefits?	Statewide records for the universe of applicants, surveys to a subset of these applicants, data on schools characteristics (U-DISE) and school fees, and representative population data from the National Family Health Survey 2019–21 and the National Sample Survey 75th Round	Benefit-incidence calculations using socioeconomic characteristics and school fees
How does a quota seat change school choice?	Lottery-exposed applicants: 16,703 targeted for the outcome survey, with outcomes observed among survey respondents	Lottery ITTs and IV estimates of securing an RTE seat, conditional on assignment-probability strata.
Why do households not apply, and does assistance help?	1,059 Raipur households, including nonapplicants; 914 in the randomized intervention.	Descriptive data from the survey and randomized effects of information and application assistance.

3.1 Statewide data on RTE applications in 2019

Our analysis of RTE applications focuses on the 2019 application cycle. We use data from three sources: (i) application data provided by parents in 2019; (ii) two rounds of phone-based survey data collected by the research team to study enrollment; and (iii) administrative data on school characteristics. We describe each of these sources below.

3.1.1 Administrative data on applications

We obtained data for all eligible applications submitted in 2019 through the online allocation system to implement the RTE in Chhattisgarh. The data include parents' school rankings, the assigned school (if any), and limited household characteristics, including phone numbers. In addition to data on students' applications, we also received data on the number of seats available in each grade at each private school registered in the administrative system. Parents applied to schools in March–April 2019 and were notified of the school assignment in May. Enrollment decisions were finalized by July 2019.

In 2019, valid applications were received from 54,676 eligible students, 7,217 of whom were not allotted an RTE seat in the first round (see Table A.1; Panel A). Nearly half (48%) of the applicants were female and 56% lived in a rural area. More than 50% of applicants have only one school on their preference list, and 92% have at most three preferences.[19] There were 5,863 schools in the lottery, with an average of 10 seats available but with 15 students applying. Schools are more likely to have seats available in Nursery than in Grade 1 (see Table A.1, Panel C).

[19]Table A.1 provides further details on the characteristics of applicants and schools.

The availability of applications, combined with the availability of seats in each school/grade combination, allows us to classify students by lottery risk. Specifically, we simulate the allocation 10,000 times and classify students into two groups: (i) those who are always allotted to some private school with certainty ("deterministic applicants", N=37,973), and (ii) those who face a positive probability of being left unallotted ("stochastic applicants", N=16,703). One-third of the stochastic applicants were left unallotted (see Table A.1, Panel B). The proportion of girls in the subsample and the number of schools applied to are similar to those of the full set of applicants. However, since urban areas have more oversubscribed schools, the subsample is more urban and, relatedly, has a lower proportion of Scheduled Castes and Scheduled Tribes.

3.1.2 Primary data from phone surveys

All applicants were required to provide a phone number to apply.[20] We used these phone numbers to survey parents on the household's schooling choices, socioeconomic background, and educational inputs. Data from these phone surveys are central to our analysis of incidence and the effects of winning the lottery.

First, between August and September of 2020, we attempted to call all individuals (N=16,703) with lottery-based variation in whether they received an RTE offer. We collected information about which school the applicant eventually enrolled in for the 2019–20 school year, along with basic school characteristics (e.g., medium of instruction and fee level) and household characteristics (parental education and occupation). We made up to five attempts to reach each household and completed interviews with about 44% of the targeted households.

Between November 2020 and January 2021, we conducted a further round of phone surveys to collect detailed information on household assets, which we use to compare the socioeconomic status of applicants to that of the eligible population. We attempted to recontact all households interviewed in the first phone survey and completed interviews with 59% of them. Further, since the original phone survey sample only focused on applicants with lottery-based variation, we also attempted to interview a random sample of 1,203 applicants from the non-lottery sample (who had a probability of one of being assigned to a private school). Of these 1,203 students, 462 completed our phone survey.

A concern is whether, given the non-response rates, respondents in our phone surveys are representative of the full applicant population.[21] We can assess this directly for the limited set of characteristics that are available in the administrative data (see Table 2). Table A.2 explores this further separately for applicants in the oversubscribed (lottery)

[20] According to the National Family Health Survey of 2019–2021, almost all urban households (97%) and most rural households (92%) in India have a mobile phone.

[21] Non-response mainly reflected an inability to connect to a respondent, with the most common reasons for incomplete surveys being that the respondent did not pick up the call, the number does not exist, or that the phone is switched off or outside the coverage area; refusal to participate in the survey, conditional on picking up the call, was rare (∼2%). These numbers are typical of phone surveys in this context.

and undersubscribed (deterministic) samples. Respondents in our phone survey do not differ from the full set of applicants in the administrative data in gender or age. However, we have higher attrition among applicants in rural areas and, relatedly, lower coverage of Scheduled Tribe households (which are overwhelmingly rural). We discuss the implications of this non-response for the representativeness of our results in Sections 4 and 5.1.1. Briefly, for the incidence results, we reweight our sample to account for non-response rates in several ways, all yielding similar results. Importantly, the patterns of self-selection we show in statewide phone-survey data are also replicated in household-survey data collected via in-person household listing and interviews in Raipur district (which does not exhibit similar non-response patterns) — see Appendix B for more details. For the results on the effects of winning a lottery seat, all results are robust to restricting the analysis to low attrition strata and to Lee (2009) bounds.

3.1.3 Administrative data on school characteristics

We use the U-DISE (Unified District Information System for Education) database, an annual census of all recognized (public and private) schools in the country, which contains information on school enrollment, infrastructure, and staffing.[22] We use data from the 2017–2018 school year, the most recent for which data were available at the time of writing. We merge the U-DISE data with a separate dataset on school fees (for non-quota students) for recognized private schools.

3.2 Representative data on the statewide population

Understanding the incidence of quota benefits requires representative data on eligible students, not just those who applied. We use population-level data from three sources.

3.2.1 National Family Health Survey 2019–21

To establish a representative benchmark for the eligible population, we use household-level microdata from the fifth round of the National Family Health Survey (NFHS-5), conducted between 2019 and 2021. The survey is representative at the state level, and the timing coincides with the 2019–20 academic application cycle.

The NFHS-5 allows us to construct an asset-based wealth index using Principal Component Analysis (PCA), which serves as the "ground truth" for the state's economic distribution. By mapping RTE applicants' asset ownership and housing characteristics to state-specific wealth quintiles derived from NFHS-5, we can estimate the policy's incidence.

3.2.2 National Sample Survey 75th Round (Education survey)

We use data from the 75th round of the National Sample Survey, a nationally representative official survey with detailed questions on education, to link regressive

[22] The U-DISE dataset does not include unrecognized (i.e., unlicensed/unauthorized) private schools (Kingdon, 2020). This is not relevant in our setting, as the policy applies only to recognized private schools.

selection to muted effects on school choice. Unlike the NFHS, which does not collect data on whether a child attends a private school, we can observe both household consumption and private-school attendance for the same individuals. The 75th round, conducted between July 2017 and June 2018, is closest to our administrative data.

3.2.3 SHRUG Data on village characteristics

We characterize the local economy and supply-side educational environment using the Socioeconomic High-resolution Rural-Urban Geographic Platform for India (SHRUG v1.5). The SHRUG dataset aggregates administrative units into stable geographic identifiers (shrids) derived from 2011 Population Census polygons (Asher et al., 2021). This stable spatial framework lets us link each applicant's residence to a fixed unit of analysis, enabling integration of village-level covariates such as literacy rates and Scheduled Caste and Tribe population shares. This also lets us assess whether supply-side factors (e.g., the absence of private schools in some areas) drive our results.

3.3 Household-level data from in-person surveys in Raipur district

Investigating barriers to applications requires representative data on eligible students, not just those who applied. We collected these data in urban and rural areas of Raipur, the state's most populous district. This district accounts for ∼8.4% of the state's population, 15.9% of the population of private school students and 12% of the total eligible applicants for RTE seats in 2019. It also includes the state capital. An estimated 43–47% of the population of the district was below the official poverty line (World Bank, 2016), 16.6% belong to Scheduled Castes and 4.3% belong to Scheduled Tribes (Government of India, 2011).

To draw our sample, we first selected a random set of 20 locations each in urban and rural areas of the district.[23] In these locations, we interviewed all households that had a child aged between 3–7 years of age (N=1,059, identified based on a door-to-door listing of 12,225 households). These households were administered an extensive questionnaire to directly study how parental preferences and application constraints vary across the SES distribution. Socioeconomic status is proxied by an index created from household ownership of assets, consumer durables, and housing quality (Filmer and Pritchett, 2001).

We compare the characteristics of the population in Raipur district in the NFHS, restricted to those with at least one child aged 4–7 years, with those in our primary sample in Table 3. Households in Raipur have, as expected, more assets and higher levels of parental education than the state average. Our in-person sample matches the NFHS sample well in the overall assets index and education levels also appear

[23] We drew a sample of squares on Google Maps of 1km x 1km in rural areas and 300m x 300m in urban areas. After verifying that these included habitations (e.g., excluding agricultural land), we identified a government early childhood care center (anganwadi) near the center of the square and re-centered the square around it.

broadly similar, although our sample has a slightly higher share of parents with secondary education than the NFHS sample for Raipur.[24]

4 Quantifying the incidence of program benefits

The first part of our analysis aims to directly quantify the progressivity of benefit incidence in practice.[25] We focus on assessing three channels that may determine the benefit incidence of the RTE quotas. The first is *policy targeting*: by design, the population groups eligible for the program may be more deprived than the overall population. This was the intention of the RTE quotas, but the effectiveness/sensitivity of the specific indicators used to identify disadvantage remains an empirical matter. The second is *selection in policy take-up:* applicants self-select out of eligible groups and may be richer or poorer than the eligible population or the general population. The third is *selection in benefit allocation:* even conditional on applying, the allocation process may change the incidence of benefits if application patterns differ substantially across economic groups (e.g., if richer households apply to more schools or to more expensive schools).

Table 4 and Figure 1 compare all quota applicants in our phone survey to households in the statewide NFHS with at least one child aged between 4–7 years.[26] We compare three groups: (a) all NFHS households with a child of eligible age, (b) NFHS households which are likely to be eligible for RTE seats, and (c) the actual population of applicants for quota seats.

The first relevant pattern, comparing Columns 1 and 2, is that households likely to be eligible — defined here as either belonging to Scheduled Castes/Tribes or below the district-specific poverty rate in terms of the distribution of assets and consumer durables — have fewer assets, with parents less likely to have completed secondary schooling, than the overall population. Thus, at least the *intent* of the policy to be redistributive seems borne out in the choice of eligibility criteria: targeted groups are, in fact, more disadvantaged than the overall population.

Second, however, actual applicants are substantially more advantaged than the population of eligible households or, indeed, the average for all households in the state with children of eligible ages. In Figures 1 and A.2, we confirm this across the full distribution of socioeconomic status, as measured by the asset index: the eligible population is poorer than the statewide population, which in turn is more deprived

[24]There are also some significant differences in specific assets: in general, it appears households in our sample are more likely to report owning a house but have lower reported ownership of some consumer durables such as furniture, electronics and bicycles/motorcycles.

[25]This follows a long tradition of quantifying the benefit incidence of individual programs. See, for example, Lustig (2018); Blanchet et al. (2022).

[26]We use inverse probability weights to account for (i) the differential probability of being surveyed between oversubscribed (lottery) applicants and undersubscribed (deterministic) applicants, and (ii) the probability of non-response to the phone survey. Survey completion in our phone surveys is predicted from household characteristics and location, as observed in administrative data on student applications.

than the population of applicants. Conditional on applying, though, there are no visible differences between all applicants and lottery winners. Thus, the benefit incidence is determined by the self-selection of eligible households in applying for these voucher seats: *actual* benefit incidence, as opposed to policy intentions, is regressive.

Figure 2 summarizes the extent of regressivity across the socioeconomic distribution. Panel (a) first investigates the proportion of households in each quintile that would be eligible on either caste or income grounds from the scheme: eligibility is near universal in the bottom two quintiles but also substantial (at 30–40%) in the top two quintiles since caste-based eligibility does not have an income or assets criterion for exclusion. The distribution of applications, however, follows the opposite pattern: less than 10% of applications come from the bottom quintile while close to 25% come from the top quintile (Panel (b)). Admissions closely track applications and retain the regressivity from the application stage. This, in turn, carries over to actual expenditures (Panel (c)): the full bar represents the market value of seats received by students who won quota seats (i.e., the private benefit), while the blue bars represent government reimbursements (capped at INR 7,000 per student). This regressivity is even more pronounced since students from more advantaged backgrounds apply to schools with higher fees (and thus, if they receive a quota seat, receive a benefit of higher monetary value).

In summary, the realized pattern of benefit incidence is regressive, favoring households at the top of the socioeconomic distribution. Regressivity is consistent with self-selection from the eligible household population to the pool of actual applicants.[27] We investigate the consequences of this selection in the following section and the sources of this selection (and potential solutions) in the next sections.

Appendix C extends this incidence analysis along the joint caste-income dimension and documents that the policy is regressive both across and within caste groups. Across caste groups, Scheduled Tribes — the poorest group in the state — are sharply under-represented in the applicant pool, accounting for 21% of applicants versus 32% of the state population. Among Scheduled Castes and Scheduled Tribes, applicants are wealthier and have more educated parents than the average household in these groups. The within-caste pattern is structurally possible because caste-based eligibility

[27] Differential attrition in our phone surveys is unlikely to explain this pattern. As explained above, we account for attrition by reweighting our sample. The results are robust to different methods to estimate the probability a household attrites in our sample (see Tables A.3-A.4 and Figures A.3-A.4, which use linear probability models and random forest, as opposed to probit models, to estimate the likelihood a household attrites. The incidence results are also consistent with comparisons between the full set of applicants (using administrative data) and representative state data from the NFHS-V. Although the set of characteristics is more limited, it also suggests that the program's incidence is regressive. In short, 56% of applicants come from rural areas (compared to 78% of the state population with kids in the eligible age range). Households from Scheduled Tribes are considerably poorer in general than those from the Scheduled Castes, yet while 21% of applicants are from Scheduled Tribes, 32% of the state population is. Finally, Appendix B confirms these patterns in the Raipur household sample and shows, through back-of-the-envelope counterfactual simulations, that the regressive incidence is driven by differential take-up: if all eligible households applied, the distribution of benefits would be progressive.

has no income ceiling: combining NFHS-V data with district-level poverty rates, we estimate that roughly 44% of SC and ST households are above the poverty line and qualify for the quota only via the caste criterion.

5 Effects of winning the quota seat on school choice

The regressive incidence documented above is potentially important not only for distributional concerns but also because the effects of receiving a free seat may differ by income. A key premise of the policy was that poorer households are financially constrained and thus cannot attend private schools; however, if most actual quota beneficiaries tend to be from upper-income households, who *can* attend private schools by paying regular fees, it is possible that receiving a free seat does not relax any constraints for private school attendance.

We investigate this using lotteries embedded in the allocation mechanism. These lotteries only provide identification for students who applied to oversubscribed schools. Therefore, our analysis proceeds in three steps. First, we estimate the causal effect of receiving a free seat in our oversubscribed sample. Second, we examine whether treatment effects vary by the ex-ante probability of winning a seat, exploiting the fact that applicants who were just below the margin of facing lottery risk should be informative about the counterfactual for deterministic applicants. Third, we use both the reweighting procedure proposed by Angrist and Fernández-Val (2010) and generalized random forests (Wager and Athey, 2018; Athey et al., 2019) to extrapolate treatment effects from the lottery sample to the deterministic sample based on observable characteristics.

5.1 Empirical specification and validity

We use the following specification to estimate the intent-to-treat (ITT) effect of being assigned a lottery seat:

$$Y_i = \alpha Z_i + \sum_x \gamma_x d_i(x) + v_i, \tag{1}$$

where Y_i indicates the outcome for child i and Z_i indicates winning the lottery for an RTE seat in a private school. This offer (Z_i) is randomly assigned conditional on applicants' school rankings, but not unconditionally. Therefore, we condition on a vector of dummy variables $d_i(x)$ to account for the application choices of each student i ("randomization strata" or risk sets). Our coefficient of interest, α, is the ITT effect of being offered an RTE seat through the lottery.

Our preferred specifications adopt Abdulkadiroğlu et al. (2017)'s approach to controlling for applicant risk sets, conditioning on a vector of narrow bins (of 0.001 probability each) of being assigned to a private school. We computed these probabilities by running 10,000 simulations of the assignment mechanism given the applicants' preferences and

seats available at each school. The identifying assumption is that the offer of an RTE seat is conditionally exogenous, given these bins of the probability of an offer. Our results are similar when conditioning on the full vector of preference sets.[28]

Ex-post, some lottery losers may be assigned RTE seats in schools that still have available seats.[29] Since the policy variable is offering an RTE seat, we estimate and focus on the local average treatment effect (LATE) of being allocated an RTE seat.[30] We estimate the LATE by instrumenting an RTE seat assignment with winning the lottery. Specifically, we estimate the following equations via two-stage least-squares:

$$T_i = \beta Z_i + \sum_x \gamma_x d_i(x) + u_i, \tag{2}$$

$$Y_i = \delta \widehat{T}_i + \sum_x \gamma_x d_i(x) + \varepsilon_i, \tag{3}$$

where T_i indicates being assigned an RTE seat, and everything else is as in Equation 1. Here, δ is the effect of securing an RTE seat (through any means) on the outcome. The implicit exclusion restriction is that, conditional on applying, winning a lottery seat only affects downstream outcomes through being allocated an RTE seat.[31] We compare the LATE estimate to the control mean for compliers (Imbens and Rubin, 1997).[32]

We examine effects on two sets of outcomes. First, we examine whether receiving a lottery seat changes the probability of enrollment in private schools and in formal schooling (i.e., the extensive margin of school choice). Next, we examine effects on the characteristics of institutions attended (i.e., the intensive margin of school choice).

5.1.1 Validity

We examine the validity of our identification strategy in Table 5. Conditional on strata fixed effects, we do not reject the balance on observed pre-treatment characteristics across lottery winners and losers in the administrative data or either of our phone surveys.

There is moderate differential attrition across treatment groups, with students offered a lottery seat being slightly more likely — by 2.1 percentage points (over a base of 45%) in the first round and by 2.8 percentage points (over a base of 26%) in the second round — to answer phone surveys than those who were not.

[28] This latter strategy is inefficient, as limiting comparisons to exact matches discards much of the available variation. Our results are similar in magnitude and statistical significance across both procedures.

[29] This can be through subsequent lottery rounds, as well as through local education authorities. Our estimates rely solely on lottery-based offers from the first lottery round.

[30] Nearly everyone ($\sim$ 95%) who is offered a seat, takes it. Thus, in practice, there is little difference between estimating the LATE of being offered a seat and enrolling in an RTE seat.

[31] If lottery losers exposed to winners are induced to seek private schooling, this would violate the exclusion restriction. Figure 3 provides evidence against this: the control group means for private school enrollment are stable across deciles of the probability of winning the lottery. In more oversubscribed bins, where each control household is mechanically exposed to a higher density of treated peers, we would expect higher control-group enrollment in the presence of spillovers.

[32] Our approach to estimation of the control complier mean is based on Abdulkadiroğlu et al. (2018). See Abadie (2002, 2003) for relevant theoretical results.

We investigate the reasons for non-response and find that this is largely due to a lower proportion of calls failing to connect.[33] We investigate the sensitivity of our results to using low differential-attrition strata and Lee (2009) bounds. Our main findings are robust to these corrections (see Table A.5).

We verify that winning the first lottery corresponds to an offer of a free seat. Nearly all lottery winners reported having been allotted a seat (95%) in the first phone survey, but so do about 22% of lottery losers (Table 5). As mentioned above, non-compliance among lottery losers (i.e., "always-takers") is expected, since education authorities attempt to fill vacant seats after the lottery-based allocation (the data we use) with unmatched parents. Compliance rates are similar in magnitude across grades (see Table A.6). We focus on the LATE of being allocated an RTE seat, using the lottery outcome as an instrument.

5.2 Effects on school choice in the lottery sample

5.2.1 Extensive margin effects on school choice

We classify students by whether they were (a) enrolled in a private preschool or school, (b) enrolled in a government school, and (c) not enrolled — and study the effects of being offered an RTE seat on each of these margins separately for the 2019–20 academic year.[34]

Nearly all applicants who were assigned an RTE seat are enrolled in private schooling; however, so are three-quarters of compliers who *did not* receive a free seat (Table 6). The estimated treatment effect on the probability of attending a private school is only around 25 percentage points. This increase is concentrated among students who had applied for a seat in preschool grades, with a substantial movement from non-enrollment to private schooling.[35] Among applicants for admission in Grade 1 — of whom 98% of lottery losers are formally enrolled — the effect on private schooling is only 13 percentage points.

This extensive-margin effect is small relative to the implicit policy assumption that eligible students would not be able to access private schools without the quota. It is also modest compared to context-specific empirical benchmarks. Muralidharan and Sundararaman (2015) show that providing (untargeted) vouchers to students in public schools boosted private sector enrollment by $\sim$ 50 percentage points in rural Andhra Pradesh, a richer, neighboring state. If financial constraints bind more

[33] A likely reason for this is that once households have a government benefit tied to a specific phone number, they are less likely to switch mobile operators and numbers (which is otherwise common in this setting).

[34] We do not distinguish between non-enrolled and government daycare centers (called *anganwadis*), because the latter provide very little early childhood stimulation in practice (see e.g., Singh (2014); Ganimian et al. (2024); Berg et al. (2025)). Nor do we distinguish between pre-primary and primary grades in private schools, since they exist in the same schools and kindergarten (preschool) classes serve as feeder grades into primary schooling (Singh, 2014).

[35] The 4–6 age group, when students apply for RTE quotas, is a period of transitioning into primary schooling from either preschool or non-enrollment. Unlike primary schooling, which is mandatory from age 6, preschool enrollment is neither universal nor compulsory. Children in this age group may be enrolled in a government childcare center or the pre-primary section of a private school, or enrolled in Grade 1 in either a government or private primary school, or not be enrolled in any preschool/school.

strongly for poorer households, a reasonable prior is that untargeted vouchers provide lower-bound estimates of the effect on private school attendance for vouchers targeted at disadvantaged groups.[36] Our estimated effect is, instead, about half the size in the overall sample and about a quarter of the size for applicants in Grade 1. A potential reason, which we examine in much greater detail later, is that the application process differs substantially. In the Andhra Pradesh voucher program, research teams conducted door-to-door visits in treatment villages, which substantially reduced information and application frictions (potentially drawing in poorer households for whom the voucher has larger effects). In contrast, the RTE quota requires households to learn about the policy independently, navigate an online application portal, assemble requisite documentation, and submit rank-ordered school preferences within a narrow window; children from such households are potentially more likely to attend private schools anyway.

5.2.2 Effects on characteristics of schools attended

We next estimate whether receiving a free seat changes the characteristics of schools that students attend (Table 7). This includes changes induced by moves *within* the private sector. We focus on applicants to Grade 1 to minimize conflating intensive-margin changes across schools with extensive-margin shifts in enrollment.

Lottery winners are more likely to attend schools that use English as the medium of instruction (rather than Hindi, the state language). These schools also have more students and slightly higher pupil-teacher ratios. However, for an initiative aimed at reducing segregation, these schools are not significantly different in the caste composition of their student bodies. The schools attended by lottery winners are also more expensive: our LATE estimates suggest an increase in the average annual school fees of ∼INR 2880, roughly 48% of the fees paid by compliers among the lottery losers (Column 7, Panel B).

Lottery winners are also 37 percentage points more likely to be enrolled in the seat they ranked as their top choice in the applications: however, 54% of students who lost the lottery *also* attend the same school (as non-RTE fee-paying students). Thus, for over half of the students who win the lottery, receiving an RTE seat does not expand school choice at all: if they had not received the RTE seat, they would have attended their top-choice school in any case. For these students, the RTE seat is, effectively, a cash transfer that displaces private spending on tuition fees but is inframarginal for school choice.[37]

We do not have direct measures of the income or asset composition of the student body, as U-DISE does not collect household-level economic data.[38] This null result on

[36] Muralidharan and Sundararaman (2015) restrict their sample to villages with at least one private school; our sample of applicants effectively imposes the same restriction.

[37] Focusing on students who attend their top-choice school without a lottery seat provides only a lower bound of the overall inframarginality of the transfer. For many students who changed the school attended in response to receiving an RTE seat, a portion of the transfer is still inframarginal since it crowds out what they would have paid in a different private school: the estimated control complier mean in Table 6, Column 7 suggests that this is 89% of the sample among students applying for Grade 1 seats.

[38] School fees provide an indirect proxy: lottery winners attend more expensive schools, suggesting

caste composition may seem at odds with Rao (2019), who finds that quota students in Delhi private schools increase exposure of wealthy students to poorer peers. However, Rao (2019) studies elite schools with annual fees exceeding INR 100,000, in which quota seats are reserved for families earning under INR 100,000 per year; fee levels alone ensure a large socioeconomic gap between quota and non-quota students. In our setting, average fees are approximately INR 6,000–7,000 per year, and three-quarters of applicants who lose the lottery attend private schools by paying regular fees. This reduces socioeconomic distance between quota and non-quota students, thereby limiting the scope for integration even when seats are allocated.

5.3 Generalizing estimates to non-lottery applicants

Our main estimates rely on lotteries in oversubscribed schools, which raises a natural question regarding external validity: are the effects of the quota different for applicants to undersubscribed schools? This distinction is potentially important, as approximately 69% of applicants in our sample applied to schools with no lottery risk (i.e., "deterministic" applicants). If the counterfactual for deterministic applicants is markedly different (e.g., none would attend private schools in the absence of the RTE quota), the implications for the policy's success in expanding school choice and integration, and for its incidence, would be very different. We investigate this in multiple ways.

Our first exercise examines whether estimated treatment effects vary by the ex-ante probability of winning an RTE seat. The intuition for this exercise is simple: given some uncertainty in the exact number of applicants/seats at the time of application, examining the counterfactual for applicants who were just below the margin of facing lottery risk (say, where the probability of an offer was 90%) should also be informative about the counterfactual for deterministic applicants who faced no lottery risk (i.e., a 100% probability of getting an offer).

Figure 3 plots the effect of receiving an RTE seat on private school enrollment across different bins of the probability of winning the lottery. Control means are high across the distribution. If we focus on compliers (Figure 3b), the control mean is slightly lower for marginally oversubscribed schools, but remains above 67%. Except for very competitive schools (those with a probability of winning the lottery under 30%), the treatment effects are similar and we cannot reject that they are equal (for the ITT the p-value of equality is 0.800, for the LATE the p-value of equality is 0.127). Even under the most conservative counterfactual for deterministic applicants, this would imply that $\sim$67% would attend a private school in the absence of the lottery. Likewise, if we focus on Grade 1 applicants, over 40% of lottery losers (as well as compliers who lose the lottery) still attend their top choice (see Figure A.5) regardless of how oversubscribed the school is, and there is

exposure to peers from families that can afford higher fees. However, because around half of compliers among lottery losers attend the same school and pay full fees, the net change in peer composition is limited for a substantial share of applicants.

a negligible effect on nominal fees for students in the least oversubscribed schools (see Figure A.6). This suggests that, in the absence of an RTE seat, most applicants who did not face lottery risk would also likely have attended private schools, many at their top-ranked school, and that most program expenditure is inframarginal to school choice.

Our second exercise uses the re-weighting procedure proposed by Angrist and Fernández-Val (2010) to extrapolate treatment effects estimated in the lottery-based sample to deterministic applicants based on observable characteristics (specifically caste, application grade, and village characteristics).[39] The estimated average treatment effect, based on extrapolation for the deterministic sample, is 24 percentage points for private school enrollment and 3,135 INR for nominal fees paid (for Grade 1 students) — very similar to the point estimates from the stochastic sample of 25 percentage points for private enrollment and 2,880 for nominal fees.

In a complementary exercise, we use conditional average treatment effect (CATE) estimation to study treatment-effect heterogeneity and extrapolate treatment effects to the deterministic sample. Specifically, we use the partialing-out estimator with generalized random forests (Athey et al., 2019; Wager and Athey, 2018), see Appendix E for details. The overall treatment effect (the average treatment effect for the whole sample), estimated by extrapolating treatment effects using machine learning, is 17.8 percentage points for private school enrollment and 2,031 INR for nominal fees paid (for Grade 1 students), similar to the ITT effects on the stochastic sample (of 18 percentage points and 2,059 INR).

All three exercises suggest that most applicants, even deterministic ones, would attend private schools regardless of whether the RTE quota is in place, and that most of the program's expenditure is inframarginal to school choice.[40]

5.4 The importance of incidence for effectiveness

We have established that (i) eligible households with high socioeconomic status disproportionately select into applying for RTE voucher seats (Figures 1 and 2), and (ii) applicants to the RTE vouchers are very likely to attend private schools even in the absence of the free RTE seat (Figure 3 and Tables 6 and 7). To establish

[39] The full approach we take is explained in Appendix E. In addition to the basic extrapolation, we also conduct tests in the spirit of Bisbee et al. (2017) to assess the validity of this exercise.

[40] Appendix G presents a simple model that can account for similar schooling responses at different lottery probabilities. Households apply when the expected value of a voucher (the offer probability times the gain from a free seat) exceeds the cost of applying. Suppose application costs are distributed according to a power function with the same exponent for all household types, the gain from a voucher never exceeds that ceiling, and all applicants to a school face the same offer probability. Then changes in the offer probability scale every typeâs application probability by the same factor. Applicant composition, and hence average potential schooling responses among applicants, therefore remain unchanged. With logit preferences and application frictions common across schools, the model also has the following testable implication: at a given enrollment share, schools with higher fees offer a more valuable voucher, attract more applications per quota seat, and so have lower offer probabilities, which we see in the data (see Figure G.1 and Table G.1). The model's other prediction, that schools with larger enrollment shares have higher offer probabilities, does not hold in our data and we discuss possible reasons why in the Appendix.

whether these facts are related, we need to plot the joint distribution of household socioeconomic status and the rate of attending private schools without a free seat in the general population. This is, unfortunately, not feasible with data from our phone surveys (which are based on a self-selected pool of applicants) or from the NFHS (which lacks an indicator of private school attendance).

To plot this joint distribution, we use data from the National Sample Survey's 2017 Round, which includes both household expenditure and private school attendance (Figure 4). Private school attendance of 6–12 year-olds in Chhattisgarh rises sharply with household per capita consumption from about 4.5% in the bottom decile to 64% in the top decile. This pattern also appears to be true in neighboring states like Odisha, Jharkhand, and Andhra Pradesh, and does not appear to be a consequence of the RTE policy itself.[41] Thus, the over-representation of students from richer households in the applicant pool (i.e., regressive incidence) directly increases the degree of inframarginality. Note that the regressivity of selection into the program also makes overall public spending on education less progressive: for the vast majority of applicants, who would have enrolled in private schools anyway *and* belong to richer quintiles, the government now incurs the cost of the voucher whereas, in the counterfactual, they would have paid these expenses out of pocket.[42]

In principle, this over-representation could result from either low parental demand for private schooling or barriers to applying. A lack of demand is unlikely to fully rationalize application patterns. In Appendix D, we report results from a stated choice experiment in our in-person household survey sample in Raipur district, which, following Delavande and Zafar (2019), elicits parental preferences across educational options with and without financial constraints. The results suggest that parental demand does not explain undersubscription or the over-representation of above-median households.

Further, the proportion of students who attend private schools in the absence of receiving an RTE seat in our sample of applicants (>85% for Grade 1) is higher than we would expect solely from their position in the socioeconomic distribution (as shown in Figure 4). It is not surprising that nearly all parents who put in the effort to apply for the program accept a seat when offered one. However, even in the top three quintiles, parents who would like to attend but are financially constrained are disproportionately unlikely to apply for the program. This suggests that barriers to applying potentially extend across the economic distribution (although they may be more severe for poorer households).

Finally, spatial supply constraints — whether eligible households even live near a

[41] In 2017, Andhra Pradesh did not implement the RTE quota at all, and the program was nascent in Odisha with a total of 20,650 students enrolled under the quota in 2018–2019, out of 5,732,582 students enrolled in primary (Dhariwal and Middha, 2023).

[42] That a universal public schooling system, in a setting that allows households to opt out to private provision, would be progressive and redistributive by default was noted by Besley and Coate (1991); the empirical importance of this channel, especially in developing countries, is underscored by Gethin (2025a). The RTE vouchers, as administered, serve to blunt the redistributive impact of public education spending.

recognized private school — set an upper bound on what improving take-up could achieve. In Appendix F, we document that private schools are systematically less likely to exist in communities with higher Scheduled Castes and Tribes or poverty shares (Figure F.1), and we simulate how the public-private gap in Scheduled Caste and Tribe representation would respond to changes in take-up under within-neighborhood reallocation (Table F.1). With full take-up of unfilled seats by Scheduled Caste and Tribe students, the public-private gap could roughly halve, and reach near zero within postcodes.

6 Investigating barriers to application

Barriers to application appear to be key to understanding who applies for, and thereby who is allotted, the quota seats. This section investigates these concerns.

6.1 Survey evidence on information and application constraints

The simplest explanation for undersubscription is that many individuals do not know about the program or find it too complex to apply.[43] If these constraints affect poorer households more, that could also explain the regressive selection into the applicant pool.

Survey data from in-person interviews in Raipur district, collected from a sample identified using comprehensive household listings of sampled neighborhoods and villages, suggest such constraints are important in this setting (Figure 5). First, information constraints appear both large and regressive. There is a stark SES gradient in whether households, with children between 3–7 years, have even heard of the policy: 65% of parents in the top decile have heard of the policy, while only 20% of parents in the bottom decile report the same. This result is robust to focusing on heterogeneity within rural/urban areas, and focusing on heterogeneity within each sampling unit (i.e., small geographical areas) — see Table A.7 and Figure A.7.

Second, potential frictions to submitting applications also burden poorer households differentially. Nearly all (98%) high socioeconomic status households report access to the internet at home, but only 9.1% of households in the bottom decile do so (see Figure 5). This differentially burdens poorer households from accessing the centralized application portal. Fewer low-SES households report having ever applied for government benefits online, and fewer of them can access support from others for doing so.

6.2 Experimental evidence on addressing application frictions

Our survey evidence suggests that information and application frictions constrain potential applicants. Yet, with additional frictions, relaxing these constraints alone

[43] Such frictions are consistent with evidence on the positive effect of information campaigns on the take-up of welfare programs and student aid. See, e.g., Bettinger et al. (2012); Hoxby and Turner (2013); Bhargava and Manoli (2015); Neilson et al. (2019); Dynarski et al. (2021). See also the theoretical framework in Finkelstein and Notowidigdo (2019) on how such differential burdens may explain take-up for a range of public policies.

may be insufficient to increase application rates or reduce regressivity (Kremer, 1993). We investigate this through a randomized field experiment — in the same sample as our household survey — to evaluate an intervention that relaxes information frictions and reduces application complexity. Specifically, by actively assisting households in applying, we aim to understand (i) constraints that may remain even after dealing with (apparent) first-order frictions and (ii) the extent to which this intervention might help address issues of undersubscription and regressive selection.

6.2.1 Intervention and Experiment Design

Our intervention aimed to address the twin constraints facing households: insufficient information about the policy and the lack of means to apply through online portals. We focus on both constraints, given evidence from other sectors and countries that information provision alone may not suffice to raise take-up when households also face other barriers (Banerjee et al., 2021; Finkelstein and Notowidigdo, 2019).

We used our in-person household survey conducted in February 2022 as the baseline and restricted our analysis to households that reported having a mobile phone (N=914, out of 1,059 households in total). This sample was not selected based on quota eligibility (or predictors thereof). We randomized ∼50% of households in each locality (N=459) to receive the intervention. Observed characteristics (including previous knowledge of the RTE policy) are balanced between the treatment and the control group (see Table 8).

The intervention consisted of providing detailed information about the policy and, if the household was willing to apply and eligible, assisting them in applying. The intervention was implemented between April 15 and May 2, 2022, when quota applications were open, and consisted of the following, sequential, steps:

1. Call all treatment households to offer information and potential support to apply for a quota seat (if eligible). If we could not reach households at the phone number they provided, we revisited them to obtain this information.
2. Households were asked if they had heard about the RTE quota, if they knew that the applications were currently open, if they had already thought of applying to the program, and if they were interested in receiving more information.
3. Interested households were provided with detailed information on the eligibility criteria for the policy and the documents required for demonstrating eligibility. Surveyors collected information on each document the household reported having.
4. If a household reported having all required documents, we offered application support and scheduled a visit. If a household lacked all required documents, we provided information on where they could obtain them. We provided applicants with a number to call if they obtained these documents and needed further support.
5. For interested households (which reported having all documents), an interviewer

visited them at home to help them fill out the application online.[44] They filled out identifying details, uploaded the documents, showed the households the list of schools available in the portal for their area, and ranked the schools as directed by the parents. If parents wanted, they could submit the forms immediately. If they wanted more time to consider it, the surveyor saved the form and provided the household with the relevant login details to submit applications later.

This intervention design resembles those commonly used in this setting. Thus, our estimates are informative about the effectiveness of untargeted interventions in this setting (which are common and reflect the difficulty external organizations, such as NGOs, face in verifying eligibility ex ante).[45] Our focus, as with prior experiments evaluating take-up, is on the rate of successful applications.

6.2.2 Implementation and take-up of application support

Our first set of insights comes from the *process* of delivering the intervention.

Of 459 households randomized to treatment, we were unable to reach 21% (N=95) within the short application/treatment window, despite multiple phone and home-visit attempts. Confirming high demand for private schooling, very few of the remaining households ($N = 36$) declined the offer of further information and support. A further 17% of households (N=80) reported having already filled out the form, including in previous years.

The main constraint for submitting applications was the availability of documents to certify eligibility. Of the remaining 248 households, 186 did not have the full set of documents (mostly proof of being income-poor or from a disadvantaged caste). This group conflates both those who are genuinely ineligible for the quota and those who are eligible but cannot document it.[46] However, the correlation between having had documents or having already applied and SES is instructive: richer households are much *more* likely to have documents certifying eligibility than poorer households (see Figure 6). In the bottom decile of the SES distribution in our sample, about 24% of households can document eligibility; this figure rises to about 56% for the top decile.[47] Finally, 19 households did not have a private school

[44] If the household did not have required documents during the home visit, surveyors revisited them after a week. If the household still did not have the necessary documents after the second visit, we provided a number they could call once they obtained them to receive further support.

[45] We chose an untargeted intervention for two reasons: (i) it reflects typical practice in this setting, and (ii) isolating barriers to application across the SES distribution needs broad coverage. Given the stark SES gradients in information and internet availability in the survey, we expected the intervention to raise application rates more for poorer households. We discuss the potential for targeting interventions further in Section 7.

[46] Of these 186, 33 (∼18%) are Scheduled Caste or Tribe households. Another 37 (∼20%) belong to other groups but are in the bottom tercile of the SES distribution. Thus, at least ∼37% of them are likely eligible.

[47] This is a national phenomenon affecting all means-tested programs, not peculiar to our sample, and reflects administrative burdens in obtaining documentation. Descriptively, about half of the households nationally that self-identify as belonging to Scheduled Castes or Tribes in the India Human Development

with seats in the relevant grade listed in their neighborhood.

We eventually helped 9.4% of households that were randomly assigned to treatment submit a form (43 out of 459 treated households).

6.2.3 Treatment effect on application rates

We attempted to re-interview all households in our experimental sample in July 2022 to elicit whether they applied for an RTE seat (our primary outcome), whether they secured a seat, and whether they used an RTE seat. We resurveyed ∼80% of them, with no differential attrition between the treatment and control groups (see Table 8).

The intervention boosted application rates by 9.4 percentage points (p-value .0041), which is a 43% increase over a control mean of 22% (see Table 9), a large effect in relative terms.[48] The increase in application rates is nearly identical to the proportion of randomly-assigned households that we were able to help submit applications, indicating that our intervention did not crowd out alternative sources of application help. This increase in application rates also translates to a 3.4 percentage point increase in the probability of being allocated an RTE seat (an increase of 43% over a control probability of 7.9%), although we lack sufficient statistical power to detect this effect (p-value .11).[49] We interpret these effects as primarily reflecting the combined provision of information and hands-on support in a narrow window at the time of applications. Nearly all households in the control group report having heard about RTE quotas at endline (Table 9), which reflects that we elicited this information the previous months (and that they had heard about it).

Contrary to our priors, the intervention *did not* differentially benefit poorer households (see Figure 7): we cannot reject the equality of treatment effects across terciles of socioeconomic status, and the point estimates are very similar.[50] This lack of progressivity in the effects of the intervention results from the joint incidence of multiple constraints: although poorer households *were* more constrained by information and the ability to apply online (Figure 5), constraints that our intervention relaxes, they were also less likely to have the documents needed to apply (Figure 6), which is the major

Survey II report not possessing an official caste certificate; these households are likely to be poorer, less educated, and less likely to have household members in professional or government employment (Bag et al., 2023). For an ethnographic description of the process of obtaining these documents, and the difficulties therein, see Shinde (2016). In the case of economic disadvantage, there is a further issue: the proxy means tests used by the government are noisy and, in practice, surveys have demonstrated substantial inclusion and exclusion errors. E.g. Drèze and Khera (2010) report that, in the poorest quintile of households, roughly 39% in the NFHS-3 (which used an asset-based wealth index) and 53% in the National Sample Survey (which uses consumption as the measure of living standards) report having a "Below Poverty Line" card; in both datasets, roughly 18% of top quintile households *also* report having these cards.

[48] All treatment effect estimates are intention-to-treat effects since not all households that were randomized into treatment could be contacted during the application window. Results are robust to controlling for household socio-demographic characteristics, as well as preferences for private schools.

[49] The effect on being allocated a quota seat is aligned with the effect on the application rate, given that an application in the control group has a ∼37% probability of translating into an offer for an RTE seat.

[50] We confirm the equality of treatment effects at other quantiles of socioeconomic status in Table A.8.

unsolved constraint in this setting: without documentation, no willing households could complete applications; with documentation, nearly all of them could (except for a subset further constrained by school availability in their neighborhood).[51] Thus, the aggregate effect on application rates was relatively uniform across the socioeconomic distribution.[52]

7 Discussion

Motivated by the goal of expanding educational access for disadvantaged students, quotas in Indian private schools constitute the largest voucher program in the world and one of the most ambitious attempts to reduce socioeconomic stratification in schooling in any setting. We evaluated whether the policy, as implemented in Chhattisgarh, delivers on that promise.

Our results paint a complex picture. The policy yields substantial monetary gains for applicants who receive a free place. Getting a seat leads some eligible students to attend preschool and others to attend schools they otherwise could not afford. Yet, this success is qualified. The quota is used primarily by households that would send their children to private schools anyway, and much of the expenditure is inframarginal to school choice. Thus, the policy largely serves as a monetary transfer for beneficiaries without achieving the goal of changing classroom composition by admitting financially constrained disadvantaged students. These insights are likely informative about the RTE policy across India: RTE quota seats are undersubscribed across states, and extensive margin effects (i.e., expanding access to private schools) appear to be modest at best.[53]

We show that these (muted) effects on school choice are consistent with selection into applications: poorer students are less likely to apply, even when eligible. This reflects several constraints — information, application complexity, and documentation — that disproportionately affect poorer households, rather than a lack of parental demand. There may be substantial scope to improve policy effectiveness, even under current policy rules, by addressing these constraints. However, potential interventions to address regressive selection are likely to differ in design from those focused only on improving application rates: in our setting, providing information and application assistance boosted application rates by 43%, with similar estimated effects across socioeconomic groups. This plausibly reflects two distinct features of our intervention: we addressed only a subset of constraints (e.g., leaving documentation unsolved),

[51] As expected, given the results from Section 4, the likelihood of households in the control group applying increases with socioeconomic status.

[52] Table A.9 explores heterogeneity by other baseline characteristics. The interaction between treatment and prior awareness of the RTE policy is positive but statistically insignificant, suggesting that treatment effects are, if anything, larger for already-informed households — consistent with application assistance, rather than information, driving the effect.

[53] Fill rates for RTE seats in 2018-19 ranged from 7% in Uttar Pradesh to 70% in Rajasthan, with other major states like Madhya Pradesh (43%), Delhi (56%) and Tamil Nadu (60%) in between (Indus Action, 2021). For evidence on extensive margin effects, see Damera (2017) on Karnataka, Agarwal (2024) on Mumbai, and Sahai (2025) on Madhya Pradesh.

and the intervention was not targeted. Both features are common in many settings, often due to the logistical infeasibility of solving all constraints or the unavailability of household information to target supplementary interventions.

These insights are relevant for the study of school vouchers and charter schools globally. The incidence or inframarginality of program benefits has not been a principal concern in this literature but it is likely that regressivity also affects program effectiveness elsewhere. For instance, Angrist et al. (2002), evaluating the PACES voucher scheme in Colombia, find similarly modest treatment effects on the extensive margin of private school enrollment ($\sim$15%). This program required applicants to seek and secure admission to a private school before applying; if wealthier households are more likely to satisfy this requirement, this plausibly produces similar regressivity within the population of eligible households as we document in India. More generally, there is substantial evidence that information frictions and application complexity deter applications across a range of educational settings, and that addressing these factors increases application rates (see, e.g., Bettinger et al. (2012); Dynarski et al. (2021); Agte et al. (2024)). Our results emphasize that these frictions not only suppress application rates but also change the *composition* of program applicants and beneficiaries: this selection raises important distributional concerns and reduces the average effectiveness of interventions in expanding access to educational opportunities. Plausibly, a careful analysis of their benefit incidence may also affect policy support for publicly funded voucher programs, especially where they compete directly with resources for public education (which recent evidence suggests is strongly progressive in its incidence and in downstream returns through the labor market (Gethin, 2025b,a)).

Finally, the issues of program design and take-up highlighted by our results apply more broadly to social programs, even outside education. Information constraints and application complexity deter applications across programs, and the documentation constraint also applies broadly. These constraints may jointly induce similar regressivity in benefit incidence and substantial inframarginality in program spending. Targeting household deprivation more precisely, rather than through coarser group-level proxies with self-selection, is often beyond the administrative capacity of governments. Thus, delivering supplementary interventions to households to lower application barriers is often the feasible second-best solution. The optimal design of such interventions — both in terms of bundling program components and further targeting — to improve progressivity and not just application rates remains an important area for further investigation.

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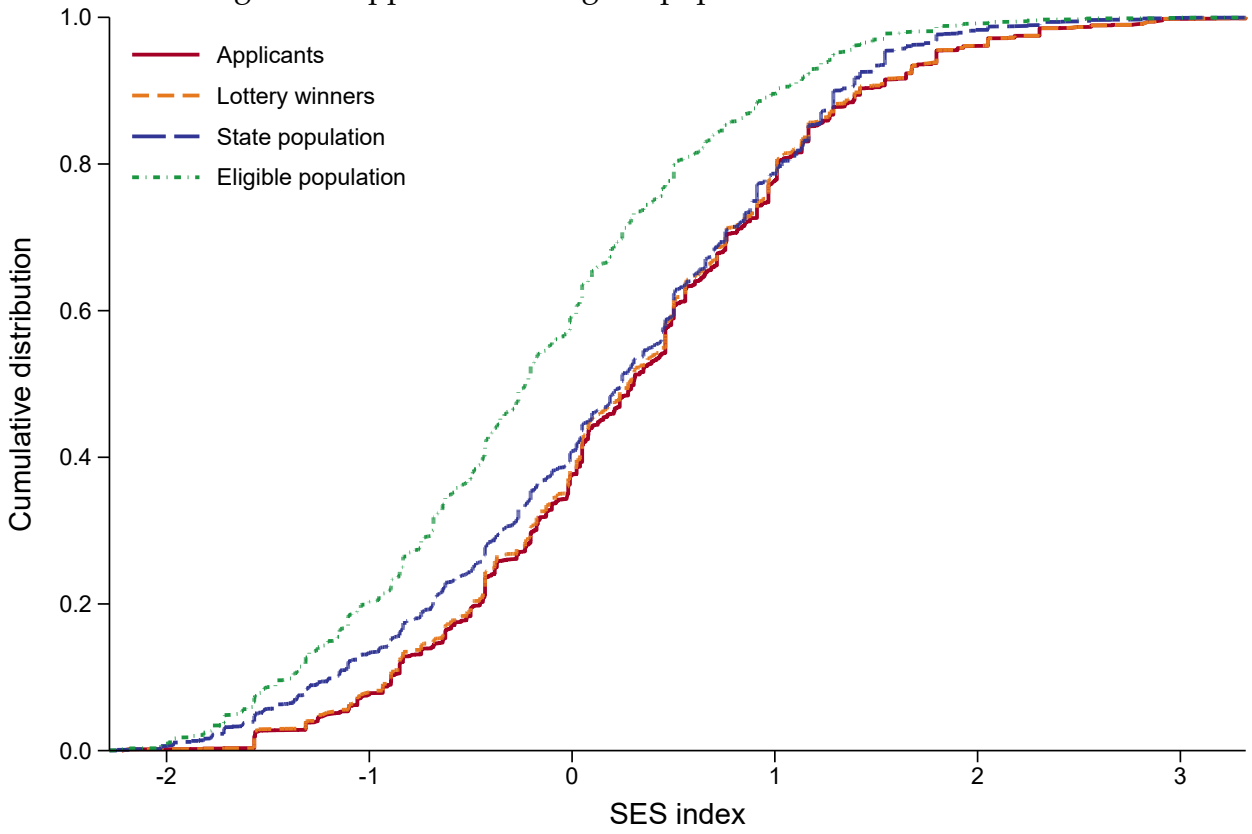
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Exhibits

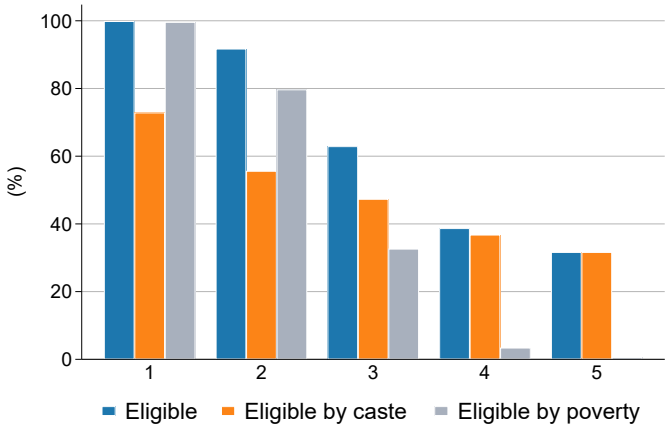
Figure 1: Applicants vs eligible population in terms of SES



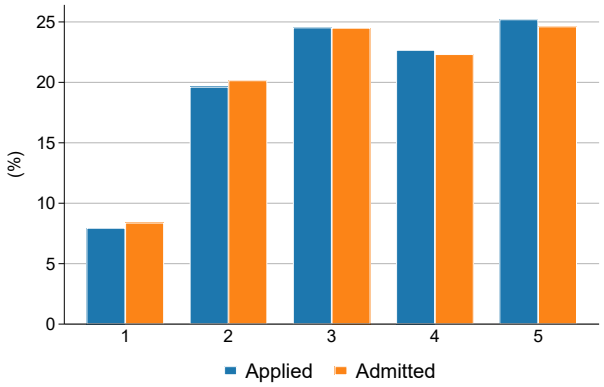
Note: This figure plots the cumulative distribution of a socioeconomic status (SES) index for four groups: (i) applicants to RTE quota seats in our phone survey sample (solid red), (ii) lottery winners among applicants (dashed orange), (iii) all households in Chhattisgarh with a child aged 4–7 in the NFHS-5 (dashed blue), and (iv) the subset of NFHS-5 households likely to be eligible for quota seats based on caste or economic criteria (dash-dotted green). The SES index is computed from household ownership of consumer durables and assets, which were collected in the representative National Family Health Survey 2019–21 for the statewide population and in our survey of applicants. NFHS-5 weights are applied to the population samples. We re-weight our sample to account for differential non-response by household characteristics. We estimate the probability of responding to our survey using a probit model that accounts for the household district, caste, and the child's age and gender.

Figure 2: Applications, admission, and expenditure by SES quintile

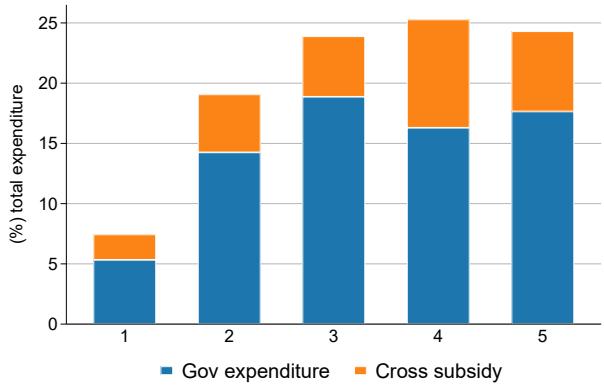
(a) Eligibility by income quintile



(b) Applications



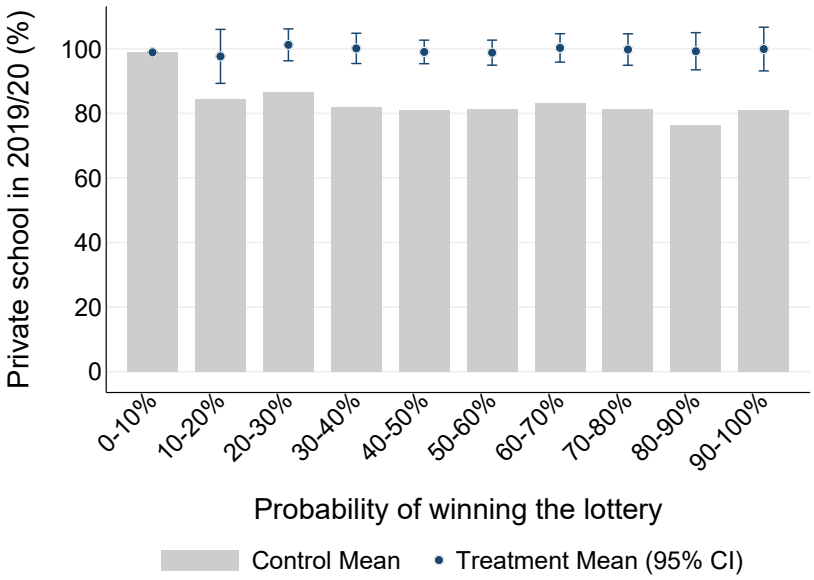
(c) Expenditure



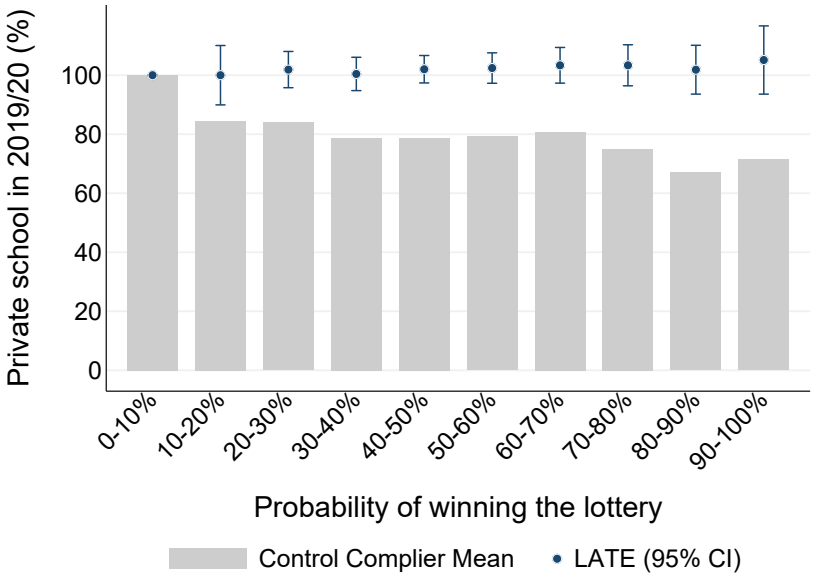
Note: Figure 2a shows the share of households in each quintile of the statewide SES distribution that are eligible for an RTE quota seat. Eligibility is defined as meeting the caste criterion (belonging to a Scheduled Caste or Scheduled Tribe) or the poverty criterion (being below the district-level poverty line, using poverty rates separately for rural and urban areas), or both. Eligibility rates and SES quintiles are computed from the National Family Health Survey 2019–21 (NFHS-V). Figure 2b shows the proportion of students who apply and are admitted for an RTE seat that belong to each quintile of the statewide SES distribution. Quintiles of socioeconomic status are computed from household ownership of consumer durables and assets, which were collected in the representative National Family Health Survey 2019–21 for the statewide population and in our survey of applicants. Figure 2c shows the proportion of the total costs of RTE quota seats (divided by the reimbursement of the fees given by the government and the cross-subsidy provided by schools if their fees are above the cap on reimbursements) that goes to each quintile of the SES distribution. We re-weight our sample to account for differential non-response by household characteristics. We estimate the probability of responding to our survey using a probit model that accounts for the household district, caste, and the child's age and gender.

Figure 3: Heterogeneity by how oversubscribed a school is

(a) Intent-to-treat

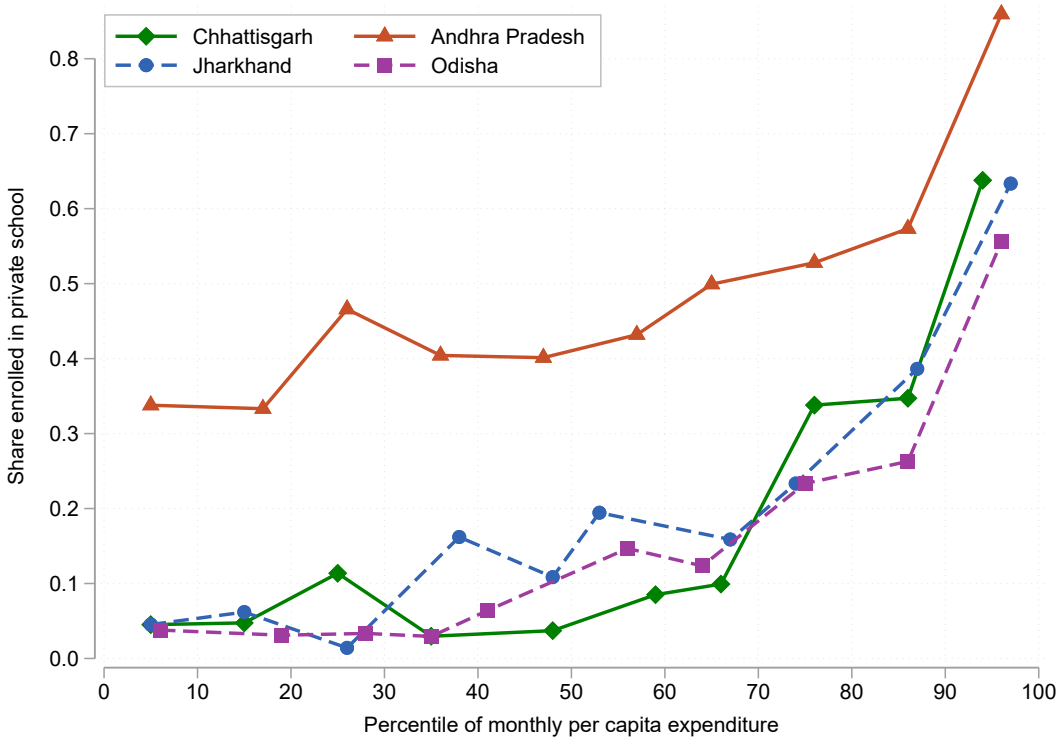


(b) LATE



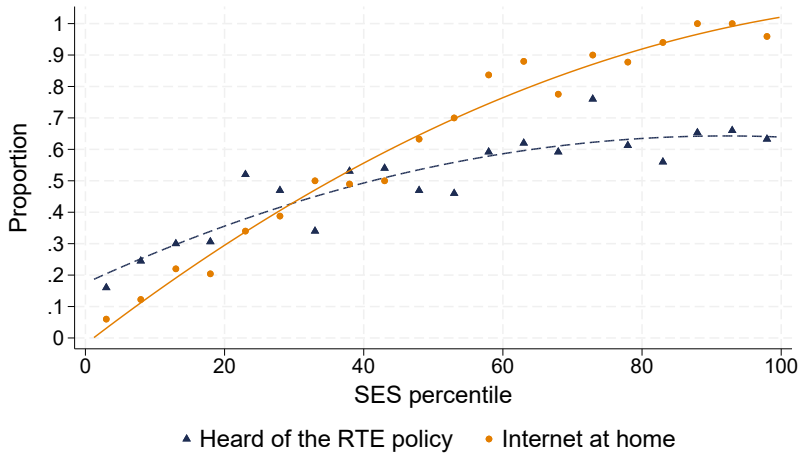
Note: This figure examines how the effects of the RTE quota on private school enrollment vary by how oversubscribed the school is (i.e., by the ex-ante probability of winning a quota seat). The x-axis divides the sample into deciles based on the ex-ante probability of winning the lottery (calculated via simulation of the assignment algorithm); lower probabilities indicate more oversubscribed schools. The y-axis shows the percentage of students enrolled in private schools for the 2019-2020 academic year. Figure 3a shows the control mean and the intent-to-treat effect (with 95% confidence intervals). Figure 3b shows the control-complier mean and the LATE effect (instrumented by securing an RTE seat via winning the lottery, with 95% confidence intervals).

Figure 4: SES gradients in private school attendance



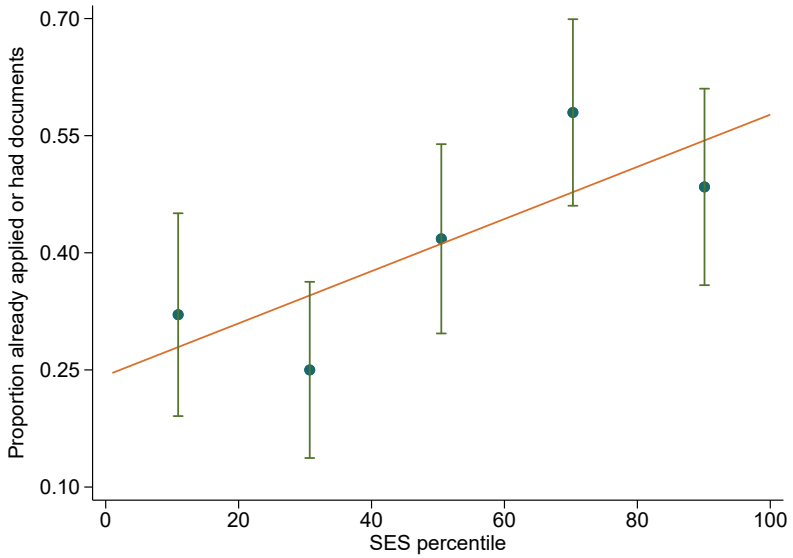
Note: This figure presents the joint distribution of private school attendance and percentiles of mean per capita consumption expenditure in four states, including Chhattisgarh, using data from the National Sample Survey (75th round, 2017). At the time of the survey, Andhra Pradesh did not implement the RTE quotas.

Figure 5: Survey evidence on application frictions



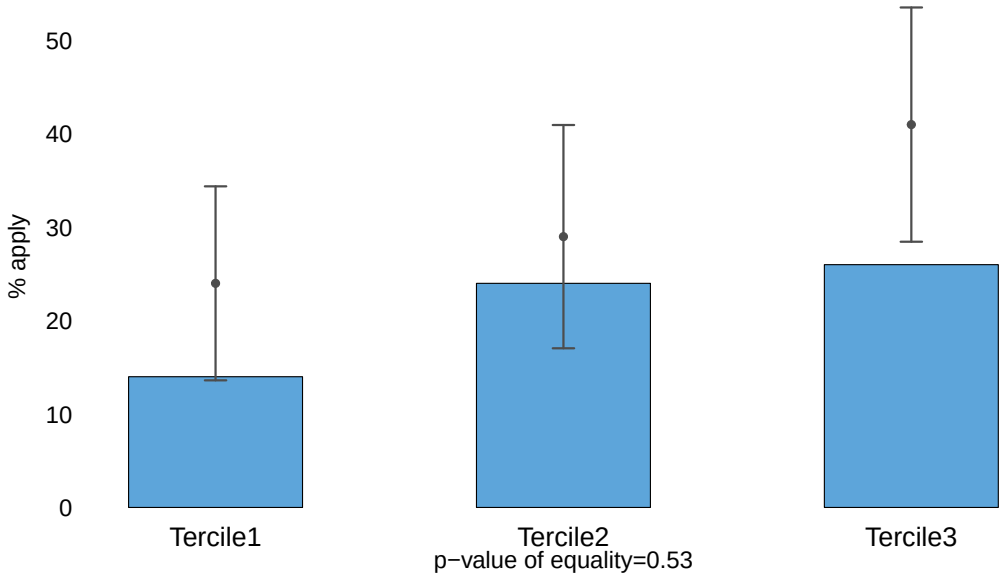
Note: This figure presents binscatter least squares estimations — following Cattaneo et al. (2024) — of the relationship between the percentiles of SES and having heard of the RTE quota policy and having internet at home. We measure socioeconomic status using an index created from household ownership of assets, consumer durables, and quality of housing using Principal Component Analysis (Filmer and Pritchett, 2001).

Figure 6: SES gradient in having all requisite documents for application: Treatment group



Note: This figure plots the proportion of households which had either applied to the RTE quota seats on their own or had all requisite documents, for applying within the randomly-selected treatment group [N=459]. We plot the mean and associated 95% confidence intervals for each quintile in the socioeconomic distribution in our data, along with a linear fit.

Figure 7: Treatment effects of application support by SES tercile



Note: This figure plots the likelihood that households apply for an RTE seat by tercile in the socioeconomic distribution. The bars correspond to the control group means in each tercile. Above each bar we present the mean in the treatment group (along with the 95% confidence interval) in each tercile.

Table 2: Full sample vs. contacted sample

		Oversubscribed					Undersubscribed		
	All applicants (1)	Admin (2)	Phone Survey #1 (3)	Phone Survey #2 (4)	Difference (3)-(2) (5)	Difference (4)-(2) (6)	Admin (7)	Phone survey (8)	Difference (8)-(7) (9)
Female	0.48	0.48	0.49	0.49	0.00	0.01	0.49	0.48	-0.00
	(0.50)	(0.50)	(0.50)	(0.50)	(0.01)	(0.01)	(0.50)	(0.50)	(0.02)
	[54,676]	[16,703]	[7,376]	[4,338]			[37,973]	[462]	
Age (Jan 1st, 2019)	4.16	4.03	3.96	3.93	-0.01	-0.01	4.21	4.24	0.03
	(0.96)	(0.94)	(0.92)	(0.92)	(0.01)	(0.01)	(0.96)	(0.92)	(0.04)
	[54,676]	[16,703]	[7,376]	[4,338]			[37,973]	[462]	
Scheduled Caste	0.23	0.18	0.18	0.17	-0.01	-0.01	0.25	0.26	0.01
	(0.42)	(0.39)	(0.38)	(0.38)	(0.01)	(0.01)	(0.43)	(0.44)	(0.02)
	[54,676]	[16,703]	[7,376]	[4,338]			[37,973]	[462]	
Scheduled Tribe	0.21	0.18	0.13	0.11	-0.05***	-0.06***	0.22	0.13	-0.09***
	(0.40)	(0.39)	(0.33)	(0.31)	(0.01)	(0.01)	(0.41)	(0.33)	(0.02)
	[54,676]	[16,703]	[7,376]	[4,338]			[37,973]	[462]	
Other Backward Class	0.47	0.52	0.56	0.58	0.04***	0.06***	0.46	0.53	0.07***
	(0.50)	(0.50)	(0.50)	(0.49)	(0.01)	(0.01)	(0.50)	(0.50)	(0.02)
	[54,676]	[16,703]	[7,376]	[4,338]			[37,973]	[462]	
Rural	0.56	0.43	0.34	0.31	-0.04***	-0.05***	0.62	0.54	-0.08***
	(0.50)	(0.49)	(0.48)	(0.46)	(0.01)	(0.01)	(0.49)	(0.50)	(0.02)
	[54,676]	[16,703]	[7,376]	[4,338]			[37,973]	[462]	
Allocated a seat	0.87	0.67	0.66	0.67	0.02**	0.03***	0.96	0.97	0.00
	(0.34)	(0.47)	(0.47)	(0.47)	(0.01)	(0.01)	(0.18)	(0.18)	(0.01)
	[54,676]	[16,703]	[7,376]	[4,338]			[37,973]	[462]	

Notes: The p-value of the null hypothesis that the differences across all the observable applicant characteristics (Column 5) are jointly zero is 0. The p-value of the null hypothesis that the differences across all the observable applicant characteristics (Column 6) are jointly zero is 0. The p-value of the null hypothesis that the differences across all the observable applicant characteristics (Column 9) are jointly zero is 0. All differences control for the probability of being assigned to a private school by the assignment mechanisms following Abdulkadiroğlu et al. (2017). Standard errors are clustered at the applicant level. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table 3: Differences
between statewide population, applicants, and our in-person survey sample in Raipur

	NFHS All	NFHS Raipur	In-person sample	(3)-(1)	(3)-(2)
	(1)	(2)	(3)	(4)	(5)
Panel A: Summary					
Asset index	0.17	0.59	0.55	0.38***	-0.04
Mother education: above primary	0.58	0.64	0.79	0.21***	0.15***
Father education: above primary	0.71	0.71	0.78	0.07***	0.07***
Panel B: Assets					
Owns house	0.64	0.73	0.88	0.24***	0.16***
number of rooms used for sleeping	2.11	2.20	2.79	0.68***	0.59***
table	0.37	0.53	0.47	0.10***	-0.06*
cot or bed	0.92	0.97	0.87	-0.06***	-0.11***
chair	0.82	0.87	0.80	-0.02	-0.07***
has electricity	0.98	1.00	0.98	-0.01	-0.02***
electric fan	0.78	0.94	0.94	0.16***	0.00
has television	0.70	0.89	0.82	0.12***	-0.07***
has refrigerator	0.22	0.35	0.35	0.13***	-0.01
has bicycle	0.72	0.77	0.57	-0.15***	-0.20***
has motorcycle/scooter	0.59	0.74	0.68	0.09***	-0.06**
Panel C: Maternal education					
no education	0.23	0.12	0.10	-0.12***	-0.02
incomplete primary	0.06	0.08	0.04	-0.02***	-0.04***
complete primary	0.13	0.15	0.07	-0.06***	-0.09***
incomplete secondary	0.41	0.38	0.47	0.06***	0.09***
complete secondary	0.10	0.10	0.19	0.09***	0.08***
higher	0.07	0.16	0.14	0.06***	-0.02
Panel D: Paternal education					
no education	0.11	0.06	0.06	-0.05***	0.00
incomplete primary	0.07	0.11	0.04	-0.02***	-0.07***
complete primary	0.12	0.12	0.11	-0.01	-0.01
incomplete secondary	0.47	0.39	0.46	-0.01	0.08**
complete secondary	0.11	0.11	0.16	0.05***	0.05**
higher	0.12	0.22	0.16	0.04***	-0.06**

Notes: This table shows the prevalence of different characteristics for households in the NFHS sample (Columns 1), households in Raipur in the NFHS sample (Column 2), and the sample of households in Raipur we interviewed in person (Columns 3). It also shows the differences between the samples and whether these differences are statistically significant (Columns 4–5). We re-weight our sample to account for differential non-response by household characteristics. We estimate the probability of responding to our (phone) survey for applicants using a probit model that accounts for the household district, caste, and child's age and gender. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table 4:
Differences between applicants, eligible households and the average household in Chhattisgarh

	NFHS All	NFHS Eligible	Applicants	(2)-(1)	(1)-(3)	(2)-(3)
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Summary						
Asset index	0.17	-0.21	0.31	-0.38***	-0.13***	-0.51***
Mother education: above primary	0.58	0.48	0.76	-0.10***	-0.18***	-0.28***
Father education: above primary	0.71	0.63	0.86	-0.07***	-0.15***	-0.22***
Panel B: Assets						
Owns house	0.64	0.61	0.89	-0.03***	-0.25***	-0.28***
Rooms	2.11	1.91	2.78	-0.20***	-0.66***	-0.86***
table	0.37	0.23	0.43	-0.14***	-0.06***	-0.20***
cot or bed	0.92	0.90	0.91	-0.02***	0.01	-0.01
chair	0.82	0.73	0.78	-0.09***	0.04***	-0.05***
has electricity	0.98	0.98	0.99	-0.01***	-0.01**	-0.01***
electric fan	0.78	0.68	0.90	-0.10***	-0.12***	-0.22***
has television	0.70	0.56	0.62	-0.13***	0.08***	-0.06***
has refrigerator	0.22	0.11	0.20	-0.11***	0.02	-0.09***
has bicycle	0.72	0.68	0.76	-0.04***	-0.04***	-0.08***
has motorcycle/scooter	0.59	0.45	0.44	-0.14***	0.15***	0.01
Panel C: Maternal education						
no education	0.23	0.29	0.09	0.07***	0.14***	0.20***
incomplete primary	0.06	0.07	0.04	0.01***	0.03***	0.04***
complete primary	0.13	0.15	0.10	0.02***	0.03***	0.05***
incomplete secondary	0.41	0.38	0.50	-0.03***	-0.09***	-0.12***
complete secondary	0.10	0.07	0.18	-0.03***	-0.08***	-0.11***
higher	0.07	0.04	0.08	-0.04***	-0.01	-0.05***
Panel D: Paternal education						
no education	0.11	0.15	0.03	0.04***	0.08***	0.12***
incomplete primary	0.07	0.08	0.02	0.02***	0.04***	0.06***
complete primary	0.12	0.13	0.08	0.02***	0.03***	0.05***
incomplete secondary	0.47	0.47	0.50	-0.01	-0.02	-0.03
complete secondary	0.11	0.09	0.22	-0.02***	-0.12***	-0.13***
higher	0.12	0.07	0.13	-0.05***	-0.01	-0.06***

Notes: This table shows the prevalence of different characteristics for households in the representative National Family Health Survey (NFHS) 2019–21 (Column 1), for eligible households in the NFHS survey, as proxied by their caste and whether they are below the district specific poverty rate when ranked by the socioeconomic index (Column 2), and all applicants (Column 3). We re-weight our sample to account for differential non-response by household characteristics. We estimate the probability of responding to our survey using a probit model that accounts for the household district, caste, and the child's age and gender. It also shows the differences between the samples and whether these differences are statistically significant (Columns 4–6). Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table 5: Balance across lottery winners and losers

	Admin data		Phone survey #1		Phone survey #2	
	Control mean (1)	Treatment differential (2)	Control mean (3)	Treatment differential (4)	Control mean (5)	Treatment differential (6)
Female	0.48	0.00	0.49	0.00	0.48	0.01
	(0.50)	(0.01)	(0.50)	(0.02)	(0.50)	(0.02)
	[5,388]	[11,024]	[2,546]	[4,481]	[1,426]	[2,591]
Age (Jan 1st, 2019)	4.06	-0.01	4.00	-0.01	3.98	-0.01
	(0.94)	(0.01)	(0.94)	(0.01)	(0.92)	(0.01)
	[5,388]	[11,024]	[2,546]	[4,481]	[1,426]	[2,591]
Scheduled Caste	0.17	-0.00	0.16	0.01	0.15	0.03*
	(0.38)	(0.01)	(0.37)	(0.01)	(0.36)	(0.01)
	[5,388]	[11,024]	[2,546]	[4,481]	[1,426]	[2,591]
Scheduled Tribe	0.16	-0.00	0.12	-0.00	0.10	-0.00
	(0.37)	(0.01)	(0.32)	(0.01)	(0.30)	(0.01)
	[5,388]	[11,024]	[2,546]	[4,481]	[1,426]	[2,591]
Other Backward Class	0.54	-0.00	0.57	-0.01	0.60	-0.01
	(0.50)	(0.01)	(0.49)	(0.01)	(0.49)	(0.02)
	[5,388]	[11,024]	[2,546]	[4,481]	[1,426]	[2,591]
Rural	0.37	0.00	0.29	0.02**	0.28	0.01
	(0.48)	(0.01)	(0.45)	(0.01)	(0.45)	(0.01)
	[5,388]	[11,024]	[2,546]	[4,481]	[1,426]	[2,591]
Surveyed			0.45	0.02**	0.26	0.03***
			(0.50)	(0.01)	(0.44)	(0.01)
			[5,388]	[11,024]	[5,388]	[11,024]
Allocated a seat			0.22	0.73***	0.18	0.77***
			(0.41)	(0.01)	(0.38)	(0.01)
			[2,487]	[4,446]	[1,305]	[2,675]

Notes: Odd columns report the control (lottery losers) mean, standard deviation of the mean (in parentheses), and number of observations in the control group (in square brackets). Even columns report the treatment effect (difference between lottery winners and losers), the standard error of the effect (in parentheses), and the number of observations in the treatment group (in square brackets). Columns 1–2 focus on the full sample. The p-value of the null hypothesis that the differences across all the observable applicant characteristics (Column 2) are jointly zero is .81. Columns 3–4 focus on those who completed the first phone survey. The p-value of the null hypothesis that the differences across all the observable applicant characteristics (Column 4) are jointly zero is .25. Columns 5–6 focus on those who answered our second phone survey. The p-value of the null hypothesis that the differences across all the observable applicant characteristics (Column 6) are jointly zero is .62. All differences control for the probability of being assigned to a private school by the assignment mechanisms following Abdulkadiroğlu et al. (2017). Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table 6: Effect on the extensive margin of enrollment

	Any school				Private school			
	Control mean	ITT	CCM	LATE	Control mean	ITT	CCM	LATE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
All	0.86	0.14***	0.83	0.19***	0.82	0.18***	0.78	0.25***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
		[7,027]		[6,933]		[6,976]		[6,890]
Nursery	0.81	0.19***	0.76	0.26***	0.78	0.22***	0.73	0.29***
	(0.01)	(0.01)	(0.02)	(0.02)	(0.01)	(0.01)	(0.02)	(0.02)
		[3,766]		[3,721]		[3,735]		[3,694]
Kindergarten	0.89	0.11***	0.86	0.16***	0.83	0.17***	0.79	0.23***
	(0.01)	(0.01)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
		[1,869]		[1,843]		[1,858]		[1,836]
Grade 1	0.98	0.02***	0.97	0.04***	0.91	0.09***	0.89	0.13***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)	(0.02)
		[1,392]		[1,369]		[1,383]		[1,360]

Notes: Columns 1 and 5 report the control (lottery losers) mean and the standard error of the mean (in parentheses). Columns 2 and 6 list the intent-to-treat (ITT) effect (difference between lottery winners and losers), the standard error of the effect (in parentheses), and the number of observations used to estimate the effect (in square brackets). Columns 3 and 7 report the control complier mean (CCM) — the mean outcomes for lottery loser compliers — and the standard error of the CCM (in parentheses). Columns 4 and 8 list the local average treatment effect (LATE) of being assigned an RTE seat (instrumented by winning the lottery), the standard error of the effect (in parentheses), and the number of observations used to estimate the effect (in square brackets). All differences control for the probability of being assigned to a private school by the assignment mechanisms following Abdulkadiroğlu et al. (2017). Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table 7: Effect on the characteristics of the school a child attends

	English medium (1)	% students ST & SC (2)	Facility index (3)	Enrollment (4)	Teachers (5)	PTR (6)	Fees (7)	Top choice (8)
Panel A: ITT								
Lottery seat	.057*	-.9	.062	47***	.64	3**	2,059***	.37***
	(.03)	(1.5)	(.044)	(16)	(.58)	(1.2)	(419)	(.031)
N. of obs.	1,373	971	982	915	954	903	1,046	1,156
Control mean	0.56	31.09	0.72	312.14	13.24	23.07	6,649.84	0.54
Control mean | enrolled	0.57	32.08	0.74	323.13	13.66	23.88	6,859.43	0.55
% Enrolled (Control)	98.04	96.90	96.96	96.60	96.90	96.63	96.94	97.16
Panel B: LATE								
Allocated an RTE seat	.086**	-1.1	.082	65***	.78	4.7***	2,880***	.52***
	(.043)	(2.2)	(.063)	(23)	(.82)	(1.8)	(579)	(.049)
N. of obs.	1,353	960	971	904	943	892	1,034	1,144
CCM	0.50	33.15	0.71	323.67	14.21	22.54	5,945.74	0.58
CCM | enrolled	0.52	34.19	0.73	336.29	14.75	23.33	6,246.41	0.61
% Enrolled (CCM)	97.01	96.58	96.66	96.40	96.47	96.49	95.66	96.00

Notes: Panel A presents the intent-to-treat (ITT) effects of winning a seat through the lottery on different characteristics of the school the child is enrolled in. The sample is restricted to students applying for Grade 1 seats. Panel B presents the local average treatment effect (LATE) of being allocated an RTE (instrumenting with the outcome of the lottery) on different characteristics of the school the child is enrolled in. CCM denotes the mean outcomes for lottery loser compliers. In Column 1, the outcome is whether the child attends an English-medium school or not. In Column 2, the outcome is the percentage of enrollment taken by Scheduled Castes and Tribes in the school the child attends. In Column 3, the outcome is a principal component analysis (PCA) facility index based on whether the school has computer-assisted learning, a homeroom, electricity, a library, a playground, a solid building, a boundary wall, functioning toilets, and solid classrooms. In Columns 4-6 the outcomes are enrollment, number of teachers, and the pupil-teacher ratio (PTR). All columns control for a full set of probability-bin fixed effects of being assigned to a private school by the assignment mechanisms following Abdulkadiroğlu et al. (2017). Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table 8: Balance in the application assistance experiment

	Control mean (1)	Treatment differential (2)
Parental edu: Completed primary	0.71	0.01
	(0.46)	(0.03)
	[455]	[459]
Parental edu: No formal education	0.04	0.01
	(0.20)	(0.01)
	[455]	[459]
Improved water source	0.38	0.04
	(0.48)	(0.03)
	[455]	[459]
SES Index	0.09	-0.01
	(0.94)	(0.06)
	[455]	[459]
SC/ST	0.15	0.04
	(0.36)	(0.02)
	[455]	[459]
Private school (when all free)	0.78	-0.03
	(0.42)	(0.03)
	[455]	[459]
Heard about RTE (baseline)	0.52	0.01
	(0.50)	(0.03)
	[455]	[459]
Follow-up interview	0.80	0.00
	(0.40)	(0.03)
	[455]	[459]

Notes: Column 1 presents the control mean, standard deviation of the mean (in parentheses), and the number of observations in the control group (in square brackets). Column 2 reports the treatment effect, the standard error of the effect (in parentheses), and the number of observations in the treatment group (in square brackets). All treatment estimates control for strata (cluster) fixed effects. Standard errors are clustered at the household level. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table 9: Treatment effects from application assistance intervention

	(1)	(2)	(3)	(4)
Panel A: Heard of RTE				
Treatment (ITT)	.057***	.057***	.061***	.061***
	(.016)	(.016)	(.016)	(.016)
N. of obs.	733	733	733	733
Control mean	.93	.93	.93	.93
Panel B: Knows application window				
Treatment (ITT)	.18***	.18***	.18***	.18***
	(.029)	(.029)	(.03)	(.029)
N. of obs.	733	733	733	733
Control mean	.69	.69	.69	.69
Panel C: Applied to RTE this year				
Treatment (ITT)	.094***	.096***	.093***	.093***
	(.032)	(.032)	(.033)	(.033)
N. of obs.	733	733	733	733
Control mean	.22	.22	.22	.22
Panel D: Secured RTE seat				
Treatment (ITT)	.034	.034	.036*	.035*
	(.021)	(.021)	(.021)	(.021)
N. of obs.	733	733	733	733
Control mean	.079	.079	.079	.079
Panel E: Used RTE seat				
Treatment (ITT)	.032	.033	.034	.033
	(.021)	(.021)	(.021)	(.021)
N. of obs.	733	733	733	733
Control mean	.077	.077	.077	.077
SES controls	No	Yes	Yes	Yes
Preferences over private schools	No	No	Yes	Yes
RTE Knowledge (baseline)	No	No	No	Yes

Notes: The outcome in Panel A is whether the household had heard of the RTE policy, in Panel B is whether they knew the right dates for the application window, in Panel C is whether the baseline child applied this year, in panel D is whether that child secured an RTE seat, and in Panel E is whether that child enrolled in an RTE seat. Column 1 does not include any additional controls. Column 2 includes socioeconomic status controls (parental education, access to improved water access and sanitation, SES index, and caste). Column 3 includes socioeconomic status controls, as well as controls for preferences over private schools (baseline shares of children in government and private schools and preferences over private schools in the fictitious scenarios discussed in Section D). Column 4 also controls for knowledge of the RTE policy (whether they had heard of the policy before and whether they had applied for an RTE seat before). All estimations are done via ordinary least squares, controlling for strata (village) fixed effects and clustering standard errors at the household level. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Online Appendix for "The benefit incidence of school vouchers: Evidence from private schools in India" by Romero and Singh

A Appendix Figures

Figure A.1: Timeline of data collection and applications/enrollment

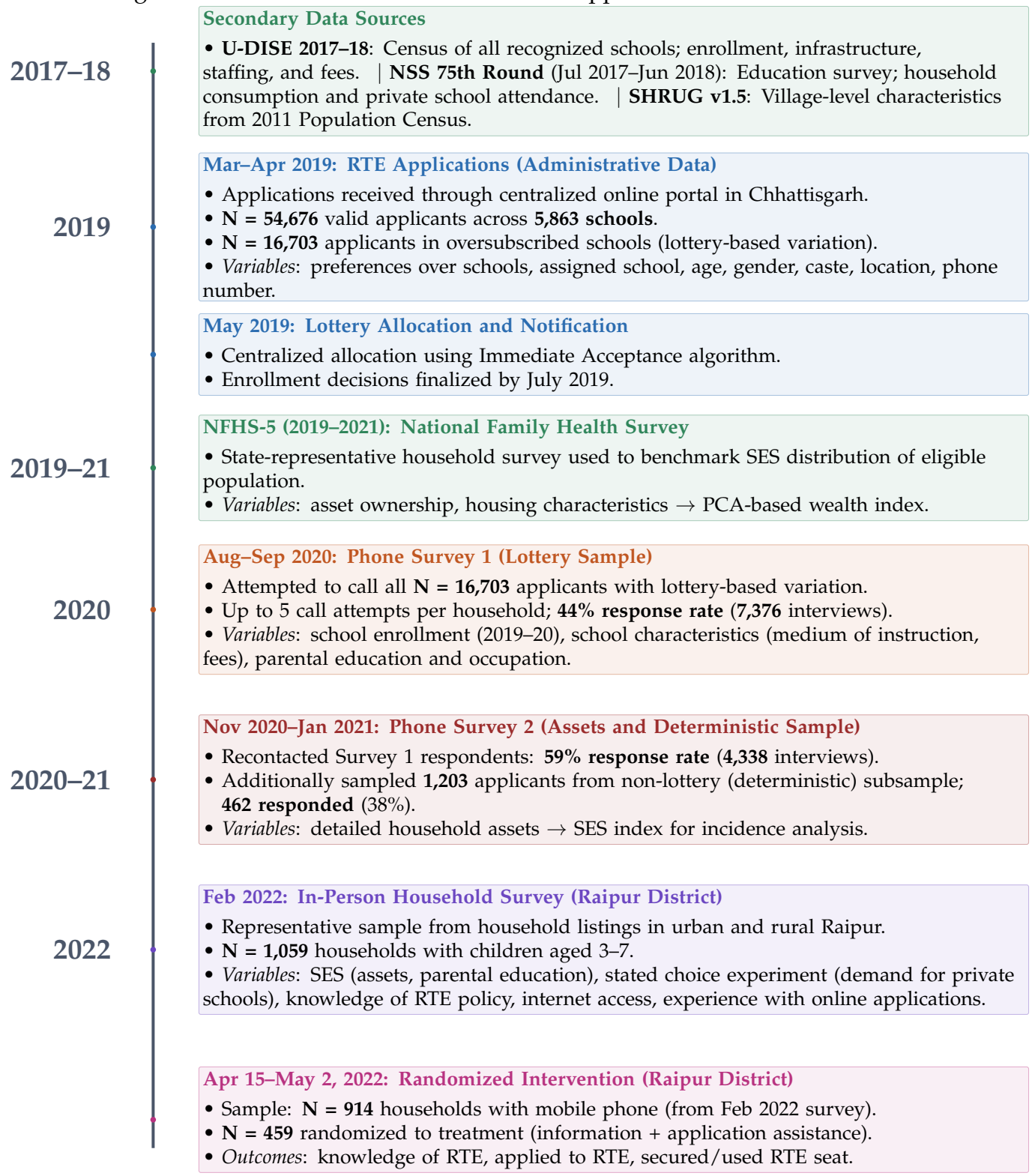
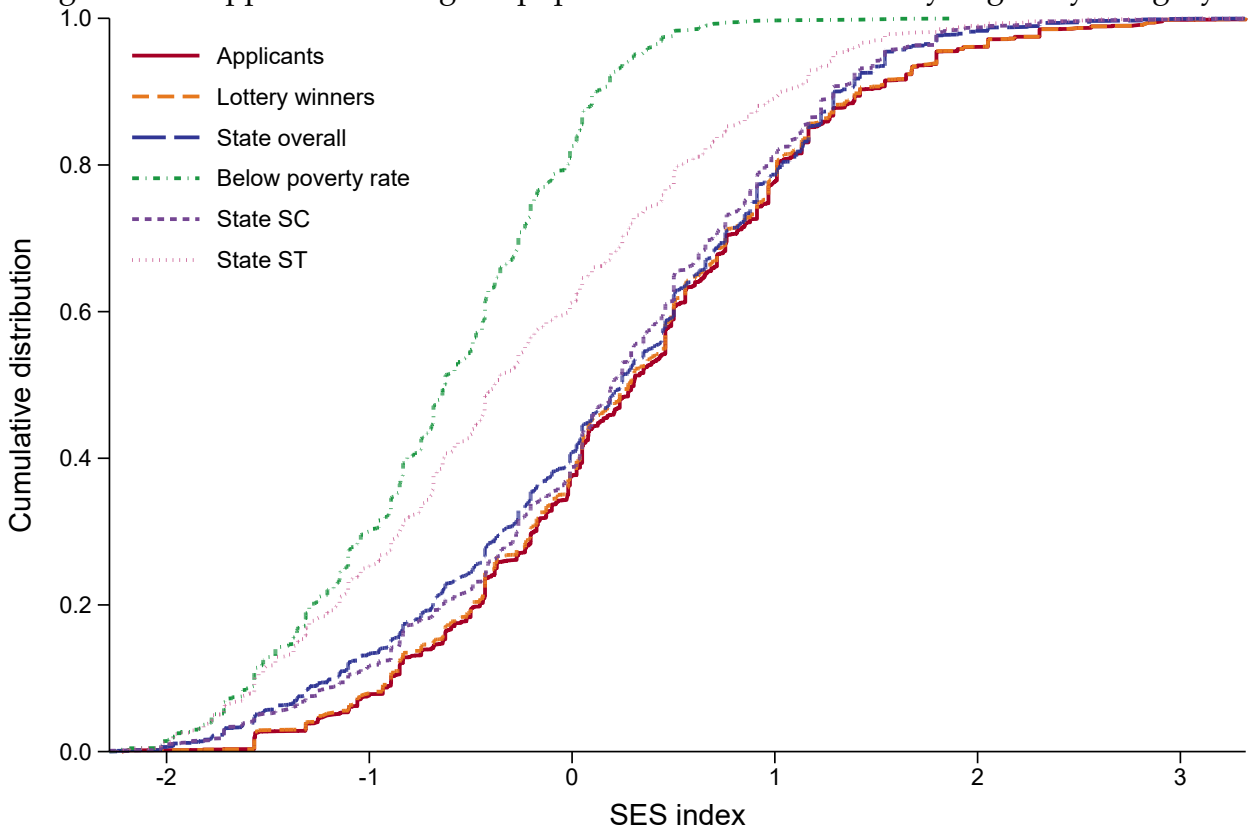
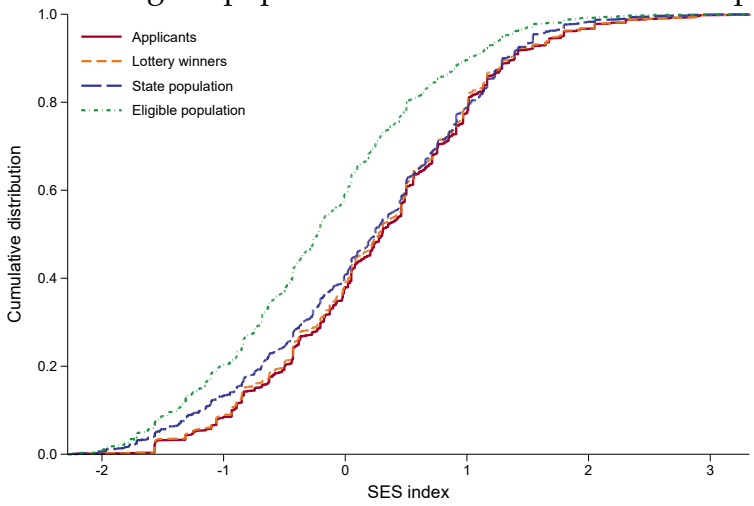


Figure A.2: Applicants vs eligible population in terms of SES by eligibility category



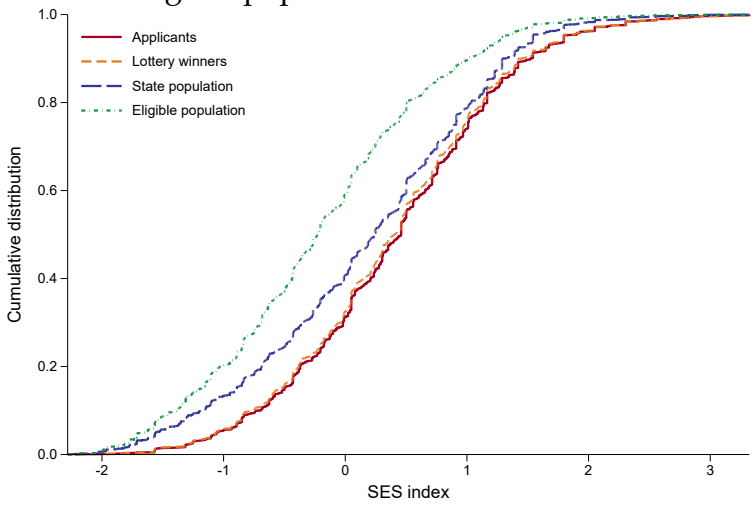
Note: This figure plots the cumulative distribution of a socioeconomic status (SES) index for six (overlapping) groups: (i) applicants to RTE quota seats in our phone survey sample (solid red), (ii) lottery winners among applicants (dashed orange), (iii) all households in Chhattisgarh with a child aged 4–7 in the NFHS-5 (long-dashed blue), (iv) the subset of NFHS-5 households below the district-specific poverty rate in the distribution of assets and consumer durables, who qualify via the income track (dash-dotted green), (v) Scheduled Caste households in the NFHS-5, who qualify via the caste track (short-dashed purple), and (vi) Scheduled Tribe households in the NFHS-5, who qualify via the caste track (dotted pink). The SES index is computed from household ownership of consumer durables and assets, which were collected in the representative National Family Health Survey 2019–21 for the statewide population and in our survey of applicants; the index is the first principal component estimated on the NFHS-5 Chhattisgarh sample and standardized, and applicants are mapped to the same index using the NFHS-5 PCA loadings so that all six series share a common horizontal axis. NFHS-5 weights are applied to the population samples. We re-weight our applicant sample to account for differential non-response by household characteristics; we estimate the probability of responding to our survey using a probit model that accounts for the household district, caste, and the child's age and gender.

Figure A.3: Applicants vs eligible population in terms of SES — linear probability weights



Note: This figure plots the cumulative distribution of a socioeconomic status (SES) index for four groups: (i) applicants to RTE quota seats in our phone survey sample (solid red), (ii) lottery winners among applicants (dashed orange), (iii) all households in Chhattisgarh with a child aged 4–7 in the NFHS-5 (dashed blue), and (iv) the subset of NFHS-5 households likely to be eligible for quota seats based on caste or economic criteria (dash-dotted green). The SES index is computed from household ownership of consumer durables and assets, which were collected in the representative National Family Health Survey 2019–21 for the statewide population and in our survey of applicants. NFHS-5 weights are applied to the population samples. We re-weight our sample to account for differential non-response by household characteristics. We estimate the probability of responding to our survey using a linear probability model that accounts for the household district, pincode, rurality, caste, and the child's age and gender.

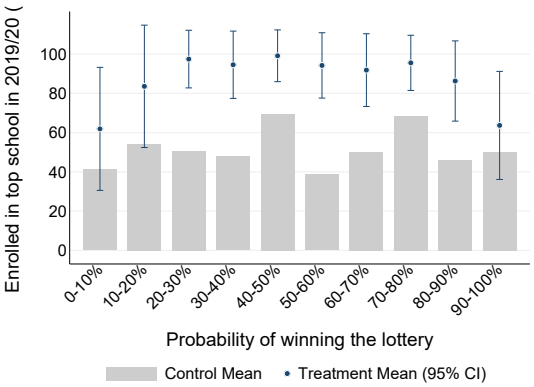
Figure A.4: Applicants vs eligible population in terms of SES — random forest weights



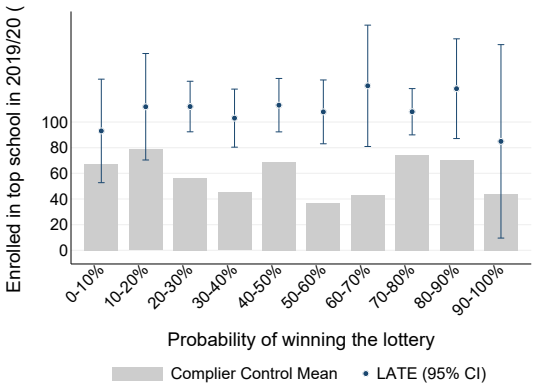
Note: This figure plots the cumulative distribution of a socioeconomic status (SES) index for four groups: (i) applicants to RTE quota seats in our phone survey sample (solid red), (ii) lottery winners among applicants (dashed orange), (iii) all households in Chhattisgarh with a child aged 4–7 in the NFHS-5 (dashed blue), and (iv) the subset of NFHS-5 households likely to be eligible for quota seats based on caste or economic criteria (dash-dotted green). The SES index is computed from household ownership of consumer durables and assets, which were collected in the representative National Family Health Survey 2019–21 for the statewide population and in our survey of applicants. NFHS-5 weights are applied to the population samples. We re-weight our sample to account for differential non-response by household characteristics. We estimate the probability of responding to our survey using a random forest model that accounts for the household district, rurality, class, the number of schools to which they applied, the likelihood of winning the lottery, caste, and the child's age and gender.

Figure A.5: Heterogeneity
by how oversubscribed a school is on the probability of attending the top choice

(a) Intent-to-treat



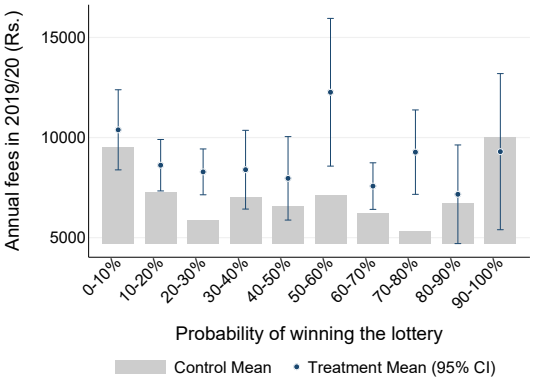
(b) LATE



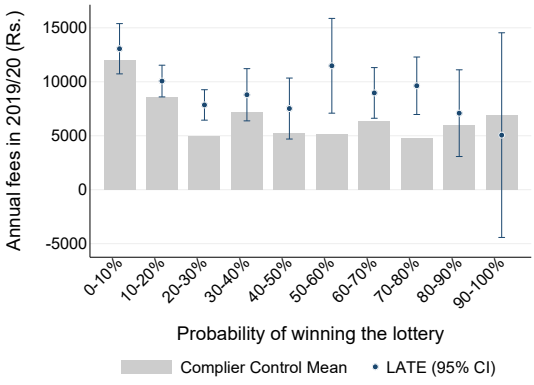
Note: This figure examines how the effects of the RTE quota on enrolling in the top choice vary by how oversubscribed the school is (i.e., by the ex-ante probability of winning a quota seat). The sample is restricted to students applying to Grade 1. The x-axis divides the sample into deciles based on the ex-ante probability of winning the lottery (calculated via simulation of the assignment algorithm); lower probabilities indicate more oversubscribed schools. The y-axis represents the percentage of students enrolled in their top choice in the 2019-2020 academic year. Figure A.5a shows the control mean and the intent-to-treat effect (with 95% confidence intervals). Figure A.5b shows the control-complier mean and the LATE effect (instrumented by securing an RTE seat via winning the lottery, with 95% confidence intervals).

Figure A.6:
Heterogeneity by how oversubscribed a school is on the nominal price of the school

(a) Intent-to-treat



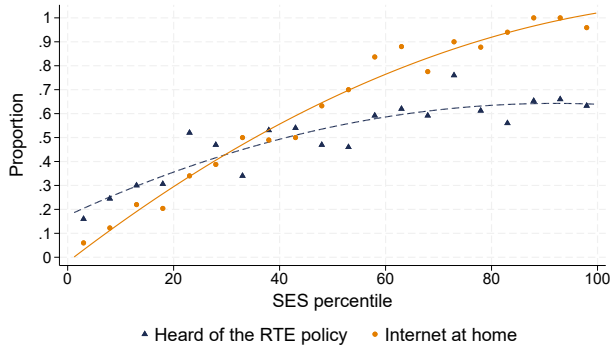
(b) LATE



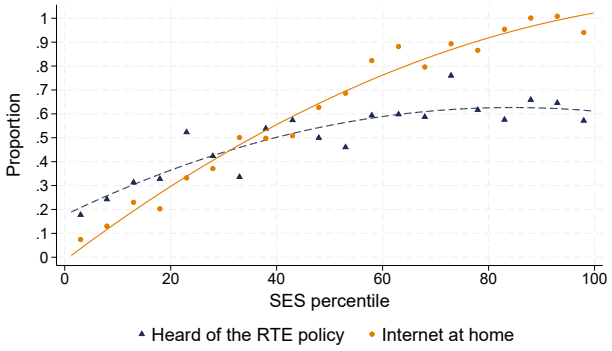
Note: This figure examines how the effects of the RTE quota on the nominal fees of the enrollment option vary by how oversubscribed the school is (i.e., by the ex-ante probability of winning a quota seat). The sample is restricted to students applying to Grade 1. The x-axis divides the sample into deciles based on the ex-ante probability of winning the lottery (calculated via simulation of the assignment algorithm); lower probabilities indicate more oversubscribed schools. The y-axis represents the nominal fees of the enrollment option. Figure A.6a shows the control mean and the intent-to-treat effect (with 95% confidence intervals). Figure A.6b shows the control-complier mean and the LATE effect (instrumented by securing an RTE seat via winning the lottery, with 95% confidence intervals).

Figure A.7: Socioeconomic status and familiarity with the RTE policy

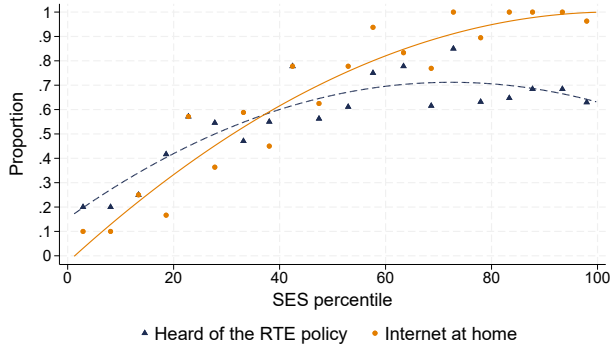
(a) Overall variation



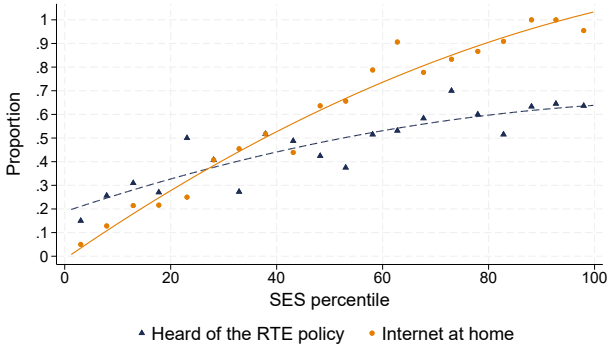
(b) Variation within sampling unit



(c) Urban areas



(d) Rural areas



Note: This figure presents binscatter least squares estimations — following Cattaneo et al. (2024) — of the relationship between the percentiles of SES and: a) having heard of the RTE quota policy and having internet at home. Figure A.7a shows the overall relationship. Figure A.7b shows the relationship within sampling units. Figure A.7c shows the relationship within urban sampling units. Figure A.7d shows the relationship within rural sampling units.

Appendix Tables

Table A.1: Summary statistics

	Mean (1)	Median (2)	Std. Dev. (3)	Min (4)	Max (5)	N. of obs. (6)
Panel A: All applicants						
Unalloted	0.12	0.00	0.33	0	1	54,676
Rural	0.56	1.00	0.50	0	1	54,676
Age (Jan 1st, 2019)	4.16	3.97	0.96	2.8	6.3	54,676
Female	0.48	0.00	0.50	0	1	54,676
Scheduled Caste	0.23	0.00	0.42	0	1	54,676
Scheduled Tribe	0.21	0.00	0.40	0	1	54,676
Other Backward Class	0.47	0.00	0.50	0	1	54,676
No. of preferences	1.63	1.00	1.35	1	26	54,676
Panel B: Applicants in oversubscribed schools						
Unalloted	0.33	0.00	0.47	0	1	16,703
Rural	0.43	0.00	0.49	0	1	16,703
Age (Jan 1st, 2019)	4.03	3.79	0.94	2.8	6.3	16,703
Female	0.48	0.00	0.50	0	1	16,703
Scheduled Caste	0.18	0.00	0.39	0	1	16,703
Scheduled Tribe	0.18	0.00	0.39	0	1	16,703
Other Backward Class	0.52	1.00	0.50	0	1	16,703
No. of preferences	1.63	1.00	1.21	1	12	16,703
Panel C: Schools						
Seats	10.07	9.14	6.37	1	80	5,863
No. applicants	15.21	10.00	19.01	1	284	5,863
Has nursery seats	0.52	1.00	0.50	0	1	5,863
Has KG seats	0.35	0.00	0.48	0	1	5,863
Has Grade 1 seats	0.47	0.00	0.50	0	1	5,863
Hindi medium	0.52	1.00	0.50	0	1	5,863
English medium	0.41	0.00	0.49	0	1	5,863

Notes: This table presents summary statistics for all lottery applicants (Panel A) and for applicants we attempted to contact during our phone survey (Panel B). Further, we present summary statistics from schools in the lottery (Panel C).

Table A.2: Attrition by child characteristics

	Survey #1 (Lottery)	Survey #2 (Lottery)	Survey (Deterministic)
	(1)	(2)	(3)
Female	.0034	.0053	-.015
	(.0079)	(.007)	(.028)
Age (Jan 1st, 2019)	-.013	-.0074	-.019
	(.012)	(.01)	(.015)
Scheduled Caste	-.047***	-.027*	.093*
	(.016)	(.015)	(.055)
Scheduled Tribe	-.12***	-.084***	-.068
	(.016)	(.015)	(.055)
Other Backward Class	-.018	.003	.14***
	(.014)	(.013)	(.051)
Rural	-.066***	-.07***	-.062**
	(.012)	(.011)	(.031)
N. of obs.	16,703	16,703	1,203
Outcome mean	.44	.26	.38

Notes: The outcome is whether we were able to conduct the interview (=1). All columns control for a full set of probability-bin fixed effects of being assigned to a private school by the assignment mechanisms following Abdulkadiroğlu et al. (2017). Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.3: Differences between applicants,
eligible households and the average household in Chhattisgarh — linear probability weights

	NFHS All	NFHS Eligible	Applicants	(2)-(1)	(1)-(3)	(2)-(3)
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Summary						
Asset index	0.17	-0.21	0.28	-0.38***	-0.11***	-0.49***
Mother education: above primary	0.58	0.48	0.78	-0.10***	-0.20***	-0.30***
Father education: above primary	0.71	0.63	0.86	-0.07***	-0.15***	-0.23***
Panel B: Assets						
Owns house	0.64	0.61	0.89	-0.03***	-0.25***	-0.28***
Rooms	2.11	1.91	2.72	-0.20***	-0.61***	-0.81***
table	0.37	0.23	0.45	-0.14***	-0.08***	-0.22***
cot or bed	0.92	0.90	0.91	-0.02***	0.01*	-0.01
chair	0.82	0.73	0.77	-0.09***	0.05***	-0.04***
has electricity	0.98	0.98	0.99	-0.01***	-0.01***	-0.02***
electric fan	0.78	0.68	0.89	-0.10***	-0.11***	-0.21***
has television	0.70	0.56	0.62	-0.13***	0.08***	-0.05***
has refrigerator	0.22	0.11	0.20	-0.11***	0.02*	-0.09***
has bicycle	0.72	0.68	0.74	-0.04***	-0.03**	-0.06***
has motorcycle/scooter	0.59	0.45	0.44	-0.14***	0.15***	0.00
Panel C: Maternal education						
no education	0.23	0.29	0.08	0.07***	0.15***	0.22***
incomplete primary	0.06	0.07	0.04	0.01***	0.03***	0.04***
complete primary	0.13	0.15	0.09	0.02***	0.04***	0.06***
incomplete secondary	0.41	0.38	0.50	-0.03***	-0.09***	-0.12***
complete secondary	0.10	0.07	0.17	-0.03***	-0.07***	-0.11***
higher	0.07	0.04	0.11	-0.04***	-0.03***	-0.07***
Panel D: Paternal education						
no education	0.11	0.15	0.03	0.04***	0.08***	0.13***
incomplete primary	0.07	0.08	0.03	0.02***	0.04***	0.06***
complete primary	0.12	0.13	0.08	0.02***	0.03***	0.05***
incomplete secondary	0.47	0.47	0.49	-0.01	-0.01	-0.02
complete secondary	0.11	0.09	0.22	-0.02***	-0.12***	-0.13***
higher	0.12	0.07	0.15	-0.05***	-0.02***	-0.07***

Notes: This table shows the prevalence of different characteristics for households in the representative National Family Health Survey (NFHS) 2019–21 (Column 1), for eligible households in the NFHS survey, as proxied by their caste and whether they are in the bottom tercile of the socioeconomic index (Column 2), and all applicants (Column 3). We re-weight our sample to account for differential non-response by household characteristics. We estimate the probability of responding to our survey using a linear probability model that accounts for the household district, pincode, rurality, caste, and the child's age and gender. It also shows the differences between the samples and whether these differences are statistically significant (Columns 4–6). Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.4: Differences between applicants,
eligible households and the average household in Chhattisgarh — random forest weights

	NFHS All	NFHS Eligible	Applicants	(2)-(1)	(1)-(3)	(2)-(3)
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Summary						
Asset index	0.17	-0.21	0.42	-0.38***	-0.25***	-0.63***
Mother education: above primary	0.58	0.48	0.80	-0.10***	-0.21***	-0.31***
Father education: above primary	0.71	0.63	0.86	-0.07***	-0.16***	-0.23***
Panel B: Assets						
Owns house	0.64	0.61	0.88	-0.03***	-0.24***	-0.27***
Rooms	2.11	1.91	2.80	-0.20***	-0.69***	-0.89***
table	0.37	0.23	0.48	-0.14***	-0.11***	-0.25***
cot or bed	0.92	0.90	0.92	-0.02***	0.01	-0.02**
chair	0.82	0.73	0.80	-0.09***	0.02**	-0.07***
has electricity	0.98	0.98	0.99	-0.01***	-0.01***	-0.01***
electric fan	0.78	0.68	0.92	-0.10***	-0.14***	-0.24***
has television	0.70	0.56	0.70	-0.13***	-0.00	-0.13***
has refrigerator	0.22	0.11	0.24	-0.11***	-0.02**	-0.13***
has bicycle	0.72	0.68	0.71	-0.04***	0.01	-0.03**
has motorcycle/scooter	0.59	0.45	0.49	-0.14***	0.10***	-0.05***
Panel C: Maternal education						
no education	0.23	0.29	0.08	0.07***	0.15***	0.22***
incomplete primary	0.06	0.07	0.03	0.01***	0.03***	0.04***
complete primary	0.13	0.15	0.08	0.02***	0.04***	0.07***
incomplete secondary	0.41	0.38	0.51	-0.03***	-0.10***	-0.13***
complete secondary	0.10	0.07	0.19	-0.03***	-0.09***	-0.13***
higher	0.07	0.04	0.09	-0.04***	-0.02***	-0.05***
Panel D: Paternal education						
no education	0.11	0.15	0.03	0.04***	0.08***	0.12***
incomplete primary	0.07	0.08	0.02	0.02***	0.05***	0.06***
complete primary	0.12	0.13	0.07	0.02***	0.04***	0.06***
incomplete secondary	0.47	0.47	0.49	-0.01	-0.02*	-0.02*
complete secondary	0.11	0.09	0.22	-0.02***	-0.11***	-0.13***
higher	0.12	0.07	0.15	-0.05***	-0.02***	-0.07***

Notes: This table shows the prevalence of different characteristics for households in the representative National Family Health Survey (NFHS) 2019–21 (Column 1), for eligible households in the NFHS survey, as proxied by their caste and whether they are in the bottom tercile of the socioeconomic index (Column 2), and all applicants (Column 3). We re-weight our sample to account for differential non-response by household characteristics. We estimate the probability of responding to our survey using a random forest model that accounts for the household district, rurality, class, the number of schools to which they applied, the likelihood of winning the lottery, caste, and the child's age and gender. It also shows the differences between the samples and whether these differences are statistically significant (Columns 4–6). Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.5: Effect on the extensive margin of enrollment, controlling for the probability of being assigned to a private school: Lee bounds and strata with low attrition

	Strata without attrition		Low attrition strata			Lee bounds	
	ITT	LATE	Differential attrition	ITT	LATE	ITT	
						LB	UB
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: All grades							
Private school (19-20)	0.19***	0.24***	0.02	0.20***	0.26***	0.15	0.24
	(0.05)	(0.07)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
	[214]	[211]	[11,795]	[5,172]	[5,117]	[4,729]	[4,729]
Any school (19-20)	0.14***	0.19***	0.02	0.15***	0.20***	0.11	0.19
	(0.05)	(0.06)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
	[140]	[139]	[11,795]	[5,209]	[5,148]	[4,751]	[4,751]
Panel B: Nursery							
Private school (19-20)	0.26***	0.33***	0.01	0.24***	0.31***	0.17	0.29
	(0.08)	(0.11)	(0.02)	(0.01)	(0.02)	(0.01)	(0.02)
	[81]	[81]	[5,622]	[2,738]	[2,715]	[2,555]	[2,555]
Any school (19-20)	0.21***	0.28***	0.01	0.21***	0.27***	0.15	0.25
	(0.07)	(0.10)	(0.02)	(0.01)	(0.02)	(0.01)	(0.02)
	[82]	[81]	[5,622]	[2,762]	[2,736]	[2,570]	[2,570]
Panel C: Kindergarten							
Private school (19-20)	0.13	0.15	0.01	0.18***	0.24***	0.14	0.23
	(0.11)	(0.13)	(0.02)	(0.02)	(0.03)	(0.02)	(0.02)
	[17]	[17]	[3,579]	[1,455]	[1,436]	[1,323]	[1,323]
Any school (19-20)	0.13	0.15	0.01	0.12***	0.17***	0.09	0.15
	(0.11)	(0.13)	(0.02)	(0.02)	(0.02)	(0.01)	(0.02)
	[17]	[17]	[3,579]	[1,463]	[1,441]	[1,327]	[1,327]
Panel D: Grade 1							
Private school (19-20)	0.07	0.11	0.03	0.11***	0.15***	0.08	0.12
	(0.07)	(0.10)	(0.02)	(0.02)	(0.03)	(0.02)	(0.02)
	[39]	[39]	[2,557]	[963]	[950]	[851]	[851]
Any school (19-20)	0.02	0.03	0.03	0.03***	0.04***	0.02	0.03
	(0.03)	(0.05)	(0.02)	(0.01)	(0.01)	(0.01)	(0.01)
	[41]	[41]	[2,557]	[969]	[956]	[854]	[854]

Notes: Columns 1–2 display the results restricting the sample to strata without attrition. Column 1 shows the intention-to-treat (ITT) effect of winning the lottery, and Column 2 the local average treatment effect (LATE) of being assigned an RTE seat (instrumented with winning the lottery). Columns 3–5 report the results after dropping the 25% of the strata with the most differential attrition. Column 3 shows the results of differential attrition, Column 4 the ITT effect, and Column 5 the LATE of being assigned an RTE seat. Columns 6–7 show Lee (2009) style bounds — Column 6 has the lower bound (LB), while Column 7 has the upper bound for (UB) — for the ITT effect of winning the lottery. Standard errors are in parentheses. The number of observations in the treatment effects estimates is in square brackets. All treatment estimates control for the probability of being assigned to a private school by the assignment mechanisms following Abdulkadiroğlu et al. (2017). Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.6: Compliance

	RTE seat			
	All	NU	KG	Grd 1
	(1)	(2)	(3)	(4)
Allocated a seat	0.73***	0.75***	0.72***	0.70***
	(0.010)	(0.014)	(0.021)	(0.025)
N. of obs.	6,933	3,721	1,843	1,369
Control mean	0.22	0.21	0.20	0.24

Notes: This table presents the effect of winning a lottery seat on being allotted an RTE seat. All regressions control for the probability of being assigned to a private school by the assignment mechanisms following Abdulkadiroğlu et al. (2017). Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.7: Relationship between familiarity with the RTE quota policy and socioeconomic status

	(1)	(2)	(3)	(4)
Panel A: Heard of the RTE quota seat policy				
SES Index	.14***	.13***	.13***	.14***
	(.012)	(.013)	(.013)	(.014)
Rural		-.1**		
		(.043)		
SC/ST				.08*
				(.04)
N. of obs.	990	990	990	990
Dep. var. mean	.5	.5	.5	.5
Panel B: Internet access				
SES Index	.3***	.29***	.3***	.3***
	(.012)	(.013)	(.013)	(.014)
Rural		-.049		
		(.029)		
SC/ST				.084**
				(.036)
N. of obs.	990	990	990	990
Dep. var. mean	.62	.62	.62	.62
Location fixed effects	No	No	Yes	Yes

Notes: This table shows the relationship between socioeconomic status (measured by a proxy means test), the likelihood that families have heard of the RTE policy (Panel A), and whether families have internet connection at home (Panel B). Column 1 is a simple univariate regression, Column 2 controls for whether the family lives in an urban or a rural location, Column 3 includes sampling location fixed effects, and Column 4 controls for whether the household belongs to one of the eligible caste groups (Scheduled Castes and Tribes). In rural areas, each sampling location has an area of 100 hectares. In urban areas, each sampling location has 9 hectares. Standard errors are clustered at the sampling location level for all regressions. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.8: Heterogeneity of treatment effects from application assistance intervention by household socioeconomic status

	Applied (1)	Secured seat (2)
Panel A: Above/below median		
Treatment (ITT) × SES - Below median	.076*	.018
	(.046)	(.029)
Treatment (ITT) × SES - Above median	.13**	.062*
	(.051)	(.034)
N. of obs.	684	684
p-value (H_0: all equal)	.46	.3
Panel B: Terciles		
Treatment (ITT) × SES - Tercile 1	.1*	.036
	(.053)	(.035)
Treatment (ITT) × SES - Tercile 2	.05	.0089
	(.061)	(.039)
Treatment (ITT) × SES - Tercile 3	.15**	.078*
	(.064)	(.044)
N. of obs.	684	684
p-value (H_0: all equal)	.53	.5
p-value(H_0:Extremes are equal)	.58	.46
Panel C: Quintiles		
Treatment (ITT) × SES - Quintile 1	.12*	.045
	(.068)	(.04)
Treatment (ITT) × SES - Quintile 2	.049	-.014
	(.073)	(.051)
Treatment (ITT) × SES - Quintile 3	.012	-.00055
	(.078)	(.049)
Treatment (ITT) × SES - Quintile 4	.28***	.2***
	(.083)	(.062)
Treatment (ITT) × SES - Quintile 5	.058	-.023
	(.082)	(.051)
N. of obs.	684	684
p-value (H_0: all equal)	.14	.028
p-value(H_0:Extremes are equal)	.54	.28

Notes: The outcome in Column 1 is whether the household applied this year and in Column 2 is whether they secured an RTE seat. All regressions include socioeconomic status controls, as well as controls for preferences over private schools (baseline shares of children in government and private schools and preferences over private schools in the fictitious scenarios discussed in Section D), as well as controls for knowledge of the RTE policy (whether they had heard of the policy before and whether they had applied for an RTE seat before). All estimations are done via ordinary least squares, controlling for strata (village) fixed effects and clustering standard errors at the household level. We exclude observations without all the necessary data to generate the SES index. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table A.9: Heterogeneous
treatment effects by baseline characteristics from the application assistance intervention

	Applied to RTE this year			
	Heard of RTE	Knows pvt. school nearby	Rural	Parent completed primary
	(1)	(2)	(3)	(4)
Treatment (ITT)	.036	.096	.096*	.018
	(.045)	(.068)	(.057)	(.06)
Treatment (ITT) × Baseline charac.	.11	-.0042	-.0053	.11
	(.066)	(.078)	(.07)	(.072)
Baseline charac.	.048	-.017	0	-.047
	(.05)	(.062)	(.)	(.051)
N. of obs.	733	733	733	733
Control mean	.53	.79	.69	.69

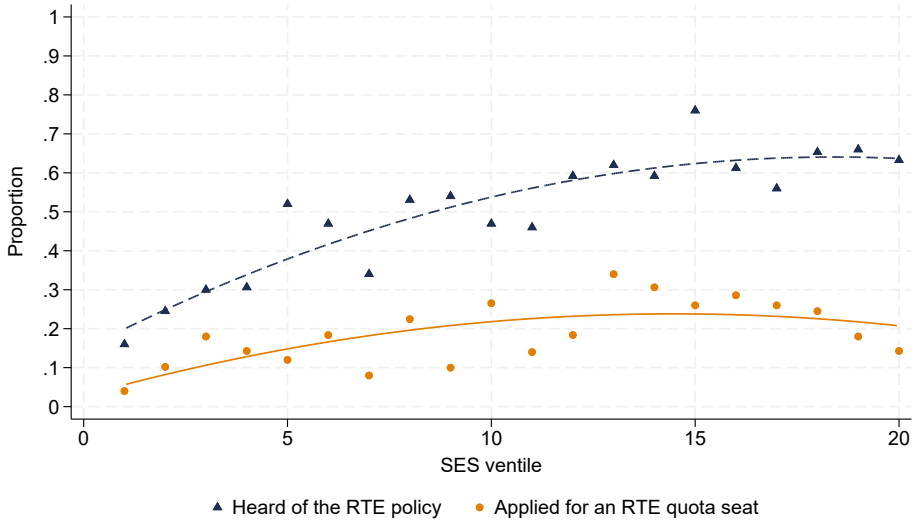
Notes: Each column reports results from a separate regression of whether the household applied for an RTE seat this year on treatment assignment, a baseline characteristic, and their interaction. The baseline characteristics are: whether the household had heard of the RTE policy (Column 1), whether the household knew of a private school within 1 km (Column 2), whether the household is in a rural area (Column 3), and whether at least one parent completed primary school (Column 4). All regressions include strata (village) fixed effects and the full set of controls from Column 4 of Table 9. Standard errors clustered at the household level in parentheses. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

B Incidence in the household data from Raipur

We confirm the main findings from Section 4 using our in-person household survey in Raipur district (N=1,059 households with children aged 3–7). This sample was drawn from comprehensive household listings in sampled neighborhoods and villages, independently of the RTE application process, and is thus informative about the broader population of eligible households (including those who never applied).

Figure B.1 shows the SES gradient in awareness and application from the full sample. The proportion of households that have heard of the RTE policy rises sharply with socioeconomic status, from below 20% in the bottom decile to above 65% at the top. Application rates follow a similar gradient: fewer than 7.1% of households in the lowest decile report having applied, compared with over 16% at the top. This pattern mirrors the regressive selection documented in the statewide applicant data.

Figure B.1: Awareness of and application to the RTE policy by SES



Note: This figure presents binscatter estimates of the relationship between ventiles of socioeconomic status and having heard of the RTE quota policy (triangles) or having applied for an RTE quota seat (circles), using our in-person household survey in Raipur district. Socioeconomic status is measured using an asset index constructed from household ownership of consumer durables and housing quality via Principal Component Analysis.

Many eligible households already send their children to private schools but have never applied for an RTE seat, thereby leaving money on the table. Among caste-eligible households with a child in a private school, 48.6% report having never applied under the RTE quota. Among poverty-eligible households in private schools, the figure is 42%.[54] These households pay full tuition fees despite qualifying for a free seat, suggesting that application and documentation frictions — rather than lack of demand for private schooling — are the binding constraint.

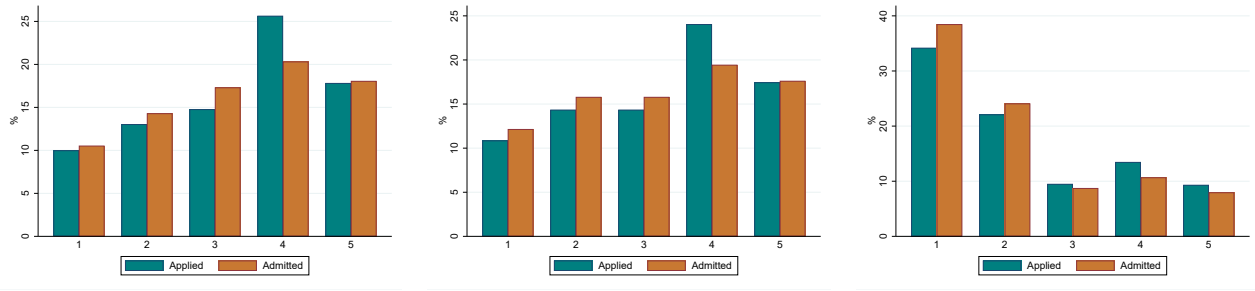
Application frictions, rather than weak targeting, drive the regressive incidence. Figure B.2, akin to Figure 2 for the in-person sample, shows the incidence of applications and admissions in our Raipur household sample. The observed distribution of applicants and admitted students is skewed towards the top quintiles of the socioeconomic distribution (Figure B.2a): the top two quintiles account for approximately 45% of all applications, whereas the bottom quintile contributes approximately 10%.

If every eligible child already enrolled in a private school applied for and used an RTE seat (Figure B.2b), the incidence barely changes relative to the status quo — this reflects that the likelihood of sending a child to private school is correlated with socioeconomic status. If, instead, all eligible households apply and enroll their children under the RTE quota (Figure B.2c), the distribution reverses: the bottom quintile would account for over 30% of beneficiaries, while the top quintile would contribute less than 10%. This confirms that the eligibility criteria are, by design, progressive. The regressive incidence documented in Section 4 is consistent with differential take-up across the socioeconomic distribution, rather than targeting design being the principal driver.

[54] Conservatively, we assume the bottom tercile of the SES distribution is eligible.

Figure B.2: Distribution of applicants across SES quintiles: Raipur household data

(a) Observed (b) Eligible in private schools (c) All eligible



Note: This figure uses the in-person household survey in Raipur district (N=1,059). Each panel shows each SES quintile's share of all applicants and admitted students. Panel (a) shows the observed distribution. Panel (b) reclassifies all eligible households (by caste or poverty criterion) whose children currently attend a private school as applicants and beneficiaries. Panel (c) reclassifies all eligible households as applicants and beneficiaries, regardless of current school type. SES quintiles are constructed from an asset index using Principal Component Analysis.

C Caste and income eligibility: joint incidence

This appendix addresses how caste-based and income-based eligibility for the RTE 25% quota interact in practice. The analysis partially addresses the concern that, although regressive in income, the policy would still achieve its primary goal of increasing the representation of Scheduled Castes and Tribes in private schools, even if it disproportionately benefits wealthier individuals within this group. Section F already highlighted that, although spatial constraints limit the equalization of caste composition, substantial reductions in segregation remain possible. We further study this along the income dimension, asking both how representative Scheduled Caste and Tribe applicants are of the Scheduled Caste and Tribe populations and how the two eligibility tracks jointly shape the applicant pool (see Table C.1).

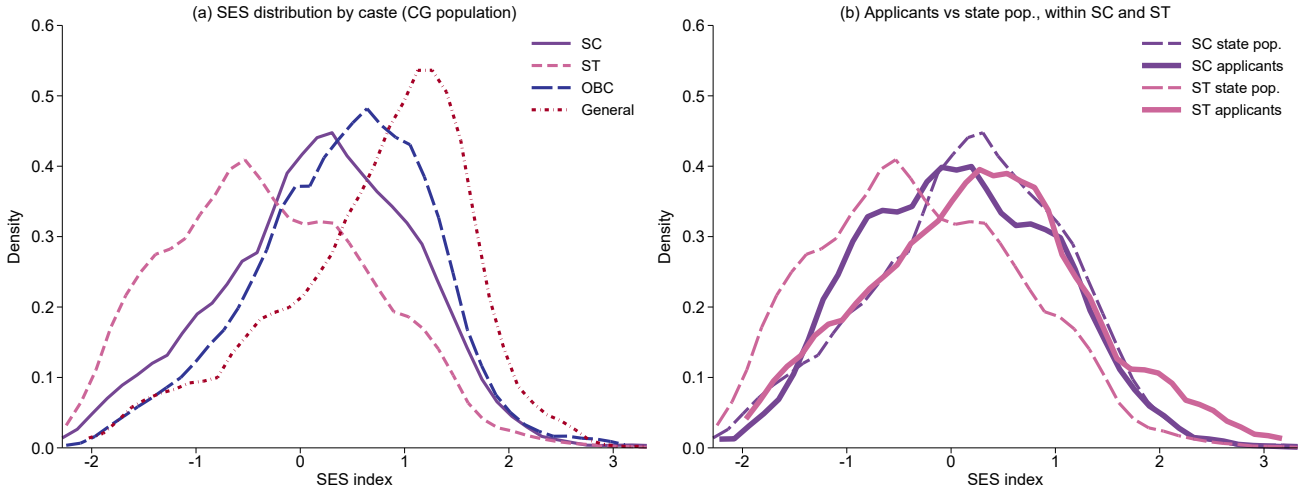
Table C.1: Population-level eligibility shares by caste (NFHS-V, Chhattisgarh)

Caste group	% below BPL	Population share (%)
SC	46.0	18.0
ST	62.0	31.2
SC + ST	56.1	49.2
OBC	34.3	43.2
General	23.5	7.6
All CG	44.2	100.0

Notes: Shares computed from the NFHS-V Chhattisgarh sample with children aged 4–7, survey-weighted. "% below BPL" is the share of each caste group classified as below the poverty line, where a household is classified as below BPL if its rank within the district-rural/urban asset distribution falls below the district SECC poverty rate. "Population share (%)" is the share of the state population in each caste group. SC and ST households are caste-eligible for the RTE quota regardless of income, so the complement of "% below BPL" (e.g., 55.0% of SC households and 38.5% of ST households) qualifies via the caste track alone, despite failing the income criterion. OBC and General households are not caste-eligible and can only qualify via the income track.

Further, the policy is also regressive within caste groups: among Scheduled Caste and Tribe applicants, those who actually benefit are positively selected on wealth and parental education relative to the population. Figure C.1, Panel (a), plots the distribution of the household SES index in Chhattisgarh by caste group. Scheduled Tribes are the poorest group and are heavily concentrated at the bottom of the statewide distribution. Scheduled Caste households are the next poorest on average, but with more spread. Panel (b) overlays, within Scheduled Castes and within Scheduled Tribes, the overall wealth distribution with the wealth distribution of RTE applicants: within both groups, applicants are strongly positively selected on wealth. Tables C.2 and C.3 confirm this pattern, documenting significant gaps in assets, parental education, and rural residence between applicants and the average Scheduled Caste and Tribe household.

Figure C.1:
Wealth distribution by caste group, and applicants vs. state population within SC and ST



Note: Panel (a) plots the kernel density of the SES index by caste group in the NFHS-V Chhattisgarh sample with at least one child aged 4–7 (survey-weighted). Panel (b) overlays, for SC and ST, the NFHS-V population density with the density of RTE applicants (IPW-weighted). The SES index is the first principal component of household assets and durables, constructed on the NFHS-V sample and standardized; applicants are mapped to the same index using the NFHS PCA loadings.

Table C.2: Differences between applicants,
Scheduled Caste households and the average household in Chhattisgarh

	NFHS Scheduled Caste	Applicants	(1)-(2)
	(1)	(2)	(3)
Panel A: Summary			
Asset index	0.17	0.07	0.10
Mother education: above primary	0.60	0.79	-0.18***
Father education: above primary	0.72	0.83	-0.11***
Panel B: Assets			
Owns house	0.60	0.90	-0.29***
Rooms	2.03	2.38	-0.35***
table	0.33	0.34	-0.01
cot or bed	0.94	0.85	0.09***
chair	0.81	0.74	0.06**
has electricity	0.99	0.98	0.00
electric fan	0.83	0.88	-0.06**
has television	0.75	0.57	0.18***
has refrigerator	0.22	0.18	0.05*
has bicycle	0.68	0.75	-0.07**
has motorcycle/scooter	0.57	0.37	0.20***
Panel C: Maternal education			
no education	0.20	0.07	0.13***
incomplete primary	0.08	0.03	0.04***
complete primary	0.12	0.10	0.02
incomplete secondary	0.44	0.54	-0.11***
complete secondary	0.09	0.16	-0.07**
higher	0.07	0.08	-0.01
Panel D: Paternal education			
no education	0.11	0.02	0.09***
incomplete primary	0.07	0.03	0.04***
complete primary	0.10	0.11	-0.01
incomplete secondary	0.48	0.40	0.08**
complete secondary	0.12	0.31	-0.19***
higher	0.13	0.12	0.00

Notes: This table shows the prevalence of different characteristics for households in the representative National Family Health Survey (NFHS) 2019–21 (Column 1) for Scheduled Caste households, and all Scheduled Caste applicants (Column 2). We re-weight our sample to account for differential non-response by household characteristics. We estimate the probability of responding to our survey using a probit model that accounts for the household district, caste, and the child's age and gender. It also shows the differences between the samples and whether these differences are statistically significant (Column 3). Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

Table C.3: Differences between applicants,
Scheduled Tribe households and the average household in Chhattisgarh

	NFHS Scheduled Tribe	Applicants	(1)-(2)
	(1)	(2)	(3)
Panel A: Summary			
Asset index	-0.28	0.27	-0.56***
Mother education: above primary	0.41	0.67	-0.26***
Father education: above primary	0.58	0.88	-0.30***
Panel B: Assets			
Owns house	0.63	0.90	-0.27***
Rooms	2.05	3.10	-1.05***
table	0.22	0.44	-0.22***
cot or bed	0.87	0.95	-0.08***
chair	0.71	0.74	-0.02
has electricity	0.97	0.99	-0.03***
electric fan	0.58	0.80	-0.22***
has television	0.48	0.45	0.03
has refrigerator	0.09	0.20	-0.11***
has bicycle	0.70	0.81	-0.11***
has motorcycle/scooter	0.44	0.45	-0.01
Panel C: Maternal education			
no education	0.36	0.17	0.20***
incomplete primary	0.06	0.03	0.03***
complete primary	0.17	0.12	0.05***
incomplete secondary	0.33	0.39	-0.06***
complete secondary	0.06	0.18	-0.12***
higher	0.03	0.11	-0.08***
Panel D: Paternal education			
no education	0.20	0.07	0.13***
incomplete primary	0.07	0.02	0.06***
complete primary	0.15	0.03	0.12***
incomplete secondary	0.44	0.53	-0.09***
complete secondary	0.08	0.19	-0.10***
higher	0.06	0.17	-0.11***

Notes: This table shows the prevalence of different characteristics for households in the representative National Family Health Survey (NFHS) 2019–21 (Column 1) for Scheduled Tribe households, and all Scheduled Tribe applicants (Column 2). We re-weight our sample to account for differential non-response by household characteristics. We estimate the probability of responding to our survey using a probit model that accounts for the household district, caste, and the child's age and gender. It also shows the differences between the samples and whether these differences are statistically significant (Column 3). Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

In sum, the two eligibility tracks compound rather than offset each other. Caste-based eligibility delivers only limited representational gains: the poorest caste group (Scheduled Tribes) is sharply under-represented in the applicant pool relative to its

population share. Among Scheduled Castes and Scheduled Tribes, applicants are wealthier and better educated than the average household in their respective groups. The interaction of caste and income eligibility, therefore, amplifies, rather than offsets, the income regressivity documented in the main text.

D Demand for private schools from low-SES households

Regressive selection is inevitable if too few poor households wish to send their children to private schools, even for free.[55] Similarly, there is limited scope to reduce inframarginality if only a few households that prefer private schools are financially constrained.

To understand preferences over different schools, we adapted the methodology of Delavande and Zafar (2019) to elicit parental preferences for private schooling across the socioeconomic distribution. Specifically, we provided parents with five (fictitious) schools — two public and three private, representing the range of choices in similar markets — from which they must choose where to enroll their child. We presented characteristics for five schools to each household:

1. **Nearby public school:** Enrollment, number of teachers/classrooms, and highest grade were fixed near the 25th percentile of school enrollment in the public sector.
2. **Distant public school:** Enrollment, number of teachers/classrooms and highest grade were fixed near the 75th percentile of school enrollment in the public sector.
3. **Lower-range private school:** Enrollment, number of teachers/classrooms and highest grade in the school benchmarked to the bottom tercile of private school fees.
4. **Mid-range private school:** Enrollment, number of teachers/classrooms and highest grade in the school benchmarked to the middle tercile of private school fees.
5. **Expensive private school:** Enrollment, number of teachers/classrooms and highest grade in the school benchmarked to the top tercile of private school fees.

We randomized the distance from each school to the household: for School 1, this was randomized to be between 250 and 750 meters in increments of 250 meters; for all other schools, this was randomized to be between 500 and 2000 meters in increments of 500 meters. Reflecting actual prevalence, the medium of instruction was fixed as Hindi for Schools 1-3 and English for School 5, and randomized between Hindi and English for School 4. The precise school fees shown to households for each school were randomly selected from a range of values. This was created using the p10-p90 spread in each tercile, with INR 1,000 added to and subtracted from the boundaries to ensure overlap in fee distributions across schools.

The specific values for each characteristic were computed separately for rural and urban areas in Raipur and are presented in Table D.1. These characteristics were presented to households using a visual stimulus (see Figure D.1) and a surveyor explained each of the characteristics in detail before eliciting responses.

[55] Low demand may result if, for example, low-SES parents expect that private schools will not adequately cater to their children's learning needs or that they will face discrimination from other students or teachers. Bau (2022) documents that private schools in Pakistan have a higher value-added for richer students.

Table D.1: School characteristics shown to respondents

Characteristic Sector	School 1 Public	School 2 Public	School 3 Private	School 4 Private	School 5 Private
			URBAN AREAS		
Enrollment	75	200	190	150	250
Num. of teachers	3	9	12	12	17
Num. of classrooms	5	8	8	8	9
Highest grade offered	5	5	8	8	12
Medium of instruction	Hindi	Hindi	Hindi	Hindi/English	English
Annual fees (in Rupees)	0	0	2000-5500	3500-8500	6500-50000
Distance	250-750 m	500-2000 m	500-2000 m	500 - 2000 m	500-2000 m
			RURAL AREAS		
Enrollment	75	150	170	140	120
Num. of teachers	3	7	11	12	11
Num. of classrooms	4	6	8	8	8
Highest grade offered	5	5	8	8	8
Medium of instruction	Hindi	Hindi	Hindi	Hindi/English	English
Annual fees (in Rupees)	0	0	2000-4500	2500-8000	6000-20000
Distance	250-750 m	500-2000 m	500-2000 m	500 - 2000 m	500-2000 m

Notes: This table shows the specific values for each characteristic shown to households in rural and urban areas. Some survey characteristics were randomized within realistic ranges observed in the data. The medium of instruction was randomized for School 4 between Hindi and English. Fees were randomized in a range within each private option, as shown above. Distance was randomized in increments of 250 meters for School 1 and 500 meters for the other 4 schools.

Parents were asked to rank these schools under three different scenarios[56]:

- **Status quo:** Public schools are free to attend, private schools charge their sticker price
- **RTE policy:** One private school (randomly selected) is made free to attend, the others charge their sticker price
- **Unconditional voucher:** All 5 schools are free to attend for the household.

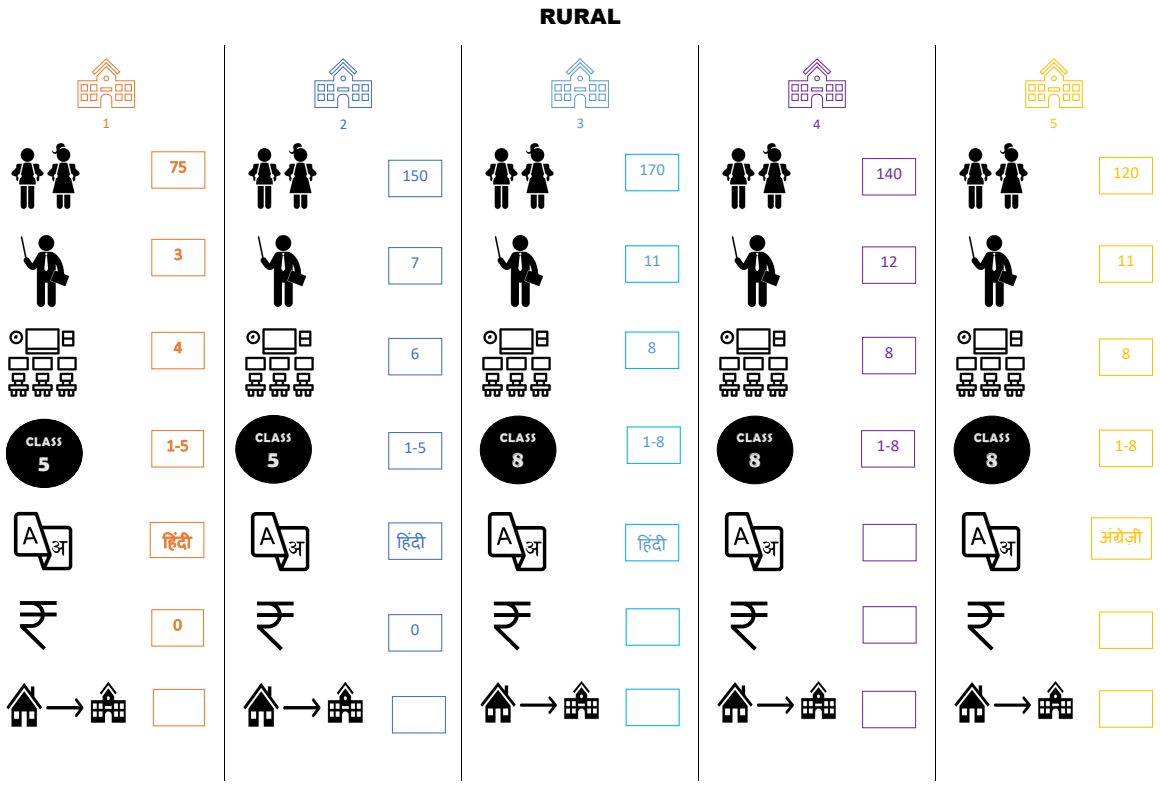
We envisaged the comparison between the "status quo" and the "unconditional voucher" scenario as a measure of financial constraints. The RTE scenario mirrors the policy's actual design, and we use it to study substitution across schools when only one school is made free.

Each school is characterized by a vector of characteristics, including the number of classrooms, total enrollment, the number of teachers, language of instruction, distance to the household, the highest class offered, and the unsubsidized fee levels. We ask parents to rank the schools from 1 (most preferred) to 5 (least preferred) in three scenarios: (a) when only public schools are free, while private schools charge posted tuition fees ("status quo"), (b) when one private school, randomly-chosen, is made

[56]In scenarios where schools have a positive price, households were also allowed to indicate that they could not afford to send their child to the school. This option was added after piloting, when it became apparent that parents wanted to distinguish this from rating a school as bottom-ranked. This distinction is not relevant to our purposes, as we focus only on the top-ranked choice in each scenario in our analysis.

free through a school-specific tuition waiver ("RTE quota") and (c) when all schools are made free through an unconditional scholarship to the student ("voucher"). The order of scenarios is randomized to avoid framing effects.

Figure D.1: Example of stimulus shown to parents for stated choice exercise



Note: This figure shows the response stimulus as shown to individuals. Fields that are empty correspond to values that were generated randomly. In addition to the information shown above, surveyors informed respondents that Schools 1 and 2 were government schools, while Schools 3-5 were private schools.

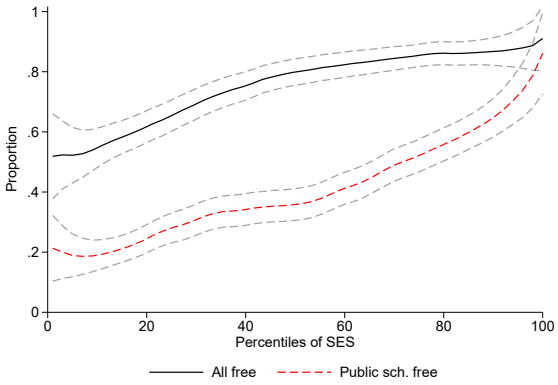
Our goal is to study parental demand for private schools by socioeconomic status and the extent of financial constraints. We are interested in two quantities. First, the proportion of households across the SES distribution who choose a private school as their top choice in the "voucher" scenario. This provides a direct measure of whether too few poor parents demand private schools, even when offered free of charge. Second, we want to compare this quantity to responses by the same parents when they have to pay the market price for private schools. This provides an estimate of the proportion of parents across the SES distribution who are financially constrained in sending their children to private schools, i.e., for whom vouchers may induce extensive-margin shifts to private schooling.

We plot these quantities non-parametrically in Figure D.2, which shows three important results. First, parental demand for private schools is high even for the poorest: when offered for free, the majority of parents across the SES distribution prefer a private school as their top choice and, in the bottom SES quintile, this figure is 53%. Second, the policy is not misguided in perceiving financial constraints as a major barrier to private school access for disadvantaged households: demand for private schools is much lower in the "status quo" scenario than the "voucher" scenario, especially for

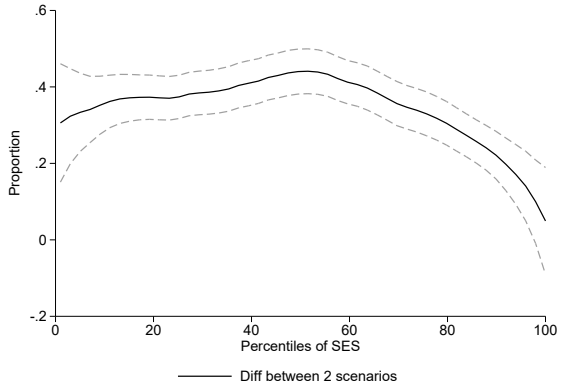
poorer households. Approximately 40% of all below-median SES households appear financially constrained on this extensive margin. Third, there *is* an SES gradient in the demand for private schools, which is steeper for below-median households. However, it cannot explain undersubscription by itself because the proportion of poor households is large and base demand is high.

Figure D.2:
Parental demand for private schools in the "voucher" and "status quo" scenarios

(a) Proportion choosing private schools



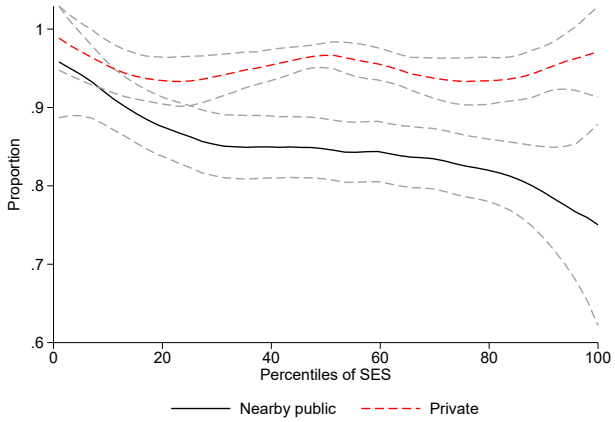
(b) Difference



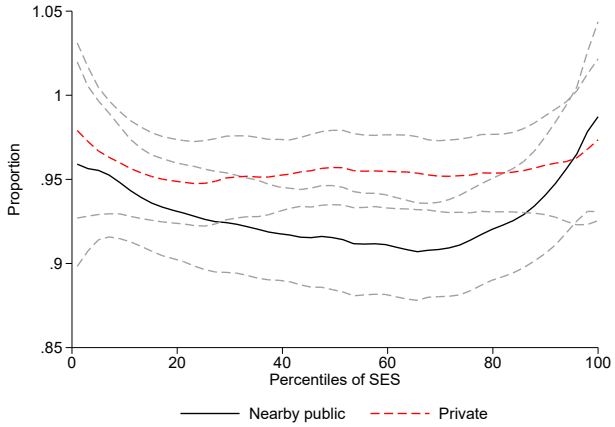
Note: Figure D.2a shows local linear regressions which plot (with 95% confidence intervals), against percentiles of SES, the proportion of parents who choose a private school as the top choice school in (a) "voucher scenario", where all schools are made free for children to attend, and (b) in the "status quo", where public schools are free but private schools charge their posted school fees. Figure D.2b shows the difference in proportion of parents choosing private schools in the two scenarios across the SES distribution. We measure socioeconomic status using an index constructed from household ownership of assets, consumer durables, and housing quality, based on Principal Component Analysis (Filmer and Pritchett, 2001). We use an Epanechnikov kernel with a bandwidth of 8 percentiles in all regressions in this plot.

Figure D.3: Parental expectations of child experience in private schools

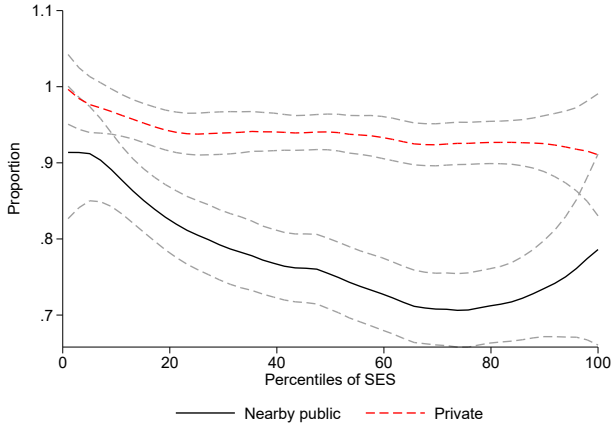
(a) Child likely to be happy



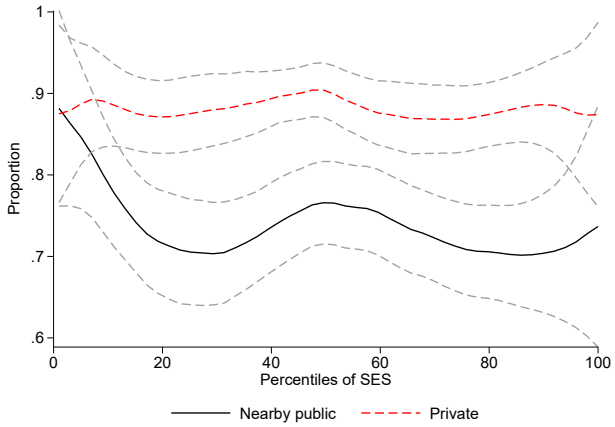
(b) Will have friends, enjoy activities



(c) Teachers will pay attention



(d) Good job at 30



Note: This figure shows output from local linear regression plots (with 95% confidence intervals) relating the percentiles of SES to whether parents responded "Likely" or "Very Likely" to four statements, asked with respect to their child's experience at the nearby government school and a (randomly-chosen) private school: (i) How likely do you think that the child will be happy at his school?; (ii) How likely do you think that the child will have friends and enjoy social activities in school?, (iii) How likely do you think that teachers will pay attention to the child?, (iv) How likely do you think the child will have a good job by the time he is 30?. Parents' responses were elicited on a 5-point scale from "Very Unlikely" to "Very likely" (with additional codes for "Don't know" and "Don't want to answer", which have been removed from the sample for these regressions). All local linear regressions use an Epanechnikov kernel with a bandwidth of 10 percentiles.

We further elicited parents' expectations for their child's experience across different schools. We asked parents on a 5-point scale to report how likely it was that their child will make friends, that the teacher will pay attention, that the child will get a good job at age 30, and that the child will be happy in the school.[57] In all dimensions, parents expect their children's experience in private schools to be better than in the nearby government school (Figure D.3). Overall, low demand among poorer households appears unlikely to fully account for the low application rates.

[57] To reduce survey fatigue, we only asked this for one government and one private school.

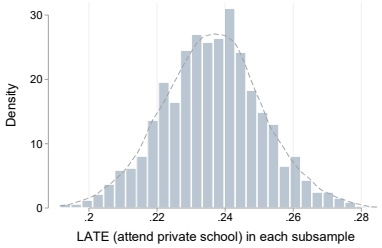
E Extrapolating treatment effects to deterministic applicants

We extrapolate LATE estimates to deterministic applicants using the reweighting procedure of Angrist and Fernández-Val (2010). The approach reweights covariate-specific LATEs from the lottery sample to match the covariate distribution in the deterministic sample, assuming that treatment effects are homogeneous conditional on observables. Covariates include caste (SC, ST, OBC, or other), rural residence, an indicator for listing more than one school, and above-median indicators of public school availability, private school density, Scheduled Caste and Tribe population shares, and poverty rate — each interacted with application grade.

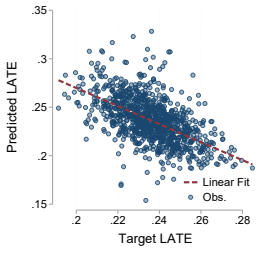
We assess the reliability of this extrapolation using the split-sample procedure (à la Bisbee et al. (2017)). In each of 1,000 simulations, we randomly select half of the experimental strata, estimate the LATE on that half, and extrapolate to the remaining strata using the reweighting procedure. We then compare the extrapolated LATE to the directly estimated LATE in the hold-out strata (Figure E.1). The distribution of directly estimated LATEs across subsamples is tight (SD = 0.015, with a mean of 0.24), consistent with limited heterogeneity across strata (Figure E.1a). Bias is low: 83.5% of predictions fall within 5 percentage points of the target LATE (Figure E.1c). However, the correlation between targeted and predicted LATEs is negative (Figure E.1b), indicating that the procedure does not predict which strata have higher or lower effects. We interpret this combination — low bias but no predictive power at the stratum level — as consistent with limited treatment-effect heterogeneity: the overall extrapolated effect is reliable, but stratum-level variation is too small to target.

Figure E.1: Extrapolating LATE estimates using complier characteristics

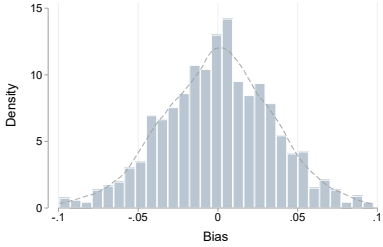
(a) Distribution of LATEs



(b) Directly estimated LATE vs extrapolation estimation



(c) Bias distribution



Note: The figure reports the results of a validity check for the extrapolation procedure for IV estimates proposed by Angrist and Fernández-Val (2010), following the split-sample approach proposed by Bisbee et al. (2017). The procedure consists of 1,000 simulations. In each simulation, we randomly select half of the experimental strata to estimate the LATE (estimation sample) and extrapolate these estimates to the remaining half of the strata (target sample) using the re-weighting procedure from Angrist and Fernández-Val (2010). The extrapolation re-weights covariate-specific LATEs to match the distribution of observable characteristics in the target sample (including gender, caste, application grade, and village attributes). Figure E.1a shows the distribution of the (directly) estimated LATEs across the 1,000 estimation subsamples. Figure E.1b compares the extrapolated (predicted) LATE against the directly estimated (target) LATE for the hold-out strata in each simulation, with the solid line representing a linear fit. Figure E.1c displays the distribution of the prediction bias, defined as the difference between the directly estimated LATE and the extrapolated LATE.

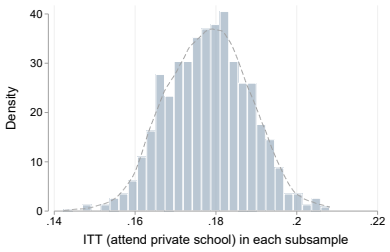
We complement this with conditional average treatment effects (CATE), which estimate

individualized treatment effects via generalized random forests (Athey et al., 2019; Wager and Athey, 2018). This nonparametric approach does not impose the conditional homogeneity assumption. The variables used to estimate individual treatment effects include caste, rural residence, number of schools listed, application grade, preference group size, and continuous village-level measures of literacy rate, poverty rate, Scheduled Caste and Tribe population shares, public school availability, school facilities, and private school density. We apply the same split-sample validation, estimating the CATE model on half the strata and predicting ITTs in the other half across 1,000 simulations (Figure E.2). Results mirror the IV reweighting findings: the ITT distribution is tight across subsamples (Figure E.2a, SD = 0.01, with a mean of 0.18), bias is low (Figure E.2c, 99.7% of predictions fall within 5 percentage points of the target), and the correlation between predicted and estimated effects is negative (Figure E.2b).

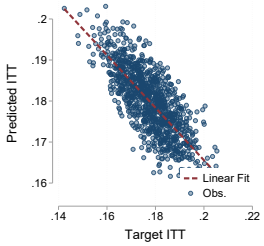
Both approaches yield extrapolated effects close to the lottery-sample estimates reported in Section 5.3. The consistency across a parametric reweighting method and a nonparametric machine learning approach reinforces the conclusion that treatment effects for deterministic applicants are similar to those in the lottery sample.

Figure E.2: Extrapolating ITT estimates using generalized random forests

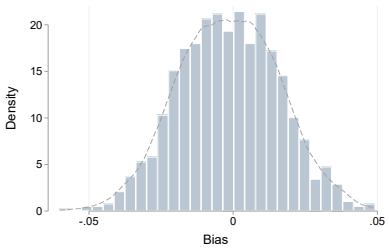
(a) Distribution of ITTs



(b) Directly estimated ITT vs predicted



(c) Bias distribution



Note: The figure reports the results of a validity check for the extrapolation procedure using conditional average treatment effect (CATE) estimation via the partialing-out estimator with generalized random forests Athey et al. (2019). We follow a split-sample approach with 1,000 simulations. In each simulation, we randomly select half of the experimental strata to estimate the CATE model (estimation sample) and use it to predict treatment effects (ITT) in the remaining half of the strata (target sample) based on observable applicant and village characteristics. Figure E.2a shows the distribution of the (directly) estimated ITTs across the 1,000 target subsamples. Figure E.2b compares the extrapolated (predicted) ITT against the directly estimated (target) ITT for the hold-out strata in each simulation, with the solid line representing a linear fit. Figure E.2c displays the distribution of the prediction bias, defined as the difference between the directly estimated ITT and the extrapolated ITT

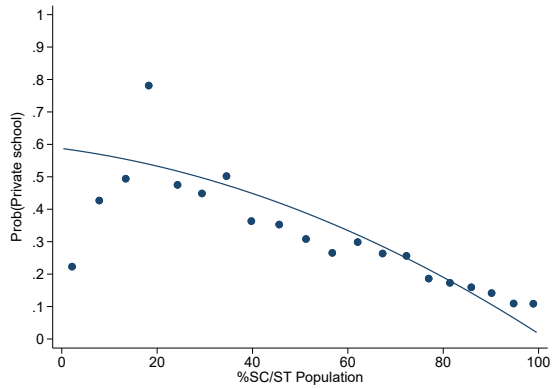
F Spatial supply-side constraints to quota take-up

Our analysis primarily examined why households do not apply for quota seats, even when private schools are available. This focus was appropriate because information constraints and application hurdles are relevant margins for policy action to improve take-up. Overall, however, a further constraint on improving school integration is whether private schools are located in communities with high proportions of disadvantaged households. If most eligible households have few private schools available in their communities, the policy would be limited in the ability to reduce stratification (see, e.g., Monarrez (2022); Campos and Kearns (2022)).

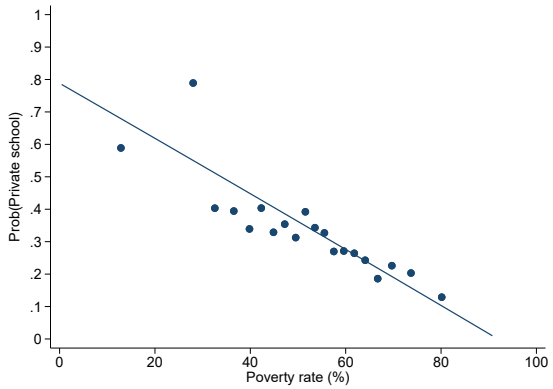
We study the extent to which such spatial constraints might matter. To this end, we assembled geolocated population and poverty rate data for individual villages and towns (Asher et al., 2021), along with GPS locations of recognized schools and administrative data on school characteristics.[58] Private schools are common in rural and urban areas, but not universal. They are less likely to be available in communities with a higher proportion of eligible disadvantaged groups (Figure F.1). In the state overall, 43.4% of Grade 1 enrollment is in communities with only public schools; this figure is 53.2% for Scheduled Caste and Tribe students and 55.1% for income-poor students.[59] Thus, the spatial distribution of private schools and disadvantaged groups is a relevant constraint for explaining statewide differences in the probability of private school enrollment for disadvantaged and non-disadvantaged households. While the proportion of students from Scheduled Castes and Tribes is 28.0 percentage points higher in public schools state-wide, 60% of this difference (17 percentage points) is within communities (see Table F.1), thus substantial segregation remains within communities with both private and public schools.

Figure F.1: Likelihood of a community
having a private school by the proportion of the eligible population in the community

(a) Scheduled Castes and Scheduled Tribes



(b) Poverty rate



Note: Figure F.1a presents binscatter least squares estimation — following Cattaneo et al. (2024) — of the relationship between the percentage of the population from Scheduled Castes and Scheduled Tribes and the likelihood of having a private school in the community. Figure F.1b presents the relationship between the poverty rate and the likelihood of having a private school in the community. Each observation is a community (defined by its SHRUG-ID) and is weighted by the total population.

F.1 Simulating potential reallocation of RTE quota seats

The analysis in Section 4 showed that regressive benefit incidence is driven by differential take-up rather than weak targeting. A related question is whether addressing undersubscription could meaningfully reduce caste-based segregation between public and private schools. We conduct two simulation exercises to understand this.

[58] We retrieved school GPS locations from the official website https://schoolgis.nic.in/ in October of 2021, which we matched to school management and enrollment data from U-DISE. We matched 99.8% of students in government schools and 95.9% of students in private schools to their locations.

[59] There is no direct poverty measure in the U-DISE data. We assume the proportion of enrolled children who are income-poor in a given neighborhood matches the proportion of income-poor households. This is a conservative estimate since fertility is typically larger in poorer households (IIPS, 2017).

The first simulation assumes that all currently used seats are occupied by Scheduled Caste and Tribe children and reallocates them to public schools in the same "neighborhood".[60] This first reallocation exercise approximates the worst-case scenario for the private-public difference in caste composition of schools if the policy had zero take-up.[61]

In the second scenario, we assign every currently unfilled quota seat in each private school to a Scheduled Caste or Tribe student from a public school in the neighborhood; this approximates the best-case scenario of addressing demand-side frictions and ending undersubscription.

In the status quo, the share of Scheduled Caste and Tribe students in the public sector is ∼28.0 percentage points higher than in the private sector. In Scenario 1, this increases to 34 percentage points, while in Scenario 2, it reduces to 16 percentage points (see Table F.1). Furthermore, the best-case scenario reduces the private-public gap within postcodes to near zero. Overall, even accounting for spatial constraints, addressing undersubscription could reduce the caste composition gap between public and private schools by half.

[60] We define "neighborhood" in two different ways. The results are qualitatively similar in both cases. First, we use postal codes to define a neighborhood. Postal codes come from the U-DISE dataset, but in India, postal codes are larger than in many other settings. The second alternative maps school coordinates to SHRUG IDs (Asher et al., 2021) and defines each ID as corresponding to a distinct neighborhood.

[61] This exercise is distinct from imagining a situation where the policy itself did not exist since we take the capacity, price and characteristics of schools as given. The introduction of a large policy of this nature likely had market-level effects on these variables, as well as private school entry and exit.

Table F.1: Proportion of Scheduled Caste and Scheduled Tribe students in public and private schools under different scenarios

	Private mean (1)	Public mean (2)	Difference (3)	Difference Block F.E. (4)	Difference Postcode F.E. (5)	Difference Shrug F.E. (6)
Panel A: Movements within postal code						
% (SC+ST)	28.21	56.37	28.17***	20.02***	19.96***	17.03***
	(21.69)	(33.13)	(0.48)	(0.40)	(0.41)	(0.47)
	[5,183]	[30,221]				
% (SC+ST-Used Seats)	23.15	57.24	34.10***	26.44***	26.60***	24.02***
	(22.99)	(32.87)	(0.49)	(0.43)	(0.44)	(0.50)
	[5,170]	[30,221]				
% (SC+ST+Available Seats)	38.59	52.56	13.96***	3.22***	2.02***	-2.23***
	(20.25)	(34.99)	(0.49)	(0.39)	(0.40)	(0.47)
	[5,183]	[30,220]				
Panel B: Movements within shrug code						
% (SC+ST)	28.21	56.37	28.17***	20.02***	19.96***	17.03***
	(21.69)	(33.13)	(0.48)	(0.40)	(0.41)	(0.47)
	[5,183]	[30,221]				
% (SC+ST-Used Seats)	23.33	57.24	33.92***	26.24***	26.21***	25.98***
	(23.02)	(32.84)	(0.49)	(0.43)	(0.44)	(0.51)
	[5,171]	[30,221]				
% (SC+ST+Available Seats)	36.09	52.08	16.00***	5.41***	5.27***	-11.20***
	(20.89)	(36.03)	(0.49)	(0.42)	(0.43)	(0.47)
	[5,183]	[30,170]				

Notes: %(SC+ST) is the percentage of Grade 1 enrollment taken by students who are from a Scheduled Caste or a Scheduled Tribe. %(SC+ST-Used Seats) estimates the percentage of Grade 1 enrollment taken by students who are from a Scheduled Caste or a Scheduled Tribe in the absence of the RTE policy, assuming all used RTE seats are taken by children from those groups and that these students would otherwise go to a public school. %(SC+ST+Available Seats) estimates the percentage of Grade 1 enrollment accounted for by students from Scheduled Castes or Scheduled Tribes, assuming all (additionally available) RTE seats are taken by children from those groups (and these students come from public schools). Column 1 shows the mean in private schools (standard deviation in parentheses), while Column 2 shows the mean in public schools (standard deviation in parentheses). Column 3 presents the difference (with its standard error in parentheses), Column 4 presents the difference with block fixed effects (with its standard error in parentheses), Column 5 presents the difference with postal code fixed effects (with its standard error in parentheses), Column 6 presents the difference with village/town fixed effects, as defined by the SHRUG-ID created by Asher et al. (2021) (with its standard error in parentheses). The estimates are weighted by total enrollment. Panel A assumes that all student movements occur within postal codes, whereas Panel B assumes they occur within villages/towns, as defined by the SHRUG-ID (Asher et al., 2021). The data is restricted to schools with at least some enrollment in Grade 1. Statistical significance at the 1, 5, 10% levels is indicated by ***, **, and *.

G Applications, allocation, and schooling responses

In Section 5.3, we showed that the average response to an offer is roughly the same across all values of the lottery risk in the data. In this Appendix, we sketch a model and conditions under which the composition of applicants is roughly the same across lottery-risk levels, which is consistent with this finding.

G.1 Setup

We assume there are N households of mass 1 with one child each. School fees, quality, and spots are fixed. All households can apply for a voucher to a private school j. All households can choose to send their children to public school, to private school, or not to school.

We let θ describe household type (which includes preferences, taste shocks, and socioeconomic characteristics). We let $B_\theta \geq 0$ describe the gain in optimized utility from receiving a voucher, and we let g_θ be the probability that a household of type θ can apply (that they are eligible, aware of the program, have the correct documents, etc.).[62] A household will apply if the cost of applying $C \leq qB_\theta$, where q is the probability that the household will be offered a voucher. Because the lottery has no priorities, all households applying to the same school face the same q.

Conditional on type, costs are drawn from

$$F_\theta(c) = (c/\bar{c}_\theta)^\kappa \quad \text{for } 0 \leq c \leq \bar{c}_\theta, \qquad \kappa > 0, \quad B_\theta \leq \bar{c}_\theta. \tag{4}$$

In other words, each household draws an application cost c, in utility units, from a distribution ranging from zero to a ceiling $\bar{c}_\theta$. The exponent κ is common to all types.[63] Note the support starts at zero, so there is no fixed cost that every applicant must pay.

The ceiling $\bar{c}_\theta$ and the probability of being able to apply at all (g_θ) can differ by type, for example by caste or wealth. $B_\theta \leq \bar{c}_\theta$ specifies that the gain never exceeds the cost ceiling, so even when an offer is certain ($q = 1$), some households of every type still find applying not worth it.

Finally, let $J_\theta \in \{0, 1\}$ indicate whether the voucher moves the child into school j, and $M_\theta \in \{0, 1\}$ indicate whether it moves the child into the private sector. Note $0 \leq M_\theta \leq J_\theta \leq 1$ since the voucher improves only the utility of attending j. A household that switches from another private school has $J_\theta = 1$ and $M_\theta = 0$.

G.2 Applications and offers

Given the setup above, type θ applies with probability

$$P(apply|\theta) = g_\theta F_\theta(qB_\theta) = q^\kappa g_\theta (B_\theta/\bar{c}_\theta)^\kappa$$

That is, the probability a household can apply times the probability its application cost is below its gain. The number of households that apply for a school with probability q of producing an offer is:

[62] A voucher can only increase the household's utility at j because the household can keep going to j and save on fees if it receives a voucher, and it can always choose not to accept a voucher (or apply for one).

[63] With $\kappa = 1$ the distribution is uniform. With $\kappa > 1$ most households have costs near the ceiling and few have cheap applications. With $\kappa < 1$ most households find applying cheap and a few face high costs.

$$A(q) = N\mathbb{E}_\theta[g_\theta F_\theta(qB_\theta)] = Hq^\kappa, \qquad H = N\mathbb{E}_\theta[g_\theta(B_\theta/\bar{c}_\theta)^\kappa] \tag{5}$$

In short, it's the number of households (N) times the expected value (over all possible types) of the probability a household of type θ applies. Because we assumed F_θ is a power function, $F_\theta(qB_\theta) = q^\kappa \times (B_\theta/\bar{c}_\theta)^\kappa$, and since q^κ is constant, we can take it out of the expected value. Application mass (H) is then a school characteristic, which is larger when more households value the school, when access is broader, or when applying is cheaper.

With Q seats, the equilibrium offer probability solves $q = \min\{1, Q/A(q)\}$, which has the unique solution $q^* = \min\{1, (Q/H)^{1/(\kappa+1)}\}$.

A notable consequence is that applicant composition does not depend on the offer probability. Relative to the population, type θ is over-represented among applicants by the factor

$$\frac{g_\theta q^\kappa (B_\theta/\bar{c}_\theta)^\kappa}{\mathbb{E}_\theta[g_\theta q^\kappa (B_\theta/\bar{c}_\theta)^\kappa]} = \frac{g_\theta (B_\theta/\bar{c}_\theta)^\kappa}{\mathbb{E}_\theta[g_\theta (B_\theta/\bar{c}_\theta)^\kappa]},$$

which does not depend on q. Selection into application is present, but it is the same at every offer probability.

Hence the average response (i.e., the effect on the likelihood of joining a private school, δ_M) to an offer among applicants does not depend on q either:

$$\begin{aligned}\delta_M = \mathbb{E}[M_\theta \,|\, \text{apply}] &= \frac{\mathbb{E}_\theta[M_\theta \Pr(\text{apply}\,|\,\theta)]}{\mathbb{E}_\theta[\Pr(\text{apply}\,|\,\theta)]} = \frac{\mathbb{E}_\theta[M_\theta g_\theta F_\theta(qB_\theta)]}{\mathbb{E}_\theta[g_\theta F_\theta(qB_\theta)]} \\ &= \frac{q^\kappa \mathbb{E}_\theta[M_\theta g_\theta (B_\theta/\bar{c}_\theta)^\kappa]}{q^\kappa \mathbb{E}_\theta[g_\theta (B_\theta/\bar{c}_\theta)^\kappa]} = \frac{\mathbb{E}_\theta[M_\theta g_\theta (B_\theta/\bar{c}_\theta)^\kappa]}{\mathbb{E}_\theta[g_\theta (B_\theta/\bar{c}_\theta)^\kappa]}.\end{aligned} \tag{6}$$

The same holds for the effect on the likelihood of joining a specific school (δ_J) after replacing M_θ with J_θ.

G.3 A specific case with logit preferences

In the previous construction, we held the distribution of B_θ, $\bar{c}_\theta$, and g_θ fixed as q changed. But these can differ across schools, so δ_M and δ_J could still differ across q if schools with different offer probabilities also differ in these characteristics.

To make these differences explicit, suppose utility from option k is $V_k + \varepsilon_k$, where $V_k = v_k - \alpha p_k$. Here, v_k captures non-price attributes, p_k is the fee, and $\alpha > 0$ measures the utility cost of paying fees. The options include public schools, private schools, and not enrolling. The shocks ε_k are independent type-I extreme-value draws, known to households before applying (these shocks are part of type θ). These preferences imply that the share choosing school j without a voucher is (Train, 2009):

$$s_j = \Pr\big(V_j + \varepsilon_j \geq V_k + \varepsilon_k,\ \forall k\big) = \frac{e^{V_j}}{\sum_k e^{V_k}}$$

Let $\mathcal{P}$ contain the private schools and $\mathcal{O}$ contain public schools and non-enrollment. The total private enrollment share without a voucher is then

$$P = \sum_{k\in\mathcal{P}} s_k = \frac{\sum_{k\in\mathcal{P}} e^{V_k}}{\sum_k e^{V_k}}.$$

Thus, $1 - P$ is the share choosing public schools or not enrolling.

A full fee waiver at private school j raises utility there by $b = \alpha p_j$, leaving the other options unchanged. However, the gain depends on whether the household would

choose j with and without the voucher. In particular,

$$B_\theta = \max\left\{ V_j + b + \varepsilon_j,\ \max_{k \neq j}(V_k + \varepsilon_k) \right\} - \max_k (V_k + \varepsilon_k).$$

A household already choosing j gains exactly b. A household that switches to j gains less than b, since part of the fee saving compensates for its preference for another option. A household that would not choose j even with the voucher gains nothing.

To calculate the average gain, let $W^0 = \max_k(V_k + \varepsilon_k)$ denote the highest utility without a voucher, and let $S = \sum_k e^{V_k}$. The taste shocks have cumulative distribution function $F(\varepsilon) = \exp[-e^{-\varepsilon}]$. The highest utility is below u if utility from every option is below u. Independence of the shocks therefore gives

$$\Pr(W^0 \leq u) = \prod_k \Pr(\varepsilon_k \leq u - V_k) = \prod_k F(u - V_k) = \exp[-Se^{-u}] = F(u - \log S).$$

which is the distribution of a single taste shock plus $\log S$. If C denotes the mean of the taste-shock distribution, expected maximum utility is:

$$\mathbb{E}_\varepsilon[W^0] = \log S + C.$$

With a voucher at school j, in the sum S, the term e^{V_j} is replaced by $e^{V_j + b}$. Denoting the new sum by S^v,

$$S^v = \sum_{k \neq j} e^{V_k} + e^{V_j + b} = S - e^{V_j} + e^{V_j + b} = S + (e^b - 1)e^{V_j}.$$

Expected maximum utility with the voucher is $\log S^v + C$. The constant C is unchanged because the taste-shock distribution is unchanged.

The average gain is then:

$$\begin{aligned}
\mathbb{E}_\theta[B_\theta] &= (\log S^v + C) - (\log S + C) \\
&= \log\left(\frac{S^v}{S}\right) \\
&= \log\left[1 + (e^b - 1)\frac{e^{V_j}}{S}\right] \\
&= \log[1 + s_j(e^b - 1)],
\end{aligned}$$

where the last equality uses the baseline choice share $s_j = e^{V_j}/S$.

G.4 Schooling responses among applicants

We can now use these gains to calculate the responses among applicants. Assume uniform application costs ($\kappa = 1$), common $g > 0$, and common $\bar{c} \geq b > 0$. For $q > 0$, the probability of applying is $gqB_\theta/\bar{c}$. Conditioning on application cancels the common factor $gq/\bar{c}$, giving

$$\delta_J = \frac{\mathbb{E}_\theta[B_\theta J_\theta]}{\mathbb{E}_\theta[B_\theta]}, \qquad \delta_M = \frac{\mathbb{E}_\theta[B_\theta M_\theta]}{\mathbb{E}_\theta[B_\theta]}.$$

Households that already choose j form population share s_j. They gain b and have $J\theta = 0$. Every other household with a positive gain switches to j, so

$$\mathbb{E}_\theta[B_\theta J_\theta] = \mathbb{E}_\theta[B_\theta] - bs_j.$$

To obtain the private-sector response, we also need to know where these switchers would otherwise enroll. A household switching from another private school contributes to J_θ but not to M_θ. Let R_θ be the highest utility among options other than j, and let K_θ identify that option:

$$R_\theta = \max_{k \neq j}(V_k + \varepsilon_k), \qquad K_\theta = \operatorname*{argmax}_{k \neq j}(V_k + \varepsilon_k).$$

Then

$$M_\theta = J_\theta \mathbf{1}\{K_\theta \in \mathcal{O}\}.$$

Under logit preferences, the identity of the best alternative is independent of how much utility it provides. To see this, let $S_{-j} = \sum_{k \neq j} e^{V_k}$ and let $f(x) = e^{-x}F(x)$ be the shock density. For $K_\theta = k$ and $R_\theta = u$, utility at k equals u and all other options except j lie below u. The joint density is

$$f_{R,K=k}(u) = f(u - V_k) \prod_{\ell \neq j,k} F(u - V_\ell) = e^{V_k} e^{-u} \exp[-S_{-j} e^{-u}].$$

Summing over $k \neq j$ gives the marginal density

$$f_R(u) = S_{-j} e^{-u} \exp[-S_{-j} e^{-u}].$$

Dividing the two expressions,

$$\Pr(K_\theta = k \mid R_\theta = u) = \frac{e^{V_k}}{S_{-j}}.$$

This probability does not depend on u, so K_θ and R_θ are independent. Moreover, the pair (B_θ, J_θ) depends only on $(V_j + \varepsilon_j, R_\theta)$ and is therefore independent of K_θ since the shock ε_j is independent of (R_θ, K_θ).

The probability that this best alternative is outside the private sector is

$$\Pr(K_\theta \in \mathcal{O}) = \frac{\sum_{k \in \mathcal{O}} e^{V_k}}{\sum_{k \neq j} e^{V_k}} = \frac{1 - P}{1 - s_j}.$$

Independence therefore allows us to write

$$\mathbb{E}_\theta[B_\theta M_\theta] = \mathbb{E}_\theta[B_\theta J_\theta \mathbf{1}\{K_\theta \in \mathcal{O}\}] = \frac{1 - P}{1 - s_j} \mathbb{E}_\theta[B_\theta J_\theta].$$

Dividing by the average gain and substituting the expression for $\mathbb{E}_\theta[B_\theta]$ gives

$$\delta_J = 1 - \frac{b s_j}{\log[1 + s_j(e^b - 1)]}, \qquad \delta_M = \frac{1 - P}{1 - s_j} \delta_J. \tag{7}$$

Thus, the private-sector response equals the own-school response times the fraction of switchers coming from public schools or non-enrollment. Application selection leaves this fraction unchanged under the common feasibility and cost assumption

In short, across schools, the responses depend on s_j and the fee through b. If these characteristics vary with lottery risk, conditional responses can differ across risk groups.

G.5 Testable implications

We can also use the model to predict how these school characteristics relate to lottery risk. Substituting the expected utility gain into (5), with $\kappa = 1$, gives

$$A_j(q_j) = \frac{Ng}{\bar{c}} q_j \log[1 + s_j(e^{b_j} - 1)], \qquad b_j = \alpha p_j.$$

With quota seats equal to 25% of enrollment, $Q_j = N s_j / 4$. When capacity binds, $q_j A_j(q_j) = Q_j$. Substituting and solving for q_j, including the case where everyone receives an offer, gives

$$q_j = \min\left\{1, \sqrt{\frac{\bar{c}}{4g} \frac{s_j}{\log[1 + s_j(e^{b_j} - 1)]}}\right\}.$$

For the comparisons below, we hold α and application frictions $g/\bar{c}$ fixed across schools.

At a given fee, the logarithmic term grows less than proportionately with s_j, while quota capacity grows proportionately. Larger schools therefore attract more applications, but have fewer applications relative to their quota capacity. The model predicts that schools with larger baseline enrollment shares have higher offer probabilities and

less lottery risk. At a given share, higher fees increase the value of the voucher and application demand, lowering the offer probability. These relationships are strict where capacity binds; otherwise, $q_j = 1$.

G.6 Predictions in the data

The model predicts that larger enrollment shares raise offer probabilities, while higher fees lower them. We examine these predictions using admin data from the 2019 lotteries.

We focus on Grade 1 applicants since the admin data on enrollment from U-DISE needed to calculate enrollment shares does not include preschool grades. We calculate a school's share (s_j) as its 2017–18 enrollment divided by the enrollment in that grade at all recognized public and private schools in its PINCODE (postal code). The main sample contains 1,720 schools with shares of at most 10%, to avoid outliers driving our results.[64]

For each school, we estimate

$$P_j = \mu_{\text{PINCODE}(j)} + \beta_s(100 s_j) + \beta_f \ln p_j + u_j,$$

where P_j is the probability of receiving an offer, and p_j are the yearly fees in rupees. The coefficients of interest are β_s, the association between enrollment share and offer probability at a given fee, and β_f, the association between log fees and offer probability at a given share. The model predicts $\beta_s > 0$ and $\beta_f < 0$. We cluster standard errors by PINCODE.

In our data, $\beta_f < 0$ as predicted, but $\beta_s < 0$, contrary to the model's prediction. The model takes school characteristics as given. Schools may instead respond to the quota by changing capacity, fees, or quality. Observed enrollment shares can reflect these responses, so the cross-school association need not follow the model's comparative static. The model also assumes common application costs across schools. If applying to larger schools is easier, they can attract more applications per seat and have lower offer probabilities.

[64] Of the 2,458 schools in the application data, we remove 249 that we could not match to the 2017–18 U-DISE data and 17 that the U-DISE data did not categorize as private schools. Another 186 have missing, invalid, or zero Grade 1 enrollment. Six lack valid PINCODEs and 222 lack fee data. The 10% cutoff removes 58 schools.

Table G.1: School mean offer probabilities, prior enrollment shares, and fees

	Any offer			First-choice offer		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Main sample						
Share: +1 pp	-1.36^{***}	-0.78^{*}	-2.01^{***}	-1.48^{**}	-0.76	-2.61^{***}
	(0.51)	(0.44)	(0.69)	(0.63)	(0.55)	(0.79)
$\ln p_j$	-1.85^{**}	-2.35^{***}	-2.08^{**}	-4.02^{***}	-4.37^{***}	-3.88^{***}
	(0.89)	(0.85)	(0.85)	(1.22)	(1.17)	(1.21)
Panel B: All shares						
Share: +1 pp	0.01	0.06^{***}	-0.44^{*}	-0.00	0.07^{**}	-0.69^{**}
	(0.02)	(0.02)	(0.26)	(0.03)	(0.03)	(0.33)
$\ln p_j$	-2.01^{**}	-2.42^{***}	-2.34^{***}	-4.36^{***}	-4.53^{***}	-4.35^{***}
	(0.89)	(0.85)	(0.88)	(1.15)	(1.13)	(1.22)
Panel C: Shares at most 5%						
Share: +1 pp	-1.41^{**}	-0.80	-2.41^{***}	-1.48^{**}	-0.68	-3.12^{***}
	(0.59)	(0.53)	(0.83)	(0.75)	(0.66)	(0.95)
$\ln p_j$	-1.79^{**}	-2.43^{***}	-2.06^{**}	-4.04^{***}	-4.51^{***}	-3.92^{***}
	(0.86)	(0.79)	(0.84)	(1.19)	(1.10)	(1.21)
Fixed effects	None	District	PINCODE	None	District	PINCODE

Notes: One observation and equal weight per school. Standard errors clustered by PINCODE appear in parentheses. Columns (1)–(3) use any-offer probabilities; columns (4)–(6) use first-choice offer probabilities. Each outcome is estimated without fixed effects, with district fixed effects, and with PINCODE fixed effects. Coefficients are in percentage points. Panel A restricts shares to at most 10% (1,720 schools, 220 PINCODEs). Panel B includes all shares (1,778 schools, 253 PINCODEs). Panel C restricts shares to at most 5% (1,662 schools, 215 PINCODEs). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ from two-sided tests. Inference conditions on saved probabilities and fixed inputs.

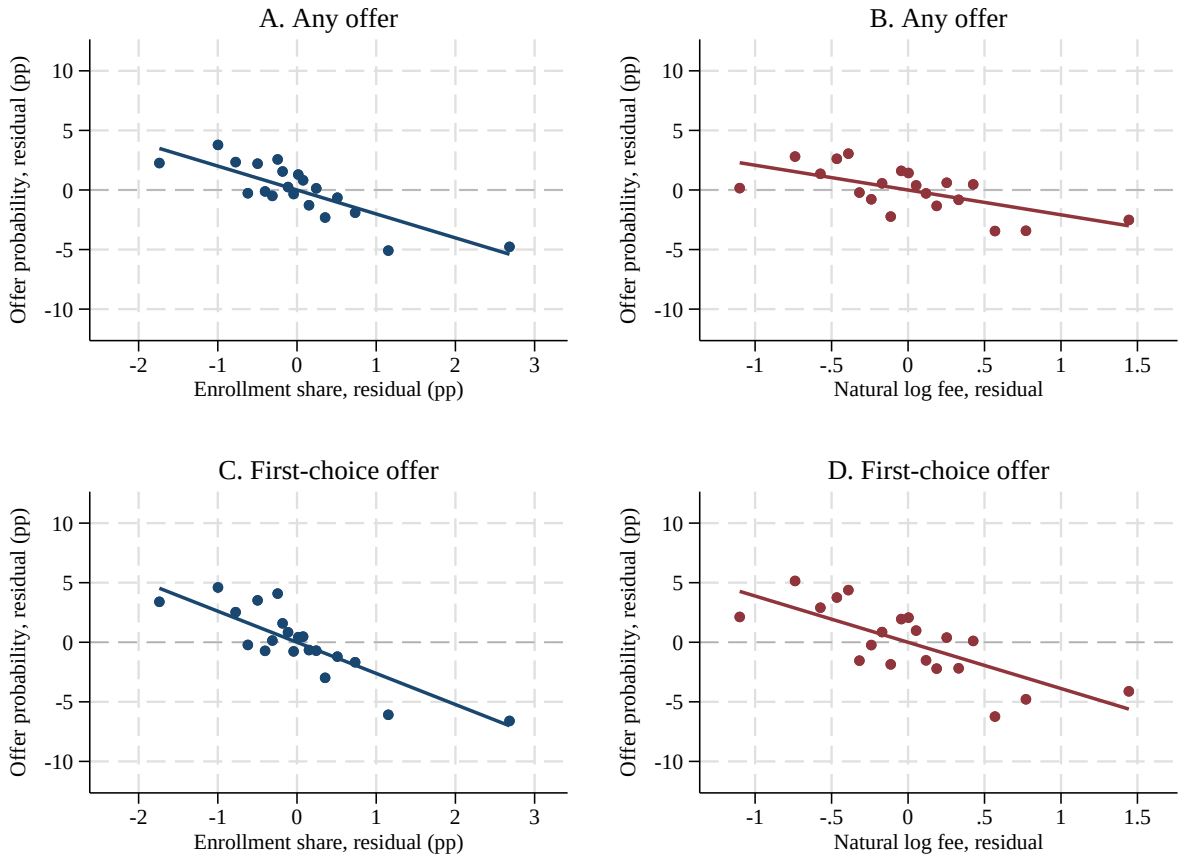


Figure G.1:
School mean offer probabilities, enrollment shares, and fees: partial associations

Notes: All panels use the same 1,720 schools. Outcome and regressor are residualized on PINCODE effects and the other regressor. Dots are means in 20 approximately equal-school-count bins. Lines fit school observations with equal weights. Their slopes equal the PINCODE fixed-effects coefficients in Table G.1. Both axes are linear.